Risk Protection and Redistribution in the Design of Social Insurance

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Abstract

We study how heterogeneity in unemployment risk and exposure to income loss shapes the social value of expanding unemployment insurance (UI). We estimate individual willingness-to-pay (WTP) for UI reforms, capturing insurance value and cross-subsidization. WTP for expanding UI is highest among lower-income households, partly because they face higher unemployment risk. An expansion of the current system yields resource-reallocation gains of $0.67 per $1 of mechanical increase in benefit expenditure, comprising $0.48 from risk protection and $0.19 from cross-subsidization, while incentive costs are $0.58. Of the resource-reallocation gain, 57% reflects efficiency gains, 19% cross-income redistribution, and 24% within-income redistribution.

Keywords: social insurance, redistribution, risk protection, consumption, unemployment insurance

JEL classification: E21, E24, H23, H31, H50, I38, J64, J65

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1. Introduction

A central role of government is to insure households against adverse shocks. In the United States, social insurance programs protect households against a wide range of risks, including unemployment, disability, and poor health. Roughly 36 percent of federal spending—about $2.4 trillion—is devoted to such programs.[1]

Designing effective programs involves a fundamental trade-off between providing insurance and limiting behavioral distortions. The canonical sufficient-statistic approach summarizes this trade-off using a small set of empirical moments: measures of the welfare gain from insurance, often based on consumption changes across states, and behavioral elasticities that summarize incentive costs (e.g., Baily 1978; Chetty 2006; Chetty and Finkelstein 2013). In standard implementations, the welfare gain from insurance is measured using average responses, such as the average consumption gap between "good" and "bad" states of the world, while incentive costs are summarized by a single behavioral elasticity.

While the benchmark approach offers powerful insights, it abstracts from heterogeneity in both the value of insurance and the cost of behavioral distortions. In practice, individuals differ widely in their exposure to shocks, ability to smooth consumption (Blundell et al. 2008), and responsiveness to incentives (Chetty 2008; Attanasio et al. 2018). Moreover, because taxes and benefits are rarely fully risk-adjusted, social insurance programs often involve implicit cross-subsidization. As a result, evaluating changes in program generosity using average responses alone may miss the welfare consequences of the way policy redistributes resources across individuals.

This paper studies the value of expanding unemployment insurance (UI) generosity in the United States, emphasizing how heterogeneity in unemployment risk and unemployment-induced consumption declines shapes the social value of reforms. We provide new evidence on individual willingness-to-pay (WTP) for UI expansions and recover the distribution of WTP across workers. Using these estimates, we decompose reform impacts into (i) the social value of resource reallocation—through risk protection and cross-subsidization—and (ii) incentive costs captured by fiscal externalities.

We use data from the Panel Study of Income Dynamics (PSID) from 1999–2019. This nationally representative panel provides detailed information on labor market outcomes, demographics, and household consumption expenditures. To motivate

[1] We sum Social Security retirement, disability, and survivor insurance; unemployment insurance; and Medicare for those age 65 and older. This share has remained relatively stable over time: Feldstein (2005) reports a value of 37% for 2003.

our analysis, we document that workers who experience an unemployment spell have lower equivalized household consumption, earnings, and household income even *before* job loss than those who remain continuously employed. Moreover, even conditional on income, workers who later become unemployed exhibit lower *in-work* consumption. These patterns point to the potential redistributive role of UI—both across and within income groups.

Our worker-level measure of WTP for UI reform has two components. The first is the individual's *personalized insurance value*: the value of a marginal transfer of resources from the employment state to the unemployment state. The second is their *net fiscal position*: the extent to which they are, ex ante, net contributors to or beneficiaries of the reform. Insurance values vary with the severity of income loss in unemployment and with self-insurance capacity. Net fiscal positions reflect the gap between actuarially fair contributions to the reform and the implicit premium embedded in non-individualized taxes and benefits, giving rise to cross-subsidization in the form of expected net transfers across individuals.

To measure the insurance component of WTP, we extend the "consumption-based" approach to valuing social insurance developed by Gruber (1997) in two ways. First, we provide new evidence on the distribution of unemployment-induced consumption declines across workers, rather than focusing only on average effects. Second, we allow for unobserved heterogeneity in both unemployment risk and consumption responses, as well as correlation between the two.

We adopt the finite-mixture approach of Lewis et al. (2026), which uncovers heterogeneity in consumption responses across latent groups without requiring strong assumptions about which observable characteristics drive that variation. We combine this model for consumption responses with a model of unemployment risk in a joint finite-mixture framework. Unemployment risk is estimated using a finite mixture of logistic regressions that allow for demographic and socioeconomic characteristics to shape baseline unemployment rates . The joint structure yields a posterior distribution over latent risk–consumption types for each worker, which we use to construct both unemployment-induced consumption declines and unemployment probabilities—even for workers we never observe experiencing unemployment.

We estimate an average unemployment-induced consumption decline of about 13.1%, broadly in line with prior studies (e.g., Gruber 1997; Hendren 2017; Kroft and Notowidigdo 2016), but find substantial heterogeneity reflecting differences in self-insurance capacity: for example, the 95th-percentile decline is roughly twice the median. We also recover large and systematic differences in unemployment risk by earnings, age, education, and race (see also Anderson and Meyer 2006). Finally,

workers who are more likely to become unemployed also tend to experience larger consumption declines conditional on unemployment, so many workers face both high unemployment risk and large consumption losses.

The cross-subsidization component of WTP depends on how reform design interacts with heterogeneity in unemployment risk. Using the estimated joint distribution of risk and consumption types, together with workers' earnings, we compute individual risk weights implied by alternative UI reforms. We compare a natural expansion of the U.S. UI system—which replaces a fraction of lost earnings up to a cap—with reforms that expand entitlement either by a flat amount or in proportion to lost earnings with no cap.

We find that WTP for expanding the generosity of the current U.S. UI system is highly heterogeneous. It is positive at the bottom of the income distribution but declines with household income, becoming negative further up the distribution. This pattern arises even though all workers value actuarially fair insurance. The decline reflects that, for most workers, the expected net tax burden of the reform exceeds its insurance value: on average, workers above the bottom quintile are net contributors who cross-subsidize others. Low-income workers benefit because they face higher unemployment risk, an effect strong enough to outweigh the link between benefit entitlements and earnings. A flat benefit reform tilts the WTP distribution further toward lower-income workers, whereas a proportional expansion shifts gains to higher earners.

We also find substantial heterogeneity in WTP within income groups, reflecting differences in unemployment risk and self-insurance capacity even conditional on income. These reform-specific patterns of winners and losers—both across and within income groups—shape the social value of alternative UI reforms and imply that workers may disagree not only about whether to expand UI, but also about which expansion they prefer.

We next consider the social value of a budget-neutral expansion of UI generosity. Since this requires interpersonal comparisons, the evaluation depends on social preferences. In our baseline implementation, we use welfare weights that (i) value redistribution across income groups at the marginal cost of achieving equivalent redistribution through the income tax system (Hendren 2020), and (ii) place greater value on within-income-group redistribution toward individuals with lower long-run equivalized household consumption, a natural measure of lifetime income and economic well-being (Poterba 1989; Meyer and Sullivan 2003). This formulation reflects the idea that broad redistribution across income groups can, in principle, be implemented through the income-tax-and-transfer system. By contrast, UI also reallocates resources within income groups, toward households that face high

unemployment risk and have low in-work consumption or limited self-insurance capacity—dimensions that are difficult to observe and target in the tax code. Our framework is designed to distinguish the redistributive value of UI along these margins from the component that could instead be achieved through changes to the income tax schedule.

We show that the social value of expanding UI can be decomposed into three components. The first is risk protection: a risk- and welfare-weighted average of individual insurance values, scaled by an aggregate welfare weight. The individual weights capture heterogeneity in the marginal social value of surplus among program beneficiaries, while the aggregate weight captures the average marginal value of transferring surplus to beneficiaries relative to the workers who finance the reform. We estimate that the risk-protection value from a budget-neutral expansion of the current UI program is \$0.48 per \$1 of mechanical increase in benefit expenditure.

The second component captures the social value of cross-subsidization, which arises from heterogeneity in net incidence rather than from insurance value itself. In representative-agent frameworks, this component is absent; in our setting, it can generate welfare gains (or losses) even when individuals are fully insured through other means. For the current UI program, we estimate that the cross-subsidization component associated with increased generosity is \$0.19 per dollar. Together with the risk-protection component, this implies total resource-reallocation gains of \$0.67. The decomposition into risk protection and cross-subsidization describes how the reform reallocates resources—across states for a given individual and across individuals through net transfers. To understand why these reallocations are socially valuable, we compare our baseline results to two benchmarks that sequentially remove within- and across-income variation in individual welfare weights. This alternative decomposition implies that about 57% of the resource-reallocation gain reflects efficiency improvements from better insurance, 19% reflects redistribution across income groups, and 24% reflects redistribution within income groups.

The third component is the fiscal externality arising from behavioral responses to program generosity. In the canonical model, these incentive costs are summarized by an aggregate elasticity. In our setting, they depend on how behavioral elasticities covary with individuals' risk and net fiscal positions (i.e., the gap between taxes paid and benefits received across states). We calibrate these heterogeneous elasticities using evidence from Chetty (2008). We estimate a direct fiscal externality of \$0.27 per dollar from higher UI payments induced by behavioral responses, plus an additional \$0.31 from lower income tax receipts. Thus, the total incentive costs of \$0.58 per dollar fall slightly below the \$0.67 resource-reallocation gains from risk protection and cross-subsidization.

Our measure of the social value of changing UI generosity depends critically on reform design. Under a flat expansion—where UI entitlements increase uniformly across workers—the value of resource reallocation rises to \$0.78 per dollar, because this reform directs a larger share of resources toward households that are less well-off and more exposed to unemployment risk. In this case, the welfare gains from increased generosity outweigh the associated incentive costs. By contrast, the standard sufficient-statistics implementation corresponds to a flat reform under social preferences that are indifferent to how surplus is distributed, yielding a much lower estimated resource-reallocation value of \$0.39. We therefore show that abstracting from redistribution across heterogeneous individuals meaningfully alters the conclusions of UI policy analysis. More broadly, these differences across reforms highlight that the social value of UI depends on program design and is therefore likely to vary across countries (Spinnewijn 2020).

We contribute to a literature that uses sufficient-statistics approaches to study the insurance-incentive trade-off in UI design (e.g., Schmieder et al. 2012; Kolsrud et al. 2018; Landais and Spinnewijn 2021).[2] Our contribution is to extend the sufficient-statistics framework for UI beyond the representative-agent setting by combining microdata on consumption and unemployment risk to recover individual-level WTP and to show how heterogeneity shapes the decomposition of the social value of UI reforms across workers. We also build on recent work that measures the value of social insurance transfer programs, including disability insurance (Deshpande and Lockwood 2022), public pensions (Kolsrud et al. 2024), and welfare programs (Rafkin et al. 2023).

Our work also relates to a literature that examines how the correlation between risk and ability shapes the joint design of income taxes and social insurance (e.g., Blomqvist and Horn 1984; Rochet 1991; Cremer and Pestieau 1996; Boadway et al. 2006). It also connects to research on how involuntary unemployment (Kroft et al. 2020) and the structure of non-linear UI benefit schedules (Ferey 2022) influence optimal tax design. This work emphasizes that redistribution across income or ability types can, in principle, be implemented through the non-linear income tax and transfer system, while social insurance instruments such as unemployment insurance play a distinct role by providing state-contingent transfers within types and across employment states. We complement this body of work by quantifying the extent to which current UI systems redistribute surplus across heterogeneous workers—both across and within income groups.

[2] A complementary strand employs dynamic structural models to analyze UI design (e.g., Acemoglu and Shimer 1999; Lentz and Tranaes 2005; Krusell et al. 2010) and highlights the redistributive implications of UI policy across heterogeneous workers (e.g., Audoly 2024; Haan and Prowse 2024).

The rest of the paper is organized as follows. Section 2 outlines our measure of individual willingness-to-pay for increased UI generosity. Section 3 introduces the data and presents preliminary evidence on negative selection into unemployment. Section 4 describes our empirical approach and presents the estimates. Section 5 develops our normative framework and reports the quantitative welfare analysis. Section 6 concludes.

2. Individual Value of a Reform

In this section, we derive an expression for individual willingness-to-pay (WTP) for an increase in the generosity of a social insurance program. We present a simple framework that captures the salient features of unemployment insurance, as well as other programs such as workers' compensation, disability insurance, health insurance, and insurance against natural disasters. The sufficient-statistics expressions we derive extend to a much richer dynamic environment (Chetty 2006; see Appendix A.1).

We build on the canonical sufficient-statistics setup (see Chetty and Finkelstein 2013) by allowing for heterogeneous individuals facing risk. Individual WTP is a key input for evaluating the social value of reform, which in our framework depends both on how WTP is aggregated across individuals and on the incentive costs generated by behavioral responses (see Section 5).

2.1. Theory

Individual's problem. There is a unit continuum of individuals indexed by i. Each individual faces uncertainty over two states: high (h) and low (l), for instance corresponding to being employed and unemployed. Individuals may differ in their preferences, incomes, and the costs associated with mitigating risk. Let y_i and z_i denote the individual's resources in the high and low state, respectively. Taxes and benefits are indexed by high-state income y_i: the government provides a benefit $\mathcal{B}(y_i)$ to those in the low state and levies a tax $\mathcal{T}(y_i)$ on individuals in the high state.[3]

Consumption in the high state is $c_i^h = y_i - \mathcal{T}(y_i)$, while consumption in the low state is $c_i^l = z_i + \mathcal{B}(y_i)$. Let $u_i^s(c)$ denote state-specific utility from consumption c; we assume $u_i^s(\cdot)$ is increasing, concave, and twice continuously differentiable. Let $\Delta c_i \equiv c_i^h - c_i^l$ denote the gap in consumption between the high and low states. For most individuals, we expect $\Delta c_i > 0$.

[3] In our empirical implementation, we also account for the role of family composition in the U.S. tax-and-transfer system. We omit this here to simplify exposition.

Individuals can control the probability of being in the high state by undertaking actions, which we model through a scalar e and refer to as effort. We normalize effort so that e equals the probability of being in the high state (and therefore $e \in [0, 1]$). Effort entails cost $\psi_i(e)$, where $\psi_i(\cdot)$ is increasing and convex. The individual solves:

$$V_i = \max_{e \in [0,1]} \; e\, u_i^h(c_i^h) + (1 - e)\, u_i^l(c_i^l) - \psi_i(e), \tag{1}$$

where V_i is the maximized value of their expected utility. Individual i's optimal effort choice, e_i, satisfies the first-order condition: $u_i^h(c_i^h) - u_i^l(c_i^l) = \psi_i'(e_i)$, i.e., the individual chooses effort so that the utility gain from the high state relative to the low state equals the marginal cost of effort.

Heterogeneity. The index i captures arbitrary heterogeneity from several sources. First, differences in earning ability—for example due to variation in innate ability, human capital, skills, or labor-market opportunities—are reflected in y_i, the individual's high-state resources. Conditional on y_i, variation in low-state resources z_i captures differences in capacity to smooth consumption, for example through savings or spousal insurance. As a result, variation in resources across states reflects both the direct financial loss from entering the low state and differences in self-insurance capacity. Together with the tax and benefit system, these factors generate heterogeneity in the consumption gap Δc_i.

Second, heterogeneity in attitudes toward work and employment opportunities (in the unemployment or disability insurance context), or in the cost of abatement investments (in health or natural-disaster insurance contexts), is captured by the effort-cost function $\psi_i(\cdot)$.

Third, variation in the marginal utility of consumption (for example, due to household composition) and in attitudes toward risk is captured by individual-specific utility functions $u_i^s(\cdot)$. This formulation also allows for heterogeneity in state dependence, which may reflect differences in the opportunity cost of time or the availability of consumption substitutes.

Each of these factors affects the individual's optimal effort choice and therefore generates heterogeneity in risk levels e_i.

Arbitrary reforms. We consider the effects of an arbitrary marginal reform to the benefit and tax schedules.[4] We parameterize the reform by $(db, d\tau)$, such that state-contingent consumption possibilities shift according to:

$$dc_i^l = \phi_{\mathcal{B}}(y_i)\, db, \qquad dc_i^h = -\phi_{\mathcal{T}}(y_i)\, d\tau, \tag{2}$$

i.e., the consumption possibilities in the low and high states change by $\phi_{\mathcal{B}}(y_i)db$ and $-\phi_{\mathcal{T}}(y_i)d\tau$, respectively, where $\phi_{\mathcal{B}}(y_i)$ and $\phi_{\mathcal{T}}(y_i)$ are positive functions of high-state income y_i. While these functions allow for reforms that vary arbitrarily with income, in practice reforms are typically simple functions—for example, a flat benefit increase ($\phi_{\mathcal{B}} = 1$) or a proportional benefit increase ($\phi_{\mathcal{B}} = y_i$), funded by a flat tax adjustment ($\phi_{\mathcal{T}} = 1$). The structure of the reform is central to individual willingness-to-pay because it governs both who receives additional resources in each state and who finances them.

Willingness-to-pay. To define the individual-level WTP for an increase in social insurance generosity, we consider the reform

$$d\theta \equiv \left(db = 1,\ d\tau = \frac{\int_i (1-e_i)\phi_{\mathcal{B}}(y_i)di}{\int_i e_i \phi_{\mathcal{T}}(y_i)di} \right).$$

This reform raises benefits and adjusts the tax schedule just enough to offset the resulting mechanical budgetary cost (holding behavior fixed).

Let $\frac{dV_i}{d\theta}$ denote the impact of this reform on individual i's expected utility.[5] We define individual i's willingness-to-pay for this reform as:

$$\begin{aligned} \text{WTP}_i \equiv & \frac{1}{(1-e_i)\phi_{\mathcal{B}}(y_i)} \frac{1}{u_i^{h\prime}(c_i^h)} \frac{dV_i}{d\theta} \\ = & \underbrace{\left(\frac{u_i^{l\prime}(c_i^l) - u_i^{h\prime}(c_i^h)}{u_i^{h\prime}(c_i^h)} \right)}_{\text{insurance}_i} + \underbrace{\left(1 - \frac{e_i \phi_{\mathcal{T}}(y_i)}{(1-e_i)\phi_{\mathcal{B}}(y_i)} \Big/ \frac{\int_{i'} e_{i'} \phi_{\mathcal{T}}(y_{i'})di'}{\int_{i'} (1-e_{i'})\phi_{\mathcal{B}}(y_{i'})di'} \right)}_{\text{cross-subsidization}_i}. \end{aligned} \tag{3}$$

[4] Focusing on marginal reforms has the advantage that, by the envelope theorem, the first-order impact of the reform on expected utility is fully captured by the change in the state-contingent budget constraint, evaluated at the pre-reform optimum. Throughout, when we write the effect of the reform in terms of dc_i^l and dc_i^h, this should be understood as the induced change in state-contingent consumption possibilities rather than as the policy directly choosing consumption. As a result, willingness-to-pay can be characterized without a full specification of the individual's optimization problem, and WTP in our simple framework aligns with that in a richer dynamic model (see Appendix A.1). The expression can also be interpreted as a first-order approximation to a non-marginal reform, or used to construct higher-order approximations by integrating along the path of a larger reform (e.g., see Kleven 2021).

[5] Throughout, derivatives with respect to θ refer to directional derivatives for the reform under consideration, combining the benefit change with the associated tax adjustment.

The scaling in the first line expresses the utility gain from the reform per \$1 of mechanical increase in individual expected benefit payments, measured in units of the marginal utility of an additional dollar in the high state.

The second line decomposes WTP into two conceptually distinct components. The insurance component captures the value to the individual of transferring a marginal unit of consumption from the high to the low state, valued using their marginal utilities. This reflects the individual-specific insurance value of social insurance and is positive when marginal utility is higher in the low state than in the high state.

The cross-subsidization component captures how the individual's net expected transfer under the reform compares to that under an actuarially fair adjustment. Under actuarial fairness, tax adjustments would be individualized: individual i's expected tax increase $e_i\,\phi_{\mathcal{T}}(y_i)\,d\tau_i^A$ would finance their expected benefit increase, $(1-e_i)\,\phi_{\mathcal{B}}(y_i)\,db$. In practice, however, the actual reform uses a common tax-schedule adjustment that typically departs from actuarial fairness. The cross-subsidization term reflects this deviation: it is positive for individuals whose actuarially fair price is higher than the implicit pooled price embedded in the reform and negative for those whose fair price is lower. It therefore measures whether the individual is a net beneficiary or net contributor relative to a fair-pricing benchmark.

Discussion. Our willingness-to-pay decomposition is related to Finkelstein et al. (2019), who show that WTP for a marginal expansion of Medicaid can be decomposed into a "pure insurance" term—capturing the covariance between marginal utility and insured medical spending across health states—and a "pure transfer" term—representing the expected reduction in out-of-pocket medical spending due to a lower out-of-pocket price. A key difference in our setting is that the pattern of transfers (cross-subsidization) arises from departures from actuarial fairness embedded in the structure of the tax and benefit reform.

An alternative interpretation of our decomposition is that the cross-subsidization component corresponds to the expected net transfer from the reform when it is valued at a common marginal utility across states, while the insurance component captures the additional value that arises because marginal utility is higher in the low state than in the high state. In this sense, the insurance term measures how much the *ex ante*, risk-adjusted value of the reform exceeds its ex post value valued at a common marginal utility. This interpretation aligns with Lieber and Lockwood's

(2019) decomposition of the individual-level value of an in-kind transfer.[6]

2.2. Implementation

The individual-level insurance term depends on the gap in marginal utility of consumption between the low and high states. In our baseline implementation, we assume that marginal utility is state independent and that individuals share a common local curvature of utility at their high-state consumption level.

ASSUMPTION 1. *Marginal utility is state independent (i.e., $u_i^{s\prime}(\cdot) = u_i'(\cdot)$ for $s \in \{l, h\}$) and relative risk aversion at high-state consumption is homogeneous across individuals (i.e., $-\frac{u_i''(c_i^h)c_i^h}{u_i'(c_i^h)} = \gamma$).*

Below we discuss the implications of relaxing both state independence and the assumption of common relative risk aversion. Following Gruber (1997), we adopt a consumption-based approach to measuring the marginal-utility gap. Under Assumption 1, a second-order Taylor approximation to the utility function around c_i^h yields:

$$\frac{u_i'(c_i^l) - u_i'(c_i^h)}{u_i'(c_i^h)} \approx \gamma \frac{\Delta c_i}{c_i^h}, \tag{4}$$

so that the individual insurance component of WTP can be expressed as the product of the percentage consumption decline between the high and low states and the coefficient of relative risk aversion at c_i^h. Our empirical strategy estimates the unconditional distribution of these consumption declines across the population.

Cross-subsidization depends both on state probabilities and the structure of the reform under consideration. In our application, we estimate how unemployment probabilities vary across individuals and consider a range of realistic reforms that differ in how benefit and tax adjustments vary with income.

Discussion. Our implementation adopts a sufficient-statistics approach, using estimates of individual insurance value and unemployment risk to construct WTP. We estimate heterogeneous consumption responses to job loss using panel data on earnings and nondurable consumption, building on Gruber (1997) and recent applications in UI (e.g., Kolsrud et al. 2018; Ganong and Noel 2019) and other social

[6] Both Finkelstein et al. (2019) and Lieber and Lockwood (2019) consider settings with potentially many states of the world and convert utility to monetary terms by scaling by the *average* marginal utility. In contrast, because our empirical application focuses on unemployment insurance, we focus on two states of the world and scale by the marginal utility in the high-income (employment) state.

insurance programs (e.g., Meyer and Mok 2019; Kolsrud et al. 2024). Under Assumption 1, these estimates map into the insurance component of WTP. We implement this using a finite-mixture regression that recovers a discrete distribution of unemployment-induced consumption declines across latent types.

We estimate unemployment risk using a finite mixture of logistic regressions for unemployment incidence as a function of demographics and a measure of permanent income constructed from a multi-year earnings history. We combine this with the consumption-response model in a joint mixture framework that allows the latent risk and consumption-response types to be correlated. We interpret the implied unemployment probabilities as summarizing heterogeneity in exposure to unemployment risk in our sample, rather than as a full structural model of the joint dynamics of wages, assets, and search.

Other approaches to valuing social insurance programs—based on search responses (Chetty 2008; Shimer and Werning 2008), differences in marginal propensities to consume across states, or revealed preference using supplemental UI purchases (Landais and Spinnewijn 2021)—recover insurance value under different assumptions and institutional and data requirements. In principle, any such method for recovering the insurance value of UI could be used in place of, or alongside, our consumption-based approach to implement equation (3) and the subsequent normative analysis.

3. Data and Setting

Our empirical analysis focuses on unemployment insurance (UI). In the United States, the federal government sets broad guidelines on UI eligibility and benefit generosity, but programs are administered at the state level, with states determining benefit duration, generosity, and eligibility criteria (see Von Wachter 2019 for a discussion). Benefit levels and entitlement are computed from workers' employment and earnings histories. Outside of temporary federal extensions during recessions, most states offer benefits for up to 26 weeks. Eligible workers receive payments in proportion to past earnings, up to a cap.

The U.S. UI system is financed through payroll taxes that are statutorily levied on employers. The federally mandated minimum for taxable earnings is $7,000 per year. While most states apply a higher threshold, in practice the tax often functions more like a per-worker charge than a proportional earnings tax. A distinctive feature of U.S. UI finance is that tax rates vary across employers due to partial experience rating (see Guo and Johnston 2021 for discussion). Recent evidence suggests that firms do not pass the firm-specific component of UI taxes through to wages (Guo

2024). Accordingly, in our quantification of a UI expansion, we assume that only the common component of UI taxes is incident on wages.[7]

Sample. We use data from the Panel Study of Income Dynamics (PSID), a nationally representative survey of the U.S. population. The PSID has surveyed households since 1968, collecting information on demographics and labor market outcomes including unemployment, earnings, and wages. We use data from the 1999–2019 biennial PSID waves, which includes comprehensive information on consumption expenditures.[8]

We drop households for which the reference person does not report educational attainment and restrict the sample to reference persons aged 23–60 who are working or looking for work. To proxy for UI eligibility, we omit households where the reference person never reports being employed, as well as observations immediately following a spell out of the labor force. This yields 50,380 observations from 10,587 families, observed for an average of 4.8 waves. Of these, 2,828 observations correspond to the reference person being unemployed; 1,862 families are observed unemployed at least once, and approximately 30% of these families are observed unemployed in multiple waves. For families that experience changes in the number of adults (e.g., divorce, marriage, or death), we construct panel identifiers for demographically stable units and omit the year of the change. Throughout, we focus on the labor-market status of the reference person, whom we refer to as the "worker."

Key variables. In our main analysis, we define current unemployment based on the employment status reported by the worker at the time of the survey interview. We measure worker earnings using reported labor earnings when employed, and define household labor income as the sum of worker and spousal earnings. For each worker, we also construct a measure of permanent income based on the worker fixed effect in a log earnings regression, conditional on life-cycle earning growth.

We use a comprehensive measure of expenditures on nondurable consumption

[7] Guo (2024) shows the firm-specific component instead induces employment responses. See Spaziani (2024) for analysis of how experience rating alters the risk borne by employers. Anderson and Meyer (2000) similarly find that firm-specific components are largely incident on firms, though they emphasize that employment responses also affect layoffs and subsequent UI claims (and denials).

[8] Before 1999 the PSID was collected annually, but it included limited data on consumption outside of food expenditures. Both Gruber (1997) and Hendren (2017), for example, use food expenditure data in studies of unemployment insurance.

and services, excluding spending on health care and housing.[9] To account for differences in household composition, we equivalize consumption expenditures using the square root of household size. We convert all financial variables to 2019 dollars. Summary statistics, details on sample selection, and the construction of key variables, including our measure of worker permanent income, are provided in Appendix B.

Preliminary evidence. We begin by presenting descriptive evidence on negative selection into unemployment. On average, equivalized household consumption among the unemployed is about three-quarters that of the employed. This reflects both negative selection—individuals who experience unemployment tend to have lower consumption levels even while working—and the direct consumption declines associated with unemployment. In this section, we focus on measuring the extent of negative selection; in the next section, we present estimates of unemployment-induced consumption declines.

Specifically, we examine how household resources during periods of employment differ between individuals who ever experience unemployment and those who never do. By restricting attention to observations in which the worker is employed, this analysis avoids mechanically induced reductions in earnings and associated consumption responses during unemployment. We estimate:

$$y_{i,t} = \delta D_i + \beta X_{i,t} + u_{i,t}, \tag{5}$$

where $y_{i,t}$ is a measure of household resources; D_i is an indicator for whether worker i (the reference person) ever experiences an observed unemployment spell;[10] and $X_{i,t}$ includes a cubic in the worker's age, household composition indicators, and year fixed effects. In some specifications, it additionally includes a cubic in household income[11] The coefficient of interest δ captures the difference in household resources associated with ever experiencing unemployment.

Figure 1 reports estimates of δ and associated 95% confidence intervals for four measures of resources: log household nondurable consumption, log reference-person labor income, log household labor income, and our standardized

[9]There are some differences over time in how consumption components are measured, including expansion of measured items and changes in the aggregation or disaggregation of certain categories. Our analysis uses the broadest set of consumption categories available in each wave. We include survey-wave fixed effects in our econometric specification to account for changes in the coverage of our consumption measure.

[10]Specifically, $D_i = 1$ if worker i reports an unemployment spell at any survey interview, including spells occurring between waves.

[11]The presence of negative selection is robust to additionally controlling for education and race (see Appendix D.1).

measure of worker permanent income. For each outcome, we report three baseline specifications: (i) without controls, (ii) with controls, and (iii) excluding post-unemployment periods to eliminate potential labor-market scarring.[12] Across these specifications, households whose reference person ever experiences unemployment have lower consumption, individual income, and household income *while employed*, as well as lower permanent income, than households whose reference person is never unemployed.

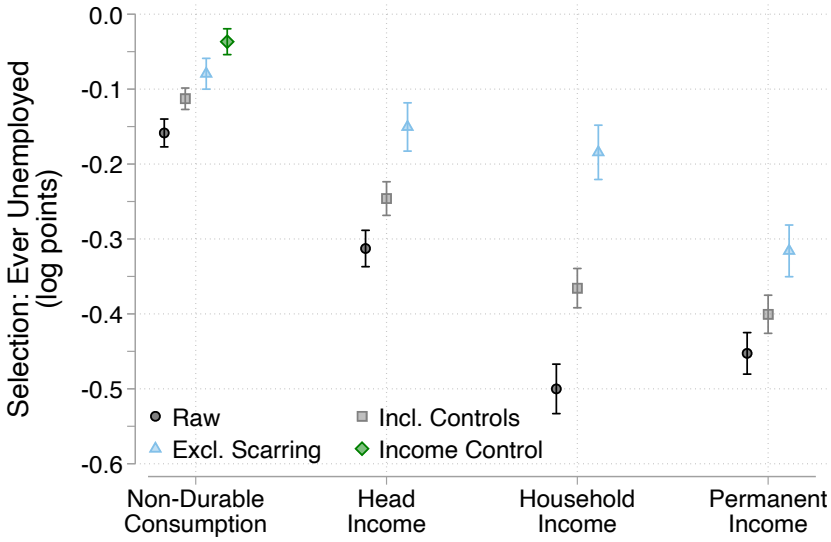


FIGURE 1. Negative Selection Among the Unemployed

Notes: All regressions drop periods during which the reference person is unemployed. For all outcomes, we report the raw, controls, and excl. scarring specifications. The raw specification includes no controls. The remaining specifications control for a cubic in age, indicators for family size and marital status, and year fixed effects. The excl. scarring specification further restricts the sample to periods before the first observed unemployment spell. For nondurable consumption only, we also report an "income controls" specification that adds a cubic in household income to the "excl. scarring" specification.

These patterns matter for the welfare analysis because UI reallocates resources toward workers who are more likely to experience unemployment. Since workers who experience unemployment tend to have lower income, this reallocation has a cross-income redistributive component, which affects the social value of UI. At the same time, redistribution across income groups can, in principle, also be achieved through the broader tax-and-transfer system. By contrast, reallocating resources within income groups—toward households with the same current income but lower consumption—is harder to implement through standard tax-and-transfer policies.

[12] For permanent income, differences reflect variation in the composition of workers observed before and after unemployment.

Moreover, because consumption may better reflect permanent income and longer-run economic well-being, redistribution based on consumption differences may be especially valuable. For household nondurable consumption, the income-controls specification (green markers in Figure 1) shows that even conditional on a flexible function of household income, households that experience unemployment have lower consumption than those that do not.

4. Measuring Willingness-to-Pay

In this section, we report estimates of heterogeneity in unemployment risk and unemployment-induced consumption declines. We detail our estimation strategy, present results, and compare them to findings in the existing literature. We then use these estimates to compute individual willingness-to-pay for UI reforms.

4.1. Empirical Framework

We allow for discrete unobserved heterogeneity in both unemployment risk and unemployment-induced consumption declines. Specifically, we assume that each worker i belongs to one of H unemployment-risk types and one of K consumption types, giving rise to $H \times K$ latent joint types. Risk types capture differences in unemployment incidence, while consumption types capture differences in the magnitude of the consumption decline following job loss. We estimate this structure using a finite-mixture framework and recover the joint distribution of risk and consumption types by maximum likelihood. For expositional clarity, we first describe the unemployment-risk block and then the consumption block; the joint likelihood and estimation algorithm are detailed in Appendix C. In our baseline specification we set $H = 2$ and $K = 3$.[13]

Our use of a finite-mixture approach is guided by the need to construct worker-level measures of WTP, which depend on worker-level estimates of unemployment-induced consumption declines—including for workers never observed unemployed—and on their correlation with unemployment risk. The joint mixture model assigns each worker posterior probabilities over latent risk–consumption types based on their history of consumption, employment and observables, which we use to construct individual unemployment-induced consumption declines and to characterize the joint distribution of unemployment risk and consumption drops across the population.

[13] We view the finite mixture as a tractable approximation to richer underlying heterogeneity. The main welfare conclusions are not driven by the exact baseline number of risk and consumption types; see Appendix H.

Unemployment risk. Patterns of program cross-subsidization depend on worker-level risk weights. To estimate these, we use a statistical model of unemployment incidence that allows for both observable and latent heterogeneity in risk. We model unemployment risk as a function of a cubic in age, indicators for gender, marital status, education, and race, and a cubic in permanent income, along with discrete unobserved risk types.[14]

Specifically, we assume that each worker belongs to one of H unemployment-risk types $h \in \{1, \ldots, H\}$ with population share $\pi(h)$. Conditional on type h, unemployment at time t follows a logit model

$$\Pr\left(U_{i,t} = 1 \mid h(i) = h, X_{i,t}^{U}\right) = g\left(\eta_0^h + X_{i,t}^{U\prime}\gamma\right),$$

where $g(\cdot)$ is the logistic function, η_0^h is a type-specific intercept, and γ is a vector of coefficients on observables $X_{i,t}^{U}$. We assume that, conditional on risk type h and observables $X_{i,t}^{U}$, unemployment shocks are independent across time and across workers. Identification of risk types uses systematic differences in unemployment incidence across workers with similar observables.

Consumption declines. The standard approach to recovering consumption declines (following Gruber 1997) estimates an average effect of unemployment, conditional on being employed in the preceding period ($t - 2$ in our case), using the following specification:

$$\begin{aligned}\Delta_{i,t}^{FD} &\equiv \ln c_{i,t} - \ln c_{i,t-2} \\ &= \delta_0 + \delta_1 U_{i,t} + X_{i,t}^{C\prime}\beta + \varepsilon_{i,t},\end{aligned} \tag{6}$$

where $U_{i,t}$ is an indicator variable denoting whether the worker is unemployed in period t and $X_{i,t}^{C}$ are demographic controls. This first-difference specification allows for arbitrary permanent individual heterogeneity in consumption levels. The parameter of interest, δ_1, measures the average proportional consumption decline that results from becoming unemployed. The identification assumption underpinning this strategy is that, absent unemployment, workers who become unemployed would have experienced the same consumption growth as those who remain employed.

We estimate the *distribution* of consumption declines using a finite-mixture

[14] An alternative approach uses idiosyncratic and subjective probabilities of job loss (Hendren 2017) and job finding (Spinnewijn 2015; Mueller et al. 2021). These measures are not directly available in our dataset, and using them would require imputing probabilities based on similar demographic variables.

approximation to the underlying distribution (e.g., Heckman and Singer 1984; Keane and Wolpin 1997). We build on Lewis et al. (2026) by adopting their finite-mixture regression for consumption responses and embedding it in a joint mixture model with unemployment risk. We assume each worker belongs to one of a finite number of latent types and augment the specification in equation (6) as follows:

$$\Delta_{i,t}^{FD} = \sum_{k=1}^{K} \mathbb{1}\left[k(i) = k\right]\left(\delta_0^k + \delta_1^k U_{i,t}\right) + X_{i,t}^{C\prime}\beta + \varepsilon_{i,t}. \tag{7}$$

We include group-specific intercepts δ_0^k, so that δ_1^k can be interpreted as the consumption decline associated with unemployment for group k. We assume that, conditional on consumption type k, observables, and unemployment status, shocks $\varepsilon_{i,t}$ are independent over time and satisfy $\mathbb{E}[\varepsilon_{i,t} \mid k(i) = k, X_{i,t}^C, U_{i,t}] = 0$, with type-specific variance σ_k^2.

To implement equation (7), we use a Gaussian mixture linear regression (Quandt 1972), which assumes normally distributed errors $\varepsilon_{i,t}$. The key identifying assumption is that, conditional on latent type k, the evolution of consumption for employed workers provides a valid counterfactual for the trend in consumption that unemployed workers would have experienced had they remained employed. Intuitively, latent types are identified by grouping households with similar distributions of consumption growth in each state.

As discussed in Lewis et al. (2026), this mixture regression can be viewed as a form of clustering regression that jointly (i) groups households with similar latent consumption dynamics during employment and unemployment and (ii) estimates the consumption decline within each group. This provides a useful interpretation for how we impute counterfactual consumption declines to workers never observed unemployed. Intuitively, the approach resembles the following steps. First, households observed in both states help identify latent consumption types through their consumption growth when employed and their consumption decline when unemployed. The model then probabilistically links always-employed households to these types using their consumption growth while employed, conditional on observables. In practice, we estimate group assignments and parameters jointly. The marginal posterior probability of each consumption type k,

$p_i(k) \equiv \sum_{h=1}^{H} p_i(k,h)$, serves as a convex imputation weight.[15]

Estimation. Estimation proceeds in two stages. In the first stage, we estimate the parameters of the unemployment-risk mixture,

$$\chi_U = \{\pi(h), \eta_0^h, \gamma\}_{h=1}^{H}$$

by maximum likelihood using workers' employment-status histories. In the second stage, conditional on these first-stage estimates, we estimate the consumption-decline parameters and the joint-type probabilities using an expectation–maximization algorithm. Let

$$\chi_C = \left\{\delta_0^k, \delta_1^k, \sigma_k^2, \beta\right\}_{k=1}^{K}$$

denote the consumption-block parameters, and let $\pi(h,k)$ denote the population share of consumption type k and risk type h. The joint-type probabilities $\pi(h,k)$ are identified from the joint distribution of unemployment histories and consumption growth paths, conditional on observables. Intuitively, when workers who experience frequent unemployment spells also exhibit large consumption declines, the model assigns higher posterior weight to high-risk/high-decline joint types.

For each worker, we then compute posterior probabilities $\hat{p}_i(k,h)$ of belonging to each latent type. We use these posteriors to construct worker-level employment probabilities and unemployment-induced consumption declines,

$$\hat{e}_{i,t} = 1 - \sum_{h=1}^{H}\sum_{k=1}^{K} \hat{p}_i(k,h)\ \Pr\left(U_{i,t}=1 \mid h(i)=h, X_{i,t}^{U}; \hat{\chi}_U\right), \quad \widehat{(\Delta c/c)}_i = -\sum_{h=1}^{H}\sum_{k=1}^{K} \hat{p}_i(k,h)\,\hat{\delta}_1^k.$$

Appendix C describes the likelihood and algorithm in detail.[16]

[15] If two consumption types experience similar consumption growth when employed but different consumption declines when unemployed, the model assigns an always-employed household with similar employed-state consumption growth a consumption decline that is a posterior-weighted average of these types. These weights depend on the household's consumption history, the variability of consumption growth within each type, observables, and the joint distribution of risk and consumption types. In this sense, the imputation uses information on both the level and variability of consumption growth, which is related to the pass-through of shocks to consumption (Blundell et al. 2008). In principle, latent consumption risk may vary over time, and our approach could be extended to allow for a regime-switching model.

[16] These expressions report posterior-weighted means of the two main empirical primitives. Because WTP and welfare expressions depend nonlinearly on unemployment risk and consumption declines, we compute them type by type and then average over joint types using posterior probabilities.

4.2. Estimates

Unemployment risk

Table 1 reports finite-mixture logit model estimates. We report the type-specific probabilities of unemployment, averaged over the empirical distribution of covariates, along with the estimated population share of each type. The estimates reveal substantial latent heterogeneity in unemployment risk. Roughly three quarters of workers belong to the low-risk group with an average unemployment probability of about 2.6%, while the remaining quarter form a high-risk group facing an average unemployment probability of close to 14%.

TABLE 1. Predicted Probabilities of Unemployment and Type Shares

	Overall	Group 1	Group 2
$\bar{P}\left(U_{i,t}=1\right)$	0.053	0.026	0.140
	(0.001)	(0.005)	(0.023)
Share ($\widehat{\pi}(h)$)		0.759	0.241
		(0.064)	(0.064)

Notes: $\bar{P}(U_{i,t}=1)$ denotes the sample-averaged predicted probability of unemployment, computed from the finite-mixture logit model and averaged over the empirical distribution of covariates. $\widehat{\pi}(h)$ denotes the estimated population shares of each unemployment-risk type. The estimation sample contains N = 50,380 person-wave observations. Bootstrap standard errors based on 400 household-level replications are shown in parentheses. Covariates include a cubic in age, indicators for gender, marital status, education, and race, and a cubic in permanent income.

We report average partial effects with 95% confidence intervals in Figure 2. The largest average partial effect is for permanent income: the marginal effect, scaled by one standard deviation of permanent income, is -4.1 percentage points.

The results show systematic differences in unemployment rates across demographic groups. Since permanent income is included as a control, these estimates compare workers with similar earnings histories. Younger workers are more likely to be unemployed: being ten years younger is associated with a 1.1 percentage point increase in the unemployment probability. We find a modest educational gradient: individuals who did not complete high school are approximately 1 percentage point more likely to be unemployed than those with higher educational attainment. This small effect is largely attributable to the direct role of permanent income. Even after controlling for education and permanent income, we find a substantial racial gap: Black individuals are 2.1 percentage points

more likely to be unemployed than White individuals. Conditional on other covariates, we find small effects for gender.

While the simple model outlined in Section 2 rationalizes differences in unemployment risk through differences in individuals' effort choices, the empirical disparities we document likely reflect broader labor-market factors. These include differences in market thickness, equilibrium congestion externalities, skill-biased labor demand, and discrimination—all of which enter the model through differences in the cost of exerting effort (i.e., in $\psi_i(\cdot)$).

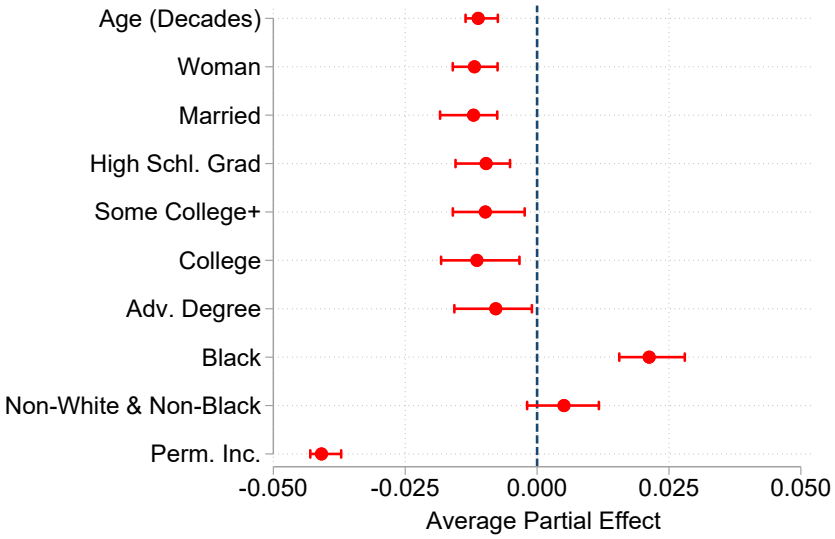


FIGURE 2. Effect of Household Characteristics on Predicted Probability of Unemployment

Notes: We estimate the finite mixture logit model described in the text using unemployment status as the dependent variable, and compute marginal effects of each covariate on unemployment probability by integrating over the empirical distribution of household characteristics and the estimated distribution of risk types. We report the marginal effect for the z-score of permanent income for easier interpretation. Each marker corresponds to the point estimate of the average partial effect (APE), and the horizontal lines span the 95% confidence intervals. The dashed line indicates zero. The estimation sample contains N = 50,380 person-wave observations, and bootstrap standard errors are based on 400 household-level replications.

While the contribution of unobserved heterogeneity is substantial, the estimated average partial effects line up with patterns documented in the broader literature. Prior work finds that unemployment rates are higher among younger workers (e.g., Choi et al. 2015); decline with higher education and other measures of human capital (e.g., Ashenfelter and Ham 1979; Nickell 1979; Cairó and Cajner 2018); and exhibit persistent racial disparities in employment outcomes (e.g., Lang and Lehmann 2012) and callback rates (e.g., Bertrand and Mullainathan 2004). We also find only a modest

unemployment gap by gender. However, our estimates pertain to women who are the PSID reference person—a group that is disproportionately single. Consistent with our small gender gap, prior work finds that labor force participation gaps between unmarried men and women are small (e.g., Borella et al. 2023).

Consumption declines

Table 2 reports our estimates from the finite-mixture approach alongside estimates from the homogeneous specification (equation (6)). The homogeneous specification yields an estimate of the average consumption decline of 13.1% associated with unemployment.

		Mixture Model		
	Homogeneous	Group 1	Group 2	Group 3
$\mathbb{1}\left[U_{i,t}=1\right]$	−0.131	−0.263	−0.113	−0.069
	(0.012)	(0.068)	(0.027)	(0.025)
Share ($\widehat{\pi}(k)$)		0.122	0.494	0.384
		(0.102)	(0.122)	(0.074)

TABLE 2. Heterogeneity in Consumption Declines ($\Delta c/c$)

Notes: The first column reports the estimated consumption decline from the homogeneous specification (equation (6)). The remaining columns report the consumption declines, δ_1^k, and marginal population shares, $\pi(k) = \sum_{h=1}^{H} \pi(h,k)$, from the Gaussian mixture linear regression framework (equation (7)). Standard errors are shown in the parentheses; in the mixture model, they account for uncertainty over both group assignment and parameter estimates conditional on group. The estimation sample contains N = 50,380 person-wave observations, and bootstrap standard errors are based on 400 household-level replications. The consumption-drop block is estimated on the first-differenced sample of N = 37,143 person-wave observations, which excludes observations without a valid lag and periods following unemployment. Covariates include a cubic in age, indicators for year, and the change in log family size.

The implied average consumption decline from the mixture approach, aggregating the group-specific consumption declines δ_1^k weighted by their marginal population shares $\pi(k)$, is slightly smaller at 11.4%.[17] This comparison of averages, however, ignores the large degree of heterogeneity in exposure to unemployment-induced consumption declines that we find. The group-specific consumption declines range from 26.3% to 6.9%. The smallest-decline group, accounting for 38% of the population, is relatively well insured against unemployment risk. A second group, with probability mass of 49%, has a

[17]The mixture-model average weights group-specific declines by their marginal population shares. As a comparison, weighting worker-level posterior-implied declines by their estimated unemployment probabilities yields a decline of 13.0%, close to the homogeneous estimate of 13.1%.

consumption decline close to the homogeneous estimate. The remaining group has a much larger consumption decline of 26.3%.

Figure 3 shows the distribution of posterior-implied consumption declines across workers, further emphasizing substantial cross-sectional heterogeneity in exposure to unemployment-included consumption losses. Using the homogeneous specification, all households are assigned the same predicted decline (the dashed vertical line). By contrast, we find that 81% of households exhibit consumption declines smaller than this estimate, while a long right tail experiences much larger declines. This variation reflects broad differences in households' ability to self-insure against unemployment risk.

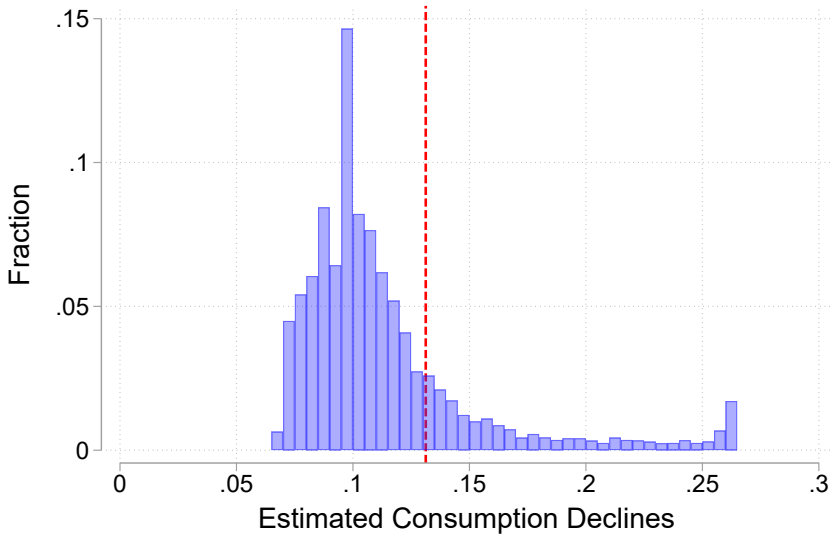


FIGURE 3. Distribution of Estimated Consumption Declines

Notes: The histogram plots worker-level unemployment-induced consumption declines implied by the Gaussian mixture linear regression, constructed by combining estimated parameters with each worker's posterior probabilities of belonging to each consumption type, $p_i(k) = \sum_{h=1}^{H} p_i(h,k)$. For each household, we compute the posterior-weighted average of the type-specific unemployment coefficients δ_1^k. The estimation sample contains $N = 50{,}380$ person-wave observations. The consumption-drop block is estimated on the first-differenced sample of $N = 37{,}143$ person-wave observations, which excludes observations without a valid lag and periods following unemployment. The homogeneous estimate (dashed vertical line) is obtained by imposing $K = 1$ in the baseline specification. Appendix D.2 reports bootstrap confidence intervals for quantiles of the worker-level consumption declines.

Comparison to existing homogeneous estimates. Our homogeneous estimate is slightly larger than those reported by the existing literature (e.g., Gruber 1997 and Hendren 2017), which find declines in food expenditures between 7 and 10%. There are three main differences between our approach and prior work.

First, we measure consumption declines by comparing consumption when unemployed with in-work consumption two years earlier. Hendren (2017) documents a statistically significant anticipatory decline of 2.7% in the year preceding unemployment; combining this with existing one-year horizon estimates reconciles the majority of the gap. Second, we use a broader measure of consumption expenditures. Our results are robust to alternative measures (see Appendix D.3). Third, our sample differs from those used in prior work and is drawn from different years. Using the PSID, East and Kuka (2015) document an increasing trend in the average decline in food expenditures following unemployment, which they attribute to factors other than changes in the survey design.[18]

Threats to identification. We assess the key identifying assumption that, conditional on type, consumption growth for workers who remain employed provides a valid counterfactual for workers who become unemployed. Relative to a homogeneous specification, conditioning on latent type may help mitigate endogenous selection into unemployment by making the counterfactual comparison within groups of workers who are more comparable in persistent consumption dynamics and unemployment risk. Following Hendren (2017), we estimate lead–lag specifications that replace the contemporaneous unemployment indicator with indicators for unemployment k periods away; the homogeneous and type-specific profiles show little evidence of differential pre-trends prior to job loss (Appendix D.4). In addition, we exploit PSID information on reasons for job separation to construct an indicator for separations due to plant closure or firm shutdown, and use this as an instrument for our baseline unemployment indicator. The resulting estimate is similar to our baseline and not statistically distinguishable from it (Appendix D.5), suggesting that selection into unemployment and violations of parallel trends are unlikely to be major drivers of our estimated consumption declines. As an additional validation of the mixture specification, we compare the consumption drop distribution implied by the finite-mixture model for unemployed workers in a training sample to quantile-regression estimates in an independent holdout sample; the two align closely across most of the distribution (Appendix D.6).

Robustness to PSID design. A limitation of the PSID's biennial structure is that it may under-represent short unemployment spells and the associated consumption

[18] Similar to our analysis, East and Kuka (2015) find modest differences across consumption measures. They attribute rising declines to a time trend rather than temporal sampling over the business cycle. In addition, our sample includes fewer married white workers compared to earlier work (see Appendix Table B.1 and Table 1 of Kroft and Notowidigdo (2016), who reproduce the estimate from Gruber (1997)). We also explore the sensitivity of our results to alternative definitions of unemployment in the two-year panel and find our results are robust.

responses. In our baseline specification, we classify a worker as unemployed at time t if they are employed at $t-2$ but unemployed at t, so that $\Delta^{FD}_{i,t}$ compares pre- and post-unemployment consumption. Unemployment spells that start and end between these interviews appear as employment at both dates and therefore do not enter our "unemployed" group, tilting the sample toward longer spells for which consumption declines may be larger. Using post-2003 questions on whether the worker experienced any unemployment in the previous 12 or 24 months, we compare our homogeneous estimate to specifications that use alternative sample restrictions based on reported unemployment timing (see Appendix D.7). The point estimates are similar, with samples that exclude more reported unemployment outside the interview window yielding slightly smaller (but statistically indistinguishable) consumption drops. This suggests that any bias from the biennial sampling frame is modest and, if anything, would lead us to slightly overstate the insurance value of UI.

Relation to evidence on heterogeneous consumption responses. Our finding of substantial heterogeneity in unemployment-induced consumption declines is consistent with a large body of evidence on how households respond to job loss. Browning and Crossley (2001) find that the marginal propensity to consume out of UI benefits is largest for those without liquid savings. Using direct evidence on binding credit constraints, Crossley and Low (2014) find that 5% of job losers face binding credit constraints and experience particularly large, welfare-reducing consumption declines at job loss. They also document smaller, yet substantial, declines among households that are likely to be unconstrained. Ganong and Noel (2019) document systematic variation in consumption responses by liquid assets using de-identified banking data. Recent evidence from linked banking and administrative records in Denmark (Andersen et al. 2023) shows that households offset income losses from job loss primarily by decumulating liquid savings, but also through added worker effects.

Recent work provides complementary evidence on the magnitude and heterogeneity of consumption responses to income shocks. Patterson (2023) and Colarieti et al. (2024) document substantial variation in marginal propensities to consume (MPCs) across households. Comparing consumption and earnings responses to unemployment in our sample implies an MPC of about 0.3, similar in magnitude to Patterson's estimates using alternative PSID consumption measures. These estimates provide a useful comparison, although our framework focuses on unemployment-induced consumption declines rather than directly on MPCs.

The key novelty in our analysis is that we construct worker-level estimates of unemployment-induced consumption declines without imposing assumptions

about which observables drive this variation. We examine how these estimated consumption declines correlate with proxies for a household's capacity to self-insure and find larger consumption declines among households with low liquid assets (see Appendix D.8).

Correlation between risk and consumption types

The joint-type framework allows us to study how unemployment risk and unemployment-induced consumption declines covary in the population. Table 3 reports the estimated joint distribution of risk and consumption types, $\pi(h, k)$, together with the conditional share of each consumption type within each risk type.

		Consumption type (k)		
	Risk type (h)	High drop	Medium drop	Low drop
$\hat{\pi}(h, k)$	High	0.045	0.131	0.064
		[0.018, 0.149]	[0.028, 0.197]	[0.028, 0.180]
	Low	0.076	0.363	0.320
		[0.050, 0.370]	[0.058, 0.449]	[0.233, 0.442]
$\hat{\pi}(k \mid h)$	High	0.189	0.545	0.266
		[0.127, 0.533]	[0.139, 0.622]	[0.166, 0.597]
	Low	0.100	0.478	0.421
		[0.069, 0.470]	[0.076, 0.556]	[0.313, 0.548]

TABLE 3. Joint and Conditional Distribution of Risk and Consumption Types

Notes: The table reports estimates of the joint and conditional type probabilities in the joint risk–consumption mixture model. The entries labeled $\hat{\pi}(h, k)$ give the estimated joint population share of risk type h and consumption type k. The entries labeled $\hat{\pi}(k \mid h)$ give the conditional share of each consumption type within risk type h, i.e. $\hat{\pi}(k \mid h) = \hat{\pi}(h, k) / \sum_{k'} \hat{\pi}(h, k')$. A Wald test using the household-bootstrap covariance matrix rejects independence between risk and consumption types ($p = 0.0013$). The high-risk type's probability of belonging to the high-drop group exceeds that of the low-risk type by 8.9 percentage points (95% bootstrap confidence interval: [3.0, 12.5]). 95% confidence intervals based on 400 household-level bootstrap replications are shown in brackets.

We find that the high-risk type accounts for about one-quarter of workers and is almost twice as likely to belong to the high-drop consumption group as the low-risk type (19% vs. 10%), and substantially less likely to be in the low-drop group (27% vs. 42%). This pattern implies that workers who face a higher probability of unemployment also tend to experience larger consumption declines when unemployment occurs, amplifying ex ante welfare losses from labor market risk. Below, we explore the implications of this correlation for the distribution of willingness-to-pay for UI reforms.

4.3. Risk Aversion and Reforms

Risk aversion. In our framework, the insurance value of a UI expansion to an individual worker is captured by the product of their consumption decline in unemployment and their coefficient of relative risk aversion (see equation (4)). We set our baseline coefficient of relative risk aversion to $\gamma = 3$, and examine sensitivity to alternative values in Appendix E.3.

Reforms. We consider three expansions of UI generosity:

(a) Proportional–capped reform: $\phi_{\mathcal{B}}(y) = \min\{y, \kappa\}$, where κ is an earnings cap.

(b) Flat reform: $\phi_{\mathcal{B}}(y) = 1$.

(c) Proportional reform: $\phi_{\mathcal{B}}(y) = y$.

All reforms are funded by a flat tax increase, $\phi_{\mathcal{T}}(y) = 1$.

Reform (a) is designed to reflect a natural expansion of the current U.S. UI system, which offers proportional UI benefits up to an earnings cap—approximately at the 41st percentile of the earnings distribution—and is funded by a payroll tax that, in most states, functions as an employment tax. Reform (b) considers a flat benefit expansion for all workers, aligning with the standard reform considered in the UI literature. Reform (c) represents a proportional expansion of UI benefits with no cap, meaning benefits rise in proportion to earnings for all workers. The interaction between the reform design and the distribution of unemployment risk determines the extent of cross-subsidization inherent in each policy.

4.4. Distribution of Willingness-to-Pay

In Figure 4, we summarize the distribution of willingness-to-pay by reform type. These WTP estimates are defined for reforms that adjust the financing tax to offset the reform's mechanical cost; Section 5 incorporates the additional financing required by behavioral responses. Panel (A) shows the insurance component of WTP, which is common across reforms and positive for all workers. Panels (B)–(D) show total WTP for each reform. Each panel plots how the median, interquartile range, and interdecile range vary across percentiles of the household income distribution. WTP is reported per \$1 of mechanical increase in an individual's expected benefit payments. This normalization is most useful for comparing how strongly different workers value a reform relative to their own expected entitlement, and for characterizing support for a reform and rankings across reforms. Higher WTP indicates greater net benefit per dollar of the worker's expected benefit

increase under a given reform, positive WTP indicates support for the reform, and comparing WTP across reforms reveals individual reform rankings.[19] These baseline estimates abstract from income gradients in UI take-up and benefit exhaustion. Lower expected benefit receipt among low-wage workers can reduce or reverse their net WTP; we return to this issue when aggregating WTP in Section 5.

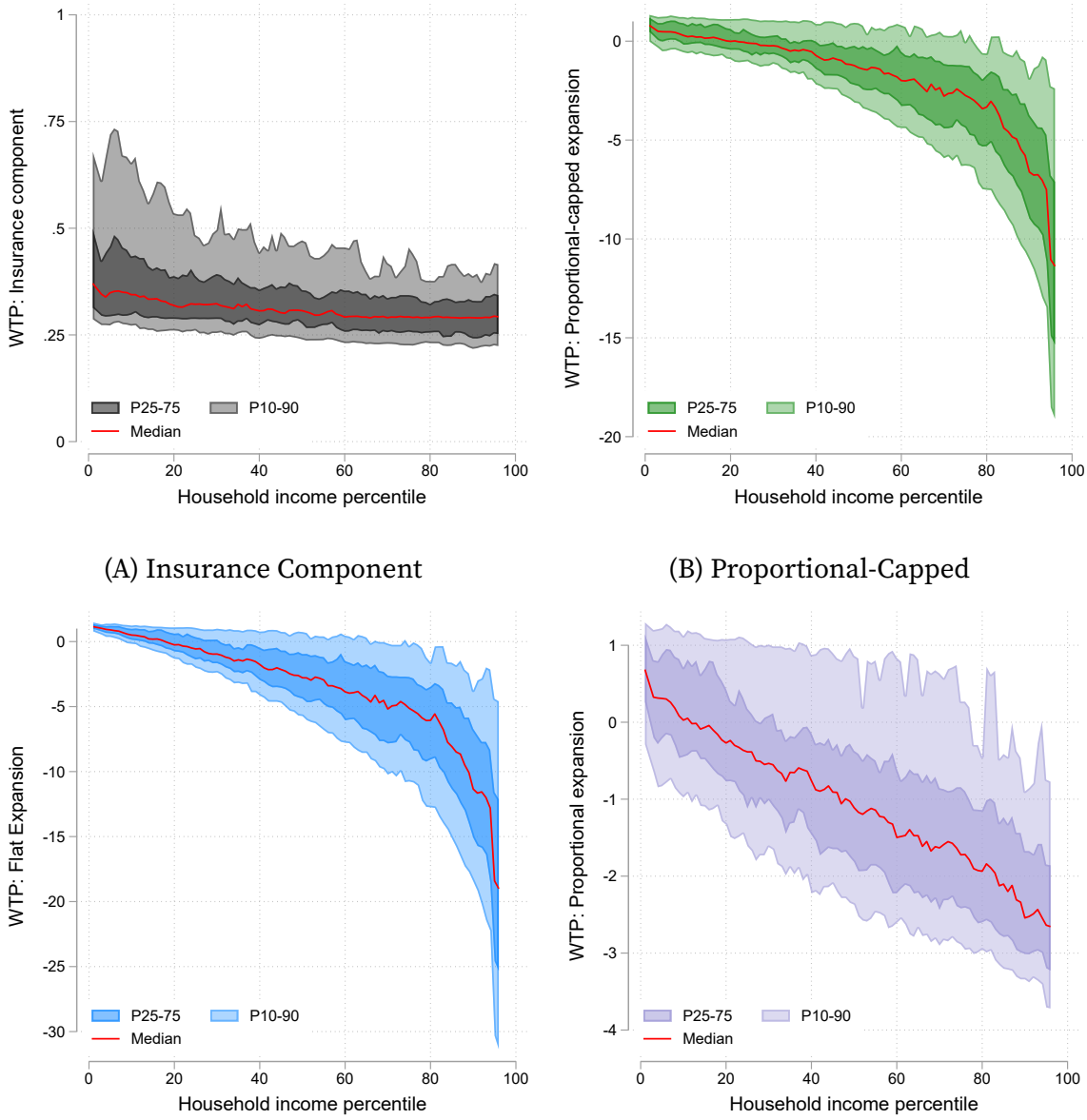
Panel (A) shows that WTP for actuarially fair insurance declines with household income and exhibits substantial within-income variation. Workers in lower-income households value actuarially fair insurance more highly than those in higher-income households. Dispersion within income groups is also sizable, reflecting heterogeneity in self-insurance capacity.

Panel (B) focuses on the proportional–capped reform, where WTP reflects both the value of insurance and the cross-subsidization embedded in the reform. The wider vertical scale relative to panel (A) highlights that much of the WTP variation by household income is driven by cross-subsidization. WTP declines with income and becomes negative over much of the upper part of the distribution, indicating that many higher-income workers are net contributors under the reform. Even conditional on income, heterogeneity remains substantial, so workers with similar earnings can differ markedly in how they value the same UI expansion.

While the insurance component is common across reforms, the reforms differ in their patterns of cross-subsidization and therefore in their resulting WTP distributions. Panel (C) shows that a flat reform tilts WTP in favor of workers in lower-income households, whereas panel (D) shows that a proportional reform tilts gains toward workers in higher-income households. The proportional–capped reform in panel (B) lies between these two cases, resembling the proportional reform at the lower end of the income distribution and the flat reform toward the upper end.

These patterns reflect the different sources of cross-subsidization across reforms. Under the flat reform, cross-subsidization arises solely from differences in unemployment risk: workers with below-average risk subsidize those with above-average risk. Since unemployment risk tends to decline with household income, WTP falls with income. Under the proportional reform, cross-subsidization reflects both unemployment risk and earnings. Conditional on unemployment risk, higher earners gain more from the expansion, but the overall decline in WTP with household income suggests that the risk gradient dominates the earnings effect.

[19] An alternative normalization defines individual surplus per $1 of aggregate mechanical increase in expected benefit payments. This rescaling combines normalized WTP with the mechanical incidence of the reform. Under the reform structures we consider, it preserves within-worker rankings across reforms; see Appendix E.1.



(A) Insurance Component

(B) Proportional-Capped

(C) Flat

(D) Proportional

FIGURE 4. Willingness-to-Pay for UI Expansion by Income and Reform Type

Notes: Each panel summarizes the distribution of workers' willingness-to-pay (WTP) for a UI expansion, reporting the median and the 10th, 25th, 75th, and 90th percentiles. We compute WTP from equation (3) for each worker type, using type-specific consumption declines and worker-specific predicted unemployment risk, and average WTP using posterior type probabilities. Worker incomes are winsorized at \$10,000 and \$200,000. Panel (A) shows the insurance component (WTP for actuarially fair insurance). Panel (B) shows results for a proportional-capped reform reflecting the current U.S. UI structure. Panels (C) and (D) show results for flat and proportional expansions, respectively. All values are expressed per \$1 of mechanical increase in an individual's expected benefit payments. Quantiles are smoothed using a uniform rolling window of ± 1 percentiles.

Beyond showing who gains and loses from each reform, the WTP estimates also imply worker-specific preference rankings over reforms. Since these rankings

depend on both unemployment risk and earnings, workers may agree on the desirability of expanding UI in general while disagreeing sharply over which reform they prefer. Our estimates imply that 82% of workers prefer expanding the proportional-capped system to a flat expansion, while 52% prefer it to a proportional expansion (see Appendix E.2). These reforms therefore create distinct winners and losers and differ in their associated fiscal externalities, so ranking their social value requires aggregating willingness-to-pay and accounting for behavioral responses.

Sensitivity to preference specification. The insurance component of WTP depends on our calibration of the coefficient of relative risk aversion and on the assumption of state-independent utility.

Varying the coefficient of relative risk aversion scales the insurance component proportionally but does not alter the shape of its distribution. We also allow for state-specific consumption prices, capturing the idea that unemployed individuals allocate more time to searching for lower prices. Assuming prices are 1.5% lower for the unemployed—based on evidence from Kaplan and Menzio (2015) and Campos and Reggio (2020)—modestly reduces the insurance component (see Appendix E.3).

As emphasized by Andrews and Miller (2013), under the sufficient-statistics mapping we use, heterogeneity in risk aversion affects aggregate insurance values only insofar as it covaries with consumption declines. Direct evidence on such heterogeneity is limited.[20] However, Figure 4 shows that cross-subsidization plays a dominant role in shaping WTP under realistic reforms, meaning that the overall WTP distribution is less sensitive to plausible changes in risk aversion than the insurance component alone.

5. Social Value of Reform

In this section, we characterize the social value of expanding a publicly funded insurance program. In the canonical setting, policy design trades off insurance gains for a representative agent against incentive costs, summarized by a single elasticity. We show that with heterogeneous individuals, the single representative-agent insurance term is replaced by a welfare-weighted sum of individual willingness-to-pay, which we decompose into the social value of risk protection and cross-subsidization. Moreover, incentive costs depend on how

[20]The 1996 PSID survey wave includes survey-elicited measures of individual risk preferences (see Kimball et al. (2009)). We link respondents to our estimation sample using permanent PSID individual identifiers. Among matched respondents, imputed risk aversion exhibits low covariance with our estimated consumption declines.

individual behavioral responses covary with risk weights and state-specific net tax liabilities.

5.1. Theory

The aggregate value of a social insurance reform depends on its fiscal cost and on how the reform's effects are weighted across individuals.

To characterize these components, we define risk-weighted expectations in each state. For any variable x_i, we denote its risk-weighted expectation in the low and high state, respectively, as:

$$\mathbb{E}^l[x] \equiv \int_i \left(\frac{(1-e_i)\,\phi_{\mathcal{B}}(y_i)}{\int_{i'}(1-e_{i'})\phi_{\mathcal{B}}(y_{i'})di'} \right) x_i\, di \qquad \mathbb{E}^h[x] \equiv \int_i \left(\frac{e_i\,\phi_{\mathcal{T}}(y_i)}{\int_{i'}\, e_{i'}\,\phi_{\mathcal{T}}(y_{i'})di'} \right) x_i\, di. \quad (8)$$

By construction, the weights in each state sum to one, so $\mathbb{E}^l[x]$ and $\mathbb{E}^h[x]$ are state-specific averages under weights proportional to each individual's mechanical contribution to the reform's fiscal impact: in the low state, expected benefit payments $(1-e_i)\,\phi_{\mathcal{B}}(y_i)$, and in the high state, expected tax payments $e_i\,\phi_{\mathcal{T}}(y_i)$.

Budgetary impact. We focus on budget-neutral expansions of social insurance. The policymaker faces the budget constraint $\int_i e_i \mathcal{T}(y_i)\,di = \int_i (1-e_i)\mathcal{B}(y_i)\,di + \bar{G}$, where $\bar{G}$ denotes an exogenous revenue requirement. The schedules $\mathcal{T}(.)$ and $\mathcal{B}(.)$ summarize the overall tax-and-transfer system in each state: $\mathcal{T}(y_i)$ denotes the net payment to the government in the high state, and $\mathcal{B}(y_i)$ the net transfer received in the low state.

Budget-neutrality implies that the reform $d\theta = (db, d\tau)$, parameterized in equation (2), must satisfy:

$$\frac{d\tau}{db} = \underbrace{\frac{\int_i (1-e_i)\phi_{\mathcal{B}}(y_i)\,di}{\int_i e_i \phi_{\mathcal{T}}(y_i)\,di}}_{\text{Mechanical effect}} \left(1 + \underbrace{\frac{1}{\int_i (1-e_i)\phi_{\mathcal{B}}(y_i)\,di} \left(\int_i \frac{d(1-e_i)}{d\theta} (\mathcal{T}(y_i) + \mathcal{B}(y_i))\,di \right)}_{\text{Fiscal externality (FE)}} \right). \quad (9)$$

The first term captures the tax increase required to cover the mechanical cost of the benefit expansion, holding behavior fixed. The second term reflects the additional tax adjustment needed to maintain budget balance in response to behavioral changes—the fiscal externality. This arises because higher benefits reduce the marginal return to exerting effort, increasing the likelihood that individuals are in the low state. The government then both forgoes taxes $\mathcal{T}(y_i)$ and pays benefits $\mathcal{B}(y_i)$, raising the required tax adjustment.

Note that the mechanical effect depends directly on reform design—through $\phi_{\mathcal{B}}(.)$ and $\phi_{\mathcal{T}}(.)$, while the fiscal externality depends on behavioral responses induced by the reform and on pre-reform schedules $\mathcal{T}(.)$ and $\mathcal{B}(.)$.

We can express the fiscal externality term as:

$$\text{FE} = \mathbb{E}^l \left[\epsilon^{(1-e,b)} \frac{\mathcal{T}(y) + \mathcal{B}(y)}{\mathcal{B}(y)} \right], \tag{10}$$

where $\epsilon_i^{(1-e,b)} \equiv \frac{1}{\phi_{\mathcal{B}}(y_i)} \frac{d(1-e_i)}{db} \frac{\mathcal{B}(y_i)}{1-e_i}$ is the individual-level elasticity of the low-state probability with respect to a benefit change $\phi_{\mathcal{B}}(y_i)\, db$, along the balanced-neutral reform path. Equation (10) makes clear that, in general, the fiscal cost from behavioral responses depends on the covariance between individual behavioral elasticities, risk weights, and state-specific tax and benefit liabilities.

Aggregate value. To capture the aggregate value of the reform, we define a weighted sum of expected utilities:

$$W(\theta) = \int_i \omega_i \, V_i(\theta) \, di, \tag{11}$$

where $\omega_i \geq 0$ is the Pareto weight assigned to individual i. As we discuss below, the inclusion of Pareto weights allows this function to accommodate a broad range of ethical positions regarding the redistributive value of social insurance.

We focus on the aggregate value of a marginal benefit adjustment, parameterized by equations (2) and (9). Analogous to our definition of individual willingness-to-pay, we convert the change in social welfare into money-metric terms by scaling the aggregate utility change $\frac{dW}{d\theta}$, expressed per \$1 of mechanical increase in benefit expenditure, by the aggregate utility value of a \$1 decrease in the tax bill. Denoting this money-metric welfare effect by $\frac{dW^{MM}}{d\theta}$, we obtain:

$$\frac{dW^{MM}}{d\theta} = \underbrace{\frac{\mathbb{E}^l[\omega u^{l\prime}(c^l)] - \mathbb{E}^h[\omega u^{h\prime}(c^h)]}{\mathbb{E}^h[\omega u^{h\prime}(c^h)]}}_{\text{Resource reallocation (RR)}} - \text{FE}. \tag{12}$$

The first term captures the gain from expanding social insurance that arises from shifting resources from high to low states. It is measured by the percentage gap between the risk- and Pareto-weighted expected marginal utility of consumption in the low and high states. The overall aggregate value of the reform equals this resource-reallocation gain net of the fiscal externality.

Insurance–cross-subsidization decomposition. We define social marginal welfare weights as $g_i \equiv \omega_i u_i^{h\prime}(c_i^h)$. The following proposition decomposes the resource-reallocation term in equation (12) into the social value of insurance—the within-individual, actuarially fair transfer of resources between states—and cross-subsidization—the across-individual net expected transfer embedded in the reform.

PROPOSITION 1. *The aggregate value of resource reallocation from a marginal, budget-neutral rise in benefits satisfies:*

$$\text{RR} = \underbrace{\bar{\lambda}}_{\substack{\text{aggregate}\\ \text{welfare weight}}} \mathbb{E}^l \left[\underbrace{\lambda}_{\substack{\text{individual}\\ \text{welfare weight}}} \left(\underbrace{\frac{u^{l\prime}(c^l) - u^{h\prime}(c^h)}{u^{h\prime}(c^h)}}_{\text{insurance}} \underbrace{+ 1 - \frac{e\,\phi_{\mathcal{T}}(y)}{(1-e)\,\phi_{\mathcal{B}}(y)} \Big/ \frac{\int_{i'} e_{i'}\phi_{\mathcal{T}}(y_{i'})\,di'}{\int_{i'}(1-e_{i'})\,\phi_{\mathcal{B}}(y_{i'})\,di'}}_{\text{cross-subsidization}} \right) \right] \tag{13}$$

where $\bar{\lambda} \equiv \frac{\mathbb{E}^l[g]}{\mathbb{E}^h[g]}$ *is the ratio of risk-weighted social marginal welfare weights across the two states, and* $\lambda_i \equiv \frac{g_i}{\mathbb{E}^l[g]}$ *is the welfare weight of individual i, normalized such that the low-state risk-weighted average is 1, i.e.,* $\mathbb{E}^l[\lambda] = 1$. *This can be equivalently expressed as:*

$$\text{RR} = \underbrace{\bar{\lambda}\mathbb{E}^l \left[\lambda \left(\frac{u^{l\prime}(c^l) - u^{h\prime}(c^h)}{u^{h\prime}(c^h)} \right) \right]}_{\substack{\text{risk protection}\\ \text{(within-individual)}}} + \underbrace{(\bar{\lambda} - 1)}_{\substack{\text{cross-subsidization}\\ \text{(across-individuals)}}} \tag{14}$$

Proof: See Appendix F.1.

Equation (13) expresses the welfare gain from social-insurance expansion via resource reallocation as a weighted average of individuals' willingness-to-pay, modulated by individual and aggregate social marginal welfare weights. The individual weight λ_i is normalized so that its low-state risk-weighted average is 1 (i.e., $\mathbb{E}^l[\lambda] = 1$). It reflects the social value of increasing individual *i*'s surplus by \$1, relative to the average value among program beneficiaries. The aggregate weight $\bar{\lambda}$ captures the relative weight placed on individuals in the low state compared to those in the high state—that is, the average social value of providing an additional \$1 of surplus to program beneficiaries relative to funders. This connects to our descriptive evidence showing households who experience unemployment have systematically lower pre-unemployment consumption and earnings, implying that reform beneficiaries differ from funders in economically meaningful ways. Importantly, $\bar{\lambda}$ is reform-specific because the risk weights depend on $\phi_{\mathcal{B}}(\cdot)$ and $\phi_{\mathcal{T}}(\cdot)$, capturing how benefit and tax changes vary with income.

Equation (14) expresses the welfare gain from the reform in terms of two components: (1) the social value of insurance, which we refer to as risk protection, and (2) the social value of the cross-subsidization embedded in the implicit pricing of social insurance. This across-individual redistribution component is captured by a single sufficient statistic—the reform-specific aggregate welfare weight $\bar{\lambda}$.[21]

These equations highlight two distinct channels through which social insurance reform affects aggregate welfare. First, a reform alters the degree of risk protection provided to individuals. All else equal, risk protection is more valuable when individual insurance values are more positively correlated with welfare weights. Moreover, its value increases with the relative weight society places on program beneficiaries compared to funders, as captured by the aggregate welfare weight $\bar{\lambda}$. Second, even absent individual demand for insurance, the reform reallocates net resources across individuals through cross-subsidization. If individuals in the low state have higher welfare weights than those in the high state (i.e., $\bar{\lambda} > 1$), expanding the program generates positive value purely from this reallocation. This cross-subsidization arises because the implicit pricing embedded in the tax and benefit adjustments typically deviates from actuarially fair terms.

Importantly, because the risk weighting in expectations depends on reform design (via $\phi_{\mathcal{B}}$ and $\phi_{\mathcal{T}}$), both the risk protection and cross-subsidization components are reform specific.

5.2. Implementation

We maintain Assumption 1, which implies state-independent marginal utility with common local curvature. In addition to individual-level insurance values and risk weights, we require two additional ingredients.

Welfare weights. The social marginal welfare weights g_i encode social preferences over interpersonal utility comparisons. As highlighted by Saez and Stantcheva (2016), these weights accommodate a wide range of normative views about how to compare welfare across individuals. In our baseline implementation, we use the following specification:

$$g_i \equiv g_{y_i}\tilde{g}_i, \qquad \tilde{g}_i \equiv (c_i^h)^{-\gamma} \Big/ \mathbb{E}[(c^h)^{-\gamma} \mid y_i], \tag{15}$$

[21] Like our willingness-to-pay expression (equation (3)), the decompositions in equations (13) and (14) focus on marginal reforms. This allows us to apply the envelope theorem and ensures consistency with the analogous conditions in a rich dynamic model (see Appendix F.2). The decomposition is general in that it applies both to marginal reforms around the current system—our focus in this paper—and to reforms evaluated at the optimum. As such, it can be combined with a structural model to help characterize optimal policy.

where g_{y_i} is the inverse-optimum weight computed in Hendren (2020), and c_i^h denotes long-run in-work (equivalized) consumption. By construction, $\mathbb{E}[\tilde{g} \mid y_i] = 1$, so the inverse-optimum weights g_{y_i} determine the *average* welfare weight placed on each income group, while $\tilde{g}_i$ reweights *within* an income group toward individuals with lower long-run consumption.[22]

Under this specification, the social value of transferring \$1 of surplus between individuals at different income levels is governed by the inverse-optimum weights g_y. These weights rationalize the existing income-tax-and-transfer system as locally optimal, meaning that marginal revenue-neutral modifications to that system have zero effect on social welfare. They can be interpreted as revealed social preferences for redistribution across income groups, or as the marginal cost to the government of providing \$1 of surplus to individuals with income y through an adjustment to the tax-and-transfer system. Using these weights to value cross-income redistribution reflects the fact that equivalent transfers could, in principle, be implemented through the tax-and-transfer system. As most married households file income taxes jointly, we evaluate g_y at household income.

Social insurance reform may also reallocate surplus among individuals with the same level of income. Our welfare-weight specification values this within-income-group surplus redistribution according to $\tilde{g}_i \propto (c_i^h)^{-\gamma}$, implying that, conditional on current income, the social value of providing an additional dollar of surplus to an individual declines with their consumption level. This is motivated by empirical evidence that consumption is a reliable proxy for both lifetime income and economic well-being (e.g., Poterba 1989; Meyer and Sullivan 2003). We measure c_i^h using panel-average in-work consumption, equivalized by household size to account for differences in needs due to household composition, and winsorize this measure at the 1st and 99th percentiles. While the income tax and transfer system redistributes across current income levels, social insurance reform may deliver additional redistributive value by directing resources to individuals with low consumption levels *within* income groups.

Fiscal externality. The fiscal externality associated with social insurance reform (equation (10)) depends on the behavioral response elasticity, $\epsilon_i^{(1-e,b)}$, and on state-specific net taxes and benefits, $\mathcal{T}(y_i)$ and $\mathcal{B}(y_i)$.

We calibrate the elasticity using evidence from Chetty (2008), who reports unemployment-duration elasticities separately by quartiles of the wealth

[22] Equivalently, we can write $g_i = \omega_i u_i'(c_i^h)$ in terms of Pareto weights ω_i and marginal utilities $u_i'(\cdot)$. Approximating $u_i'(c_i^h) \approx (c_i^h)^{-\gamma}$ and choosing $\omega_i = \frac{g_{y_i}}{\mathbb{E}[(c^h)^{-\gamma} \mid y_i]}$ implies the social marginal welfare weights in equation (15), $g_i = g_{y_i} (c_i^h)^{-\gamma} / \mathbb{E}[(c^h)^{-\gamma} \mid y_i]$.

distribution.[23] Since UI benefits expire after a fixed potential duration, we adjust the mapping from duration responses to fiscal costs to account for benefit exhaustion (see Appendix G.1). The average elasticity reported in Chetty (2008) aligns closely with the median value reported in the broader literature, as summarized by Schmieder and Von Wachter (2016).

We compute state-contingent net tax and benefit schedules using NBER TAXSIM, assuming married households file jointly (see Appendix G.1 for details). This means our measure of the fiscal externality captures the impact of behavioral responses through both higher UI payments and changes in income tax revenues.

Discussion. Our characterization of the welfare gains from resource reallocation parallels recent work on the social value of transfer programs. For example, Kolsrud et al. (2024) examine reforms that shift resources between an early and a late retiree and show that differences in their consumption levels and welfare weights capture the combined insurance and redistributive effects. A key difference in our setting is that we model heterogeneity across the entire distribution of workers, so the social value of reform depends on the joint distribution of insurance values and welfare weights, together with unemployment risk, which is central in our setting. In this respect, our approach also connects to recent work using PSID consumption data to evaluate the targeting value of disability insurance (Deshpande and Lockwood 2022) and welfare programs (Rafkin et al. 2023).

5.3. Results

Baseline

Row (a) of Table 4 presents our baseline estimate of the social value of increasing UI generosity. These estimates are based on a budget-neutral expansion of the current U.S. UI program that raises benefits in proportion to the minimum of a worker's earnings and an earnings cap, funded by a flat increase in payroll taxes.[24] Columns (1) and (2) report the social value derived from risk protection and cross-subsidization, respectively, while column (3) shows the total resource-reallocation effect—the sum

[23] Chetty (2008) estimates elasticities of unemployment exit hazards with respect to UI benefits that range from –0.642 in the lowest wealth quartile to 0.016 in the highest (Table 2, p. 204). Following Chetty (2008), we map these hazard elasticities into unemployment-duration elasticities by using the fact that, under a constant-hazard approximation, the elasticity of duration is the negative of the elasticity of the exit hazard. This duration elasticity is the dynamic counterpart of our static elasticity $\epsilon_i^{(1-e,b)}$ (see Appendix F.2). In the PSID, we assign each household the corresponding elasticity based on its wealth quartile.

[24] Our welfare results are broadly stable to alternative choices of the number of latent types, and rescaling the insurance component of WTP leaves the relative ranking of the baseline, flat, and proportional reforms essentially unchanged (see Appendix H).

of these two components. All numbers are reported per $1 of mechanical increase in benefit expenditure. We estimate a risk-protection value of $0.48 and a cross-subsidization value of $0.19, implying a total resource-reallocation gain of $0.67.

These gains must be weighed against the associated fiscal externalities. Column (4) reports the direct fiscal externality from higher UI payments, column (5) reports the additional fiscal externality due to reduced income tax receipts, and column (6) presents the total. We find that accounting for the broader tax-and-transfer system increases the fiscal externality from $0.27 (excluding income-tax revenue effects) to $0.58 (including them). Overall, the fiscal externality ($0.58) falls slightly below the resource-reallocation gains ($0.67), implying a modest positive but statistically indistinguishable from zero net welfare effect under our baseline welfare weights.

Reform structure

Rows (b) and (c) examine two alternative reforms: a flat benefit expansion (independent of earnings) and a proportional expansion without a cap. Relative to the baseline reform, the flat expansion is more progressive, allocating a larger share of additional resources to individuals with higher welfare weights. This raises the social value of both risk protection and cross-subsidization; the total resource-reallocation gain from the flat reform is $0.78 per dollar.

By contrast, the proportional reform is more regressive than the baseline, directing a larger share of resources to workers with earnings above the cap. This reduces the social value of risk protection and cross-subsidization, yielding a lower resource-reallocation gain of $0.61 for this reform.

The overall social value of the flat expansion is further enhanced by a lower fiscal externality than the baseline. This reduction reflects two offsetting forces. On the one hand, the flat reform directs a larger share of the benefit expansion toward lower earners, who tend to exhibit somewhat stronger behavioral responses; this modestly increases the direct fiscal externality. On the other hand, lower earners face smaller income-tax liabilities, which reduces the indirect fiscal externality linked to lost tax revenue—and this effect dominates. For the proportional reform, the pattern is reversed: shifting the incentives to reduce effort toward higher earners who are less responsive lowers the direct fiscal externality relative to the baseline, but this is more than offset by a higher indirect fiscal externality due to their larger tax contributions.

Taken together, these differences across reform structures underscore the importance of incidence in determining both the distributional benefits of reform and the magnitude of the associated fiscal externalities.

Social preferences

Row (a) reports baseline results under the welfare weights in equation (15); rows (d)–(f) illustrate how the social value of the reform depends on the welfare weights placed on individuals' willingness-to-pay.

Money-metric weights. Under money-metric marginal social welfare weights (g_i = 1), social preferences are indifferent to how surplus is distributed across individuals. One rationale for this approach is the Kaldor-Hicks compensation principle: if it is theoretically possible to combine the reform with individual-specific lump-sum taxes and transfers to achieve a Pareto improvement, the reform is considered desirable. Since this criterion is distribution-neutral, cross-subsidization contributes zero by construction, and risk protection is valued using common welfare weights, yielding a resource-reallocation value of \$0.38 (row (d)). A key drawback of this criterion is that it relies on a lump-sum redistribution scheme that is not feasible in practice.

Money-metric weights also clarify the connection between the budget-neutral reform that we focus on and the "unpackaged" components: (i) an unfunded benefit increase and (ii) a tax change. UI beneficiaries' average WTP for an *unfunded* UI expansion, per mechanical \$1 increase in benefits, is \$1.38—the direct \$1 plus \$0.38 in money-metric risk-protection value (Table 4, row (d)).

To unpack the budget-neutral reform into an unfunded benefit increase and a financing tax change, we attribute the fiscal externality to the benefit expansion. Then each \$1 of higher benefits costs the government \$1.58 once behavior is accounted for (the \$1 mechanical cost plus \$0.58 through both higher UI payments and lower tax receipts). The implied marginal value of public funds (MVPF) for an unfunded UI benefit expansion is therefore \$1.38/\$1.58 $\approx$ 0.87. Under the same unpacking, an unfunded payroll tax reduction has a mechanical cost of \$1, is assigned no fiscal externality, and raises recipients' surplus by exactly \$1, so its MVPF is 1. With money-metric welfare weights, this comparison implies that an additional dollar of government revenue is more valuable when used to cut payroll taxes than to expand UI benefits. When welfare weights value surplus redistribution, the joint reform can be unpacked analogously into appropriately welfare-weighted MVPFs for UI benefits and for the financing tax (Appendix F.3).

Inverse-optimum weights. Row (e) uses inverse-optimum welfare weights and assumes that social preferences are indifferent to the distribution of surplus *within* income groups—i.e., it sets within-income welfare weights $\tilde{g}_i$ = 1. Hendren (2020) motivates inverse-optimum weights as an incentive-compatible extension of the Kaldor-Hicks compensation principle that accounts for the distortionary costs of

achieving cross-income redistribution through the income tax system. Under these weights, the social value of resource reallocation is $0.51.

Cross- versus within-income redistribution. Comparing money-metric weights, inverse-optimum weights, and the baseline specification quantifies the social value of surplus redistribution.

Comparing row (d) with row (e) isolates the value of *cross*-income surplus redistribution delivered by a UI expansion. Moving from money-metric to inverse-optimum weights increases the resource-reallocation gain by $0.13, with over two-thirds of this increase coming from the cross-subsidization component and the remainder from changes in the welfare-weighted insurance term.

Comparing row (e) with the baseline results in row (a) reveals the additional social value of *within*-income-group surplus reallocation—toward individuals with lower long-run in-work consumption. This within-income effect further increases the resource-reallocation gain by $0.16, with around two-thirds of the increase arising from the cross-subsidization component. This pattern reflects that, conditional on current income, lower long-run consumption households are more likely to become unemployed and thus receive a larger share of UI transfers, consistent with evidence on targeting and selection in disability insurance (Deshpande and Lockwood 2022) and welfare programs (Rafkin et al. 2023).

Of the overall resource-reallocation gain of $0.67 in row (a), we attribute $0.38 (57%) to efficiency gains from risk protection, $0.13 (19%) to cross-income surplus redistribution, and the remaining $0.16 (24%) to within-income surplus redistribution.

Utilitarian weights. The use of inverse-optimum weights to value cross-income redistribution can be motivated either by the extended Kaldor-Hicks compensation principle or by interpreting these weights as reflecting society's revealed preferences for redistribution. One potential objection to the first rationale is that, in practice, social insurance reforms are rarely accompanied by the income-tax adjustments required to implement the implied compensation scheme. The second rationale may be challenged on normative grounds, as these weights may encode preferences for redistribution that are either too strong or too weak.

In row (f), we therefore present results based on utilitarian social welfare weights, $g_i = (c_i^h)^{-\gamma}$. These weights place greater value on cross-income redistribution than inverse-optimum weights, increasing both the value of risk protection and the cross-subsidization component of resource reallocation.

While the precise social value of surplus reallocation ultimately depends on the policymaker's chosen welfare-weighting scheme, this analysis shows that it will generally exceed the value implied by money-metric weights. In our application, as long as marginal surplus is valued more for (i) lower-income than higher-income households, (ii) lower-consumption than higher-consumption households, or (iii) some combination of both, the social value of the reform exceeds that implied by money-metric weights.

Alternative implementations

Standard implementation. The penultimate row of Table 4 reports results for a flat reform evaluated using money-metric welfare weights, approximating the standard Baily–Chetty sufficient-statistics implementation. The implied risk protection value is \$0.39 per dollar, reflecting the use of unemployment-risk-weighted averages rather than welfare weights that place additional value on surplus redistribution (as in row (b)). The fiscal externality of \$0.28 includes additional UI payments but excludes changes in income-tax revenues. The resulting net welfare effect is positive, at approximately \$0.11.

Average implementation. The final row of Table 4 presents results from an alternative implementation of a flat reform that adapts the approach of Kolsrud et al. (2024), developed in the context of retirement benefit reform. In our adaptation, we shut down within-income-group heterogeneity in welfare weights and evaluate equation (14) after replacing individual-level objects with their averages within the employed and unemployed populations. It captures redistribution between employment states, but abstracts from within-state heterogeneity in consumption losses and welfare weights. It yields a resource-reallocation gain of \$0.57 per dollar—closer to our full individual-level estimate for a flat reform (\$0.78) than the standard implementation (\$0.39).

We apply the same logic to the fiscal externality, using average elasticities, UI entitlements, and state-specific tax positions among the unemployed. In our UI application, averaging these inputs separately overstates the indirect fiscal externality (\$0.38 versus \$0.15 under the flat reform). This reflects both the negative association between behavioral elasticities and tax losses per dollar of UI and the positive association between this ratio and UI entitlements. The average implementation yields a net social value of -\$0.09, compared with +\$0.35 under the full implementation. We provide further details in Appendix G.2.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Resource reallocation			Fiscal externality			Overall effect
	Risk protection	Cross-subsidization	Total (1)+(2)	Direct	Indirect	Total (4)+(5)	(3)-(6)
(a) Baseline	0.48	0.19	0.67	0.27	0.31	0.58	0.10
	[0.38, 0.58]	[0.15, 0.22]	[0.56, 0.78]	[0.26, 0.27]	[0.29, 0.33]	[0.56, 0.60]	[-0.03, 0.20]
Reform structure							
(b) Flat	0.52	0.25	0.78	0.28	0.15	0.43	0.35
	[0.41, 0.63]	[0.22, 0.29]	[0.66, 0.89]	[0.28, 0.28]	[0.12, 0.18]	[0.40, 0.46]	[0.22, 0.45]
(c) Proportional	0.46	0.15	0.61	0.25	0.36	0.62	-0.01
	[0.36, 0.55]	[0.11, 0.18]	[0.50, 0.70]	[0.25, 0.26]	[0.34, 0.39]	[0.60, 0.64]	[-0.13, 0.08]
Social preferences							
(d) Money-metric	0.38	0	0.38	=(a)	=(a)	=(a)	-0.19
			[0.31, 0.46]				[-0.27, -0.12]
(e) Inverse-optimum	0.42	0.09	0.51	=(a)	=(a)	=(a)	-0.07
	[0.33, 0.50]	[0.08, 0.09]	[0.42, 0.59]				[-0.16, 0.01]
(f) Utilitarian	0.64	0.53	1.17	=(a)	=(a)	=(a)	0.60
	[0.50, 0.78]	[0.45, 0.59]	[1.00, 1.35]				[0.41, 0.77]
Alternative implementations							
(g) Standard implementation	0.39	-	0.39	0.28	-	0.28	0.11
(flat reform+money-metric)			[0.31, 0.48]			[0.28, 0.28]	[0.03, 0.20]
(h) Average implementation	0.44	0.13	0.57	0.28	0.38	0.66	-0.09
	[0.35, 0.53]	[0.12, 0.13]	[0.48, 0.66]	[0.28, 0.28]	[0.36, 0.40]	[0.64, 0.68]	[-0.18, 0.00]

TABLE 4. Risk Protection, Cross-subsidization, and the Fiscal Externality from UI Expansion

Notes: Column (7) reports the overall social welfare impact of a UI expansion, decomposed in columns (1)–(6). Columns (4) and (5) split the fiscal externality into the direct component from extra UI benefit payments and the indirect component from changes in income tax liabilities; Appendix G.1 provides details. Row (a) corresponds to an expansion of the current U.S. UI system under baseline social preferences; remaining rows report results for alternative reform structures or social preferences. All entries are expressed per $1 of mechanical increase in benefit expenditure. 95% confidence intervals are based on 400 household-level bootstrap replications and reported in square brackets. Confidence intervals reflect sampling uncertainty in the PSID estimates, conditional on the calibrated behavioral elasticities and benefit-duration parameters.

The average implementation also does not distinguish between reforms that differ only in their incidence within unemployed workers. Once normalized per dollar of mechanical benefit increase, the average implementation gives the same value for flat, proportional, and proportional-capped reforms.

Cross-sectional implementation. The results in Table 4 use the panel dimension of the PSID. Panel data offer two main advantages over cross-sectional data. First, they enable us to estimate individual-level insurance values and WTP, which facilitates a decomposition of the social gains from expanding UI generosity into within-individual risk protection and across-individual cross-subsidization. Second, longitudinal information allows us to measure consumption and earnings using within-household panel averages, helping to limit the influence of measurement error. However, when only cross-sectional data are available, it is still possible to compute the social value of resource reallocation by directly implementing the social marginal utility gap in equation (12).

Discussion. Our analysis advances the literature on the design of social insurance by showing how individual-level heterogeneity and reform structure shape insurance value, implicit cross-subsidization, and incentive costs—and, through these channels, the social value of UI reforms. Our implementation has several limitations that are important to keep in mind when interpreting the quantitative results.

First, our baseline calculations treat workers in our sample as covered by the UI system and abstract from heterogeneity in eligibility and take-up. In practice, many low-income workers either do not qualify for benefits or fail to claim them conditional on eligibility. While we proxy for eligibility by excluding periods immediately following time out of the labor market from our sample, this is an imperfect measure of actual eligibility. Similarly, although our implementation accounts for benefit exhaustion on average, our data do not allow us to measure how the share of unemployment duration covered by benefits varies across workers.

Our framework can accommodate these dimensions of heterogeneity by scaling the marginal benefit adjustment $\phi_{\mathcal{B}}(y_i)$ by the probability of take-up conditional on unemployment and by the expected share of unemployment time covered by benefits. Using recent evidence on gradients in UI take-up and benefit duration by pre-separation wages from Lachowska et al. (2025), we implement a simple sensitivity analysis (see Appendix H.1). Allowing for a take-up gradient reduces the baseline resource-reallocation term from \$0.67 to about \$0.62 per \$1 of mechanical increase in benefit expenditure. Allowing benefit exhaustion to vary across the wage distribution produces a similar reduction. Accounting for both heterogeneous take-up and benefit exhaustion reduces the estimate to \$0.57. These adjustments attenuate the extent of cross-income redistribution generated by the reform, but leave the qualitative conclusion unchanged: accounting for heterogeneous incidence remains important for evaluating UI expansions. Our framework can also be used to study reforms that explicitly expand coverage or take-up among currently uncovered

low-income workers, or that extend potential benefit duration, but we leave a full exploration of such policies to future work.

Second, our baseline estimates capture the impact of UI generosity on employment and earnings, and hence on UI payments and income-tax revenues, holding other programs fixed. In reality, UI interacts with the broader safety net. More generous UI may alter reliance on other transfer programs (see Mueller et al. 2016 who study interactions between UI and Social Security Disability Insurance), affecting both households' net benefit receipts and the fiscal cost of reform. At the same time, UI reforms may affect other behavioral margins that could amplify or offset these effects (see, e.g., Anderson and Meyer 1997 on UI generosity and take-up). We view a full accounting of these program interactions and behavioral responses as an important avenue for future research.

6. Conclusion

We study how heterogeneity in unemployment risk and consumption responses shapes both individual demand for unemployment insurance (UI) and the social value of expanding UI generosity. We estimate the distribution of workers' willingness-to-pay for UI reforms and show that it reflects two forces: heterogeneous insurance value, captured by unemployment-induced consumption declines, and cross-subsidization arising from pooled financing and heterogeneous unemployment risk. Because workers who experience unemployment are negatively selected—by income and, conditional on income, by in-work consumption—UI expansions tend to redistribute surplus toward less well-off households through both risk protection and cross-subsidization.

We weigh these gains against distortionary costs, including reduced income-tax revenues. Under our baseline welfare weights, a natural proportional–capped expansion of the current U.S. UI system yields a modestly positive estimated net value, statistically indistinguishable from zero. A flat expansion that targets more resources toward lower-income households yields positive net value.

Our approach is marginal and avoids estimating a full structural model. A promising direction for future work is to combine our decomposition with a structural framework to assess optimal UI design. Such work could also quantify the extent to which heterogeneity in insurance values reflects not only variation in unemployment-induced consumption declines, but also richer heterogeneity in risk aversion or state-dependent marginal utilities.

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Appendix: For online publication

Risk Protection and Redistribution in the Design of Social Insurance

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A. Willingness-to-Pay: Theory

A.1. General model

Here we extend the static model from Section 2 to a richer dynamic setting, following the framework in Chetty (2006) and derive an expression for willingness-to-pay for social insurance reform. As in equation (3), willingness-to-pay decomposes into components reflecting insurance and cross-subsidization. The difference is that, in the dynamic model, these components are evaluated as expectations over the individual's lifetime.

We consider a setting in which time is continuous and individuals live over $t \in [0, 1]$. Let $\varphi_{i,t}$ collect the exogenous shocks observed up to time t, with distribution $F_{i,t}(\varphi_{i,t})$ given the information available at $t = 0$. Assume $F_{i,t}$ is smooth with maximal support Φ for all (i, t).

Let $c_{i,t}(\varphi_{i,t})$ denote individual i's time t state-contingent consumption. Let $x_{i,t}(\varphi_{i,t})$ denote M other choices the individual makes (for instance, different dimensions of effort, actions to self-insure like borrowing from family, spousal labor supply decisions and so on). Denote the individual's flow utility function $u_i^s(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}))$ for $s \in \{l, h\}$.

Employment status is determined by the shocks $\varphi_{i,t}$ and the worker's feasible choice program, $\xi_{i,t}(\varphi_{i,t}; x_i)$. We suppress the choice argument and evaluate employment outcomes at the optimal program when computing WTP, writing $\xi_{i,t}(\varphi_{i,t}) \in \{0, 1\}$ to denote the state the individual is in at time t and given state variable $\varphi_{i,t}$: if $\xi = 1$ the individual is in the high state; if $\xi = 0$ the individual is the low state. The reform affects employment only through changes in optimal choices. Under the regularity conditions stated below, the envelope theorem eliminates the first-order utility effects of these induced behavioral adjustments, including

changes in employment.

Denote the full program of individual i's state-contingent choices:

$$c_i = \{c_{i,t}(\varphi_{i,t})\}_{t\in[0,1],\varphi_{i,t}\in\Phi},$$
$$x_i = \{x_{i,t}(\varphi_{i,t})\}_{t\in[0,1],\varphi_{i,t}\in\Phi}.$$

When in the high state the individual earns $y_i - \mathcal{T}(y_i)$ and when in the low state they receive benefits $\mathcal{B}(y_i)$. The individual can also earn additional income $f_{i,t}(x_{i,t}(\varphi_{i,t}))$. They face the flow budget constraint,

$$\dot{A}_{i,t}(\varphi_{i,t}) = \xi_{i,t}(\varphi_{i,t})\left(y_i - \mathcal{T}(y_i)\right) + (1-\xi_{i,t}(\varphi_{i,t}))\mathcal{B}(y_i) + f_{i,t}(x_{i,t}(\varphi_{i,t})) - c_{i,t}(\varphi_{i,t}),$$

with terminal condition: $A_{i,1}(\varphi_{i,1}) \geq \bar{A}_i$ for all $\varphi_{i,1}$. They also face N additional constraints in each state $\varphi_{i,t}$ at each time t:

$$g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t}) \geq 0.$$

The individual's problem is to choose the program (c_i, x_i) to solve:

$$\begin{aligned}
\max \int_0^1 \int_{\varphi_{i,t}} &\left(\xi_{i,t}(\varphi_{i,t})u^h_i(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t})) + (1-\xi_{i,t}(\varphi_{i,t}))u^l_i(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}))\right) dF_{i,t}(\varphi_{i,t})dt \\
&+ \int_0^1 \int_{\varphi_{i,t}} \lambda^A_{it}(\varphi_{i,t})\Big(\xi_{i,t}(\varphi_{i,t})\left(y_i - \mathcal{T}(y_i)\right) + (1-\xi_{i,t}(\varphi_{i,t}))\mathcal{B}(y_i) + f_{i,t}(x_{i,t}(\varphi_{i,t})) - c_{i,t}(\varphi_{i,t}) - \dot{A}_{i,t}(\varphi_{i,t})\Big) dF_{i,t}(\varphi_{i,t})dt \\
&+ \int_{\varphi_{i,1}} \lambda^A_{i,1}(\varphi_{i,1})(A_{i1}(\varphi_{i,1}) - \bar{A}_i) dF_{i,1}(\varphi_{i,1}) \\
&+ \sum_{n=1}^N \int_0^1 \int_{\varphi_{i,t}} \lambda^n_{it}(\varphi_{i,t}) g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t}) dF_{i,t}(\varphi_{i,t})dt.
\end{aligned}$$

Denote the maximum function of this problem $V_i(\theta)$, written as a function of the policy parameters θ, which parameterizes modifications to the tax and benefit schedules. We assume the following regularity conditions:

ASSUMPTION 1 (Regularity Conditions). *Assume*

i. *Total lifetime utility is smooth, increasing and strictly quasi-concave in* (c, x)

ii. *The choices* (c, x) *that satisfy the constraints are convex*

iii. $V_i(\theta)$ *is differentiable*

where (i) and (ii) ensure the individual's problem has a unique solution and (iii) ensures the envelope theorem applies.

Consider the reform, $d\theta$, to the benefit and tax schedules parameterized by:

$$\frac{d\mathcal{B}(y_i)}{d\theta} = \phi_{\mathcal{B}}(y_i)$$
$$\frac{d\mathcal{T}(y_i)}{d\theta} = \phi_{\mathcal{T}}(y_i) \times \frac{\int_{i'} \int_0^1 \int_{\varphi_{i',t}} \phi_{\mathcal{B}}(y_{i'}) \left(1 - \xi_{i',t}(\varphi_{i',t})\right) dF_{i',t}(\varphi_{i',t}) dt di'}{\int_{i'} \int_0^1 \int_{\varphi_{i',t}} \phi_{\mathcal{T}}(y_{i'}) \xi_{i',t}(\varphi_{i',t}) dF_{i',t}(\varphi_{i',t}) dt di'}.$$

This reform is budget-neutral in the absence of any behavioral responses.

Assume that the constraints $g^n_{i,t}$ satisfy the regularity conditions:

$$\frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial \mathcal{B}(y_i)} = -(1 - \xi_{i,t}(\varphi_{i,t})) \frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial c_{i,t}(\varphi_{i,t})}$$
$$\frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial \mathcal{T}(y_i)} = \xi_{i,t}(\varphi_{i,t}) \frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial c_{i,t}(\varphi_{i,t})}$$
$$\frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial c_{i,s}(\varphi_{i,s})} = 0 \quad \text{if } s \neq t,$$

for $n = 1, \ldots, N$. See Chetty (2006) for a demonstration of the mildness of these conditions and an example of when they do not hold.

The impact of the reform $d\theta$ on individual i's expected utility is:

$$\frac{dV_i}{d\theta} = -\int_0^1 \int_{\varphi_{i,t}} \left(\xi_{i,t}(\varphi_{i,t}) \lambda^A_{i,t}(\varphi_{i,t}) - \sum_{n=1}^N \lambda^n_{i,t}(\varphi_{i,t}) \frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial \mathcal{T}(y_i)} \right) dF_{i,t}(\varphi_{i,t}) dt \frac{d\mathcal{T}(y_i)}{d\theta}$$
$$+ \int_0^1 \int_{\varphi_{i,t}} \left((1 - \xi_{i,t}(\varphi_{i,t})) \lambda^A_{i,t}(\varphi_{i,t}) + \sum_{n=1}^N \lambda^n_{i,t}(\varphi_{i,t}) \frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial \mathcal{B}(y_i)} \right) dF_{i,t}(\varphi_{i,t}) dt \frac{d\mathcal{B}(y_i)}{d\theta}.$$

Under the regularity assumptions on the constraints, individual i's optimal consumption choice satisfies,

$$\frac{\partial u^s_i(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}))}{\partial c_{i,t}(\varphi_{i,t})} = \lambda^A_{i,t}(\varphi_{i,t}) - \sum_{n=1}^N \lambda^n_{i,t}(\varphi_{i,t}) \frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial c_{i,t}(\varphi_{i,t})},$$

for all t and $\varphi_{i,t}$. The first two assumptions imply for all t and $\varphi_{i,t}$:

$$\sum_{n=1}^N \lambda^n_{i,t}(\varphi_{i,t}) \frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial \mathcal{T}(y_i)} = \sum_{n=1}^N \lambda^n_{i,t}(\varphi_{i,t}) \xi_{i,t}(\varphi_{i,t}) \frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial c_{i,t}(\varphi_{i,t})}$$
$$\sum_{n=1}^N \lambda^n_{i,t}(\varphi_{i,t}) \frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial \mathcal{B}(y_i)} = -\sum_{n=1}^N \lambda^n_{i,t}(\varphi_{i,t}) (1 - \xi_{i,t}(\varphi_{i,t})) \frac{\partial g^n_{i,t}(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}); \varphi_{i,t})}{\partial c_{i,t}(\varphi_{i,t})}.$$

Hence, we can re-write $dV_i/d\theta$:

$$\frac{dV_i}{d\theta} = -\int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{T}}(y_i)\xi_{i,t}(\varphi_{i,t}) \frac{\partial u_i^h(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}))}{\partial c_{i,t}(\varphi_{i,t})} dF_{i,t}(\varphi_{i,t})dt$$
$$\times \frac{\int_{i'} \int_0^1 \int_{\varphi_{i',t}} \phi_{\mathcal{B}}(y_{i'}) \left(1 - \xi_{i',t}(\varphi_{i',t})\right) dF_{i',t}(\varphi_{i',t})dtdi'}{\int_{i'} \int_0^1 \int_{\varphi_{i',t}} \phi_{\mathcal{T}}(y_{i'})\xi_{i',t}(\varphi_{i',t})dF_{i',t}(\varphi_{i',t})dtdi'}$$
$$+ \int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{B}}(y_i)(1 - \xi_{i,t}(\varphi_{i,t})) \frac{\partial u_i^l(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}))}{\partial c_{i,t}(\varphi_{i,t})} dF_{i,t}(\varphi_{i,t})dt\,.$$

We define willingness-to-pay, scaling $\frac{dV_i}{d\theta}$ as follows:

$$\text{WTP}_i \equiv \frac{dV_i}{d\theta} \times \left[\frac{\frac{1}{\int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{B}}(y_i)(1-\xi_{i,t}(\varphi_{i,t}))dF_{i,t}(\varphi_{i,t})dt}}{\frac{\int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{T}}(y_i)\xi_{i,t}(\varphi_{i,t}) \frac{\partial u_i^h(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}))}{\partial c_{i,t}(\varphi_{i,t})} dF_{i,t}(\varphi_{i,t})dt}{\int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{T}}(y_i)\xi_{i,t}(\varphi_{i,t})dF_{i,t}(\varphi_{i,t})dt}} \right].$$

The scaling expresses the utility gains from the reform per \$1 increase in expected lifetime benefit payments, measured in units of the average of marginal utility of an unfunded tax cut (also expressed per \$1 worth of lifetime tax reduction).

Define the individual's state-specific average marginal utility of consumption:

$$\widetilde{\mathbb{E}}_i^h \left[\frac{\partial u^h(c,x)}{\partial c} \right] = \int_0^1 \int_{\varphi_{i,t}} \frac{\phi_{\mathcal{T}}(y_i)\xi_{i,t}(\varphi_{i,t})}{\int_{t'} \int_{\varphi_{i,t'}} \phi_{\mathcal{T}}(y_i)\xi_{i,t'}(\varphi_{i,t'})dF_{i,t'}(\varphi_{i,t'})dt'} \frac{\partial u_i^h(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}))}{\partial c_{i,t}(\varphi_{i,t})} dF_{i,t}(\varphi_{i,t})dt$$
$$\widetilde{\mathbb{E}}_i^l \left[\frac{\partial u^l(c,x)}{\partial c} \right] = \int_0^1 \int_{\varphi_{i,t}} \frac{\phi_{\mathcal{B}}(y_i)(1 - \xi_{i,t}(\varphi_{i,t}))}{\int_{t'} \int_{\varphi_{i,t'}} \phi_{\mathcal{B}}(y_i)(1 - \xi_{i,t'}(\varphi_{i,t'}))dF_{i,t'}(\varphi_{i,t'})dt'} \frac{\partial u_i^l(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}))}{\partial c_{i,t}(\varphi_{i,t})} dF_{i,t}(\varphi_{i,t})dt\,.$$

We can then express willingness-to-pay for the reform:

$$\text{WTP}_i = \frac{\widetilde{\mathbb{E}}_i^l \left[\frac{\partial u^l(c,x)}{\partial c} \right] - \widetilde{\mathbb{E}}_i^h \left[\frac{\partial u^h(c,x)}{\partial c} \right]}{\widetilde{\mathbb{E}}_i^h \left[\frac{\partial u^h(c,x)}{\partial c} \right]} +$$
$$1 - \left(\frac{\int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{T}}(y_i)\xi_{i,t}(\varphi_{i,t})dF_{i,t}(\varphi_{i,t})dt}{\int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{B}}(y_i)(1 - \xi_{i,t}(\varphi_{i,t}))dF_{i,t}(\varphi_{i,t})dt} \right) \Bigg/ \left(\frac{\int_{i'} \int_0^1 \int_{\varphi_{i',t}} \phi_{\mathcal{T}}(y_{i'})\xi_{i',t}(\varphi_{i',t})dF_{i',t}(\varphi_{i',t})dtdi'}{\int_{i'} \int_0^1 \int_{\varphi_{i',t}} \phi_{\mathcal{B}}(y_{i'}) \left(1 - \xi_{i',t}(\varphi_{i',t})\right) dF_{i',t}(\varphi_{i',t})dtdi'} \right),$$

where the first expression on the right-hand side captures the lifetime value of insurance and the second expression captures the lifetime value of program cross-subsidization.

A.2. State-dependent utility

Consider the simple framework outlined in Section 2. Suppose the worker has state-dependent utility, which we capture through the marginal utility shifter $\rho > 0$, defined such that $u_i^{h\prime}(c) = u_i'(c)$ and $u_i^{l\prime}(c) = \rho u_i'(c)$.

In this case, the individual's WTP is:

$$\text{WTP}_i = \left(\frac{\rho u_i'(c_i^l) - u_i'(c_i^h)}{u_i'(c_i^h)} \right) + \left(1 - \frac{e_i \phi_{\mathcal{T}}(y_i)}{(1-e_i)\phi_{\mathcal{B}}(y_i)} \Big/ \frac{\int_{i'} e_{i'} \phi_{\mathcal{T}}(y_{i'}) di'}{\int_{i'} (1-e_{i'}) \phi_{\mathcal{B}}(y_{i'}) di'} \right). \tag{A.1}$$

The state-dependence term enters the insurance component of WTP, by scaling marginal utility in the low state. In this case, the consumption-based implementation uses the approximation around c_i^h:

$$\frac{\rho u_i'(c_i^l) - u_i'(c_i^h)}{u_i'(c_i^h)} \approx (\rho - 1) + \rho\gamma \frac{\Delta c_i}{c_i^h}.$$

The state-dependence term captures factors that cause the same level of consumption expenditure to yield different marginal utilities across states. For example, leisure time may vary by state and act as a complement to or substitute for consumption. There may also be state-specific expenditures (e.g., work-related costs) that do not directly generate utility (Browning and Crossley 2001). Additionally, the extent to which individuals combine consumption expenditures with home production may differ across states (Aguiar and Hurst 2005).

The state-dependence parameter can also capture situations in which individuals face state-specific consumption prices (see Campos and Reggio 2020). Suppose the consumption price in the high state is ρ times the price in the low state, and utility is otherwise state independent. In this case, WTP takes the same form as in equation (A.1). The consumption-based implementation also leads to the same state-dependence-adjusted formula. However, it is necessary to correct observed expenditure changes to account for the state-specific price difference. Let x_i^s denote state-specific expenditures, which are observed in the data. These are related to consumption through:

$$\frac{\Delta c_i}{c_i^h} = \frac{x_i^h/\rho - x_i^l}{x_i^h/\rho} = \rho \left(\frac{\Delta x_i}{x_i^h} \right) - (\rho - 1).$$

B. Description of PSID Data

We use the Panel Study of Income Dynamics to estimate the sufficient statistics that quantify the welfare effects of unemployment insurance expansion. We use data from 1999–2019 biennial PSID waves, which include information on consumption expenditures and asset holdings. For our baseline sample, we focus on non-immigrant households.[1] We include both single and married households and do not select the sample on the basis of the reference person's gender.[2] Our sample has non-missing information on key demographics (age, education, and state of residence).

We restrict attention to households in which the reference person is aged 23–60 and classified as either working now or looking for work/unemployed. We exclude the remaining PSID employment-status categories, including the combined category for temporary layoff, sick leave, or maternity leave, as well as observations in which the reference person is retired, permanently disabled and neither working nor looking for work, in education, a homemaker, or in prison. In addition, to proxy for UI program eligibility we exclude those who never report employment, as well as the first observation following an excluded employment status. We retain observations in which the reference person is either employed or unemployed, and we do not condition on spousal employment status, since we are explicitly interested in household exposure to unemployment risk, including self-insurance through spousal labor supply (Blundell et al. 2016). We also require households to contribute more than one observation after applying these sample restrictions, since our main empirical specifications use within-household changes over time.

To reduce the influence of measurement error, we drop observations with extremely high asset values and observations that exhibit implausibly large fluctuations in key outcomes. Following Blundell et al. (2016), we exclude households with net worth greater than $20 million and drop observations with extreme "jumps" in earnings, household income, and consumption which we attribute to measurement error. A "jump" is defined as an extreme positive (negative) change from $t-2$ to t, followed by an extreme negative (positive) change from t to $t+2$. Formally, for each variable x, we compute the biennial log difference $\Delta \ln(x_t) = \ln(x_t) - \ln(x_{t-2})$ and drop the observation t if $\Delta \ln(x_{t+2}) \Delta \ln(x_t)$ lies in the bottom 0.25 percent of its wave-specific distribution. Furthermore, in constructing permanent income we exclude earnings observations when the implied hourly

[1] This is an additional sample frame beginning in the late 1990s. We do, however, include the Survey of Economic Opportunity households, which oversample low-income families. Results using survey weights are similar.

[2] Previously, the PSID referred to the reference person as the household head.

wage is below one-half of the state minimum wage. Table B.1 provides summary statistics for our sample of interest.

To measure education, we create five categories based on completed years of schooling: less than high school, high school graduate, some college (including those who dropout of four-year degrees and those who attain a community college degree or other diploma), four-year college graduate; and postgraduate education. Our measure of race is derived from the self-reported race of the household reference person, from which we define three race categories (White, Black, and other).

The majority of our empirical analysis focuses on the relationship between unemployment risk and both levels and growth rates in consumption. The PSID measures disaggregated consumption across a number of different household expenditure categories designed to cover approximately 70% of aggregate consumption. Respondents also report the reference period for the expenditure item. We convert all expenditures to annual values (e.g., multiplying weekly expenditures by 52 and monthly expenditures by 12) and treat missing values in subcategories as zeros. We focus on broad expenditure categories available throughout the sample. Our primary consumption measure captures a range of utility-relevant expenditures comprising nondurable purchases plus service. We also show that our results are robust to alternative consumption measures (including food consumption as in Gruber 1997 and Hendren 2017), which we describe below. Consumption in the PSID is measured at the household level. To account for differences in household size, we equivalize using the square root of the household size.[3]

Baseline: Nondurable Expenditure including Services. Our baseline measure includes a broad range of nondurable expenditures and services, excluding categories with an investment or durable component (e.g., vehicle maintenance). To build our baseline consumption series, we first construct a food-expenditure series by summing food consumed at home, food consumed away from home, and food purchased using Supplemental Nutrition Assistance Program (SNAP). Including SNAP-financed food expenditures (formerly the Food Stamp Program) is important because they are a relevant source of consumption financing for low-income households, and Low and Pistaferri (2015) show that food stamps can act as substitutes for social insurance.

We then construct a service-expenditure series excluding durables. We sum

[3]Our baseline uses broad expenditure categories available throughout the sample, although the wording and aggregation of some component questions change across waves. We group the available items into these categories and include survey-wave fixed effects to absorb common shifts in measured consumption.

spending on home and automobile insurance, utilities, parking, and other direct transportation costs (e.g., bus fare and taxi payments) that do not correspond to vehicle maintenance, and child care.

Finally, we combine the aggregated food series with gasoline expenditures and the services series without a durable component.

Food Expenditure. To construct a food-expenditure series that excludes other components of our baseline measure, we sum food consumed at home, food purchased to be consumed away from home, and food purchased using SNAP benefits.

Services Expenditure. To construct a series for broad services, including categories that may relate to durables or have an investment component, we combine the services measured in our baseline expenditure with a set of additional spending categories. These additional categories include health-care-related spending (out-of-pocket payments including for hospital and nursing home stays, doctor visits, and prescription drugs as well as insurance premiums), vehicle repairs, and payments for educational services or schooling costs such as school tuition.

Total (Non-Housing) Expenditure. We combine our baseline measure with the additional services described above to produce a household-level series for total non-housing expenditures. As the PSID consumption categories are not designed to have full coverage, the term "total" is a misnomer. We continue to exclude durable purchases (e.g., vehicles) and memory goods (Hai et al. 2020), such as vacations, that have durable-like properties.

Total (Including Housing) Expenditure. Our final consumption measure incorporates housing services by adding the consumption value of housing services to the total non-housing expenditures. For renters, we use reported rental expenditures. For homeowners, we approximate the rental-equivalent flow of housing services as a 6 percent yield on the house price (Poterba and Sinai 2008).

We report summary statistics for our sample in Table B.1.

B.1. Constructing a Measure of Permanent Income

We include a measure of workers' permanent income in our statistical model for unemployment risk. Here we describe how we construct that measure.

We define permanent income as the predicted time-invariant individual component from the following log earnings regression

$$\ln w_{i,t} = \beta_0 + \beta_1 age_{i,t} + \beta_2 age_{i,t}^2 + \beta_3 age_{i,t}^3 + \mu_i + \eta_t + u_{i,t},$$

where $w_{i,t}$ denotes annual labor earnings of the reference person. We control for a third-order polynomial in age and year fixed effects.[4]

	All	Employed	Unemployed
Ref. Person's Age (years)	40.05	40.14	38.41
	(10.14)	(10.13)	(10.20)
Married	0.56	0.57	0.33
	(0.50)	(0.49)	(0.47)
Number of Children	1.04	1.03	1.13
	(1.21)	(1.20)	(1.35)
Share White	0.55	0.56	0.33
	(0.50)	(0.50)	(0.47)
Share Black	0.12	0.13	0.12
	(0.33)	(0.33)	(0.33)
Ref. Person's Schooling (years)	13.83	13.89	12.84
	(2.44)	(2.43)	(2.40)
Unemployed	0.06	0.00	1.00
	(0.23)	(0.00)	(0.00)
Non-Durable Expenditure ($1,000s)	20.74	21.07	15.19
	(10.78)	(10.79)	(8.97)
Food Expenditure ($1,000s)	9.66	9.80	7.43
	(5.77)	(5.79)	(4.91)
Services Expenditure ($1,000s)	16.53	16.86	10.99
	(13.44)	(13.51)	(10.88)
Total Expenditure (Non-Housing, $1,000s)	26.11	26.56	18.49
	(16.47)	(16.51)	(13.78)
Total Expenditure (Incl. Housing, $1,000s)	39.14	39.89	26.53
	(25.10)	(25.15)	(20.54)
N	50380	47552	2828
Waves	4.76	4.49	1.52
Unique Households	10587	10587	1862

TABLE B.1. Summary Statistics for Our Sample of Interest

Notes: Computed from the 1999–2019 waves of the PSID. Table reports means, with standard deviations in parenthesis. We deflate all dollar to 2019 dollars.

[4]To resolve the collinearity between linear age, individual fixed effects, and year fixed effects, we normalize the year effects in the first and last sample periods to zero.

C. Estimation of the Joint Risk–Consumption Mixture

This appendix describes the estimation of the joint latent-type model for unemployment risk and unemployment-induced consumption declines. We first present the specifications for the unemployment-risk and consumption-drop blocks, then derive the observed-data likelihood and the conditional consumption likelihood, and finally describe our estimation procedure.

C.1. Model specification

We observe a panel of workers indexed by $i = 1, \ldots, N$ and survey waves indexed by $t = 1, \ldots, T_i$. Since survey waves are biennial, we use $t - 2$ to denote the previous observed wave, two calendar years earlier. Let $\Delta^{FD}_{i,t} \equiv \ln c_{i,t} - \ln c_{i,t-2}$ denote the two-year change in log consumption, and let $U_{i,t}$ denote an indicator for unemployment in period t. Let $X^B_{i,t}$ denote a vector of block-specific covariates, and let $o^B_{i,t}$ indicate whether (i, t) is observed in our estimation sample for block $B \in \{U, C\}$ (unemployment-risk block U and consumption block C).

Each worker belongs to one of H unobserved *unemployment-risk types* and one of K unobserved *consumption types*. We denote the risk type by $h(i) \in \{1, \ldots, H\}$ and the consumption type by $k(i) \in \{1, \ldots, K\}$. We sometimes use indicator notation

$$d^H_{i,h} \equiv \mathbb{1}[h(i) = h], \qquad d^K_{i,k} \equiv \mathbb{1}[k(i) = k].$$

Unemployment risk. The unemployment-risk block corresponds to the specification in the main text: conditional on risk type h and covariates $X^U_{i,t}$, unemployment in period t follows a logit model

$$\Pr\left(U_{i,t} = 1 \mid h(i) = h, X^U_{i,t}\right) = g\left(\eta^h_0 + X^{U\prime}_{i,t}\gamma\right), \tag{C.1}$$

where $g(\cdot)$ is the logistic function, η^h_0 is a type-specific intercept, and γ is a vector of common coefficients on $X^U_{i,t}$. We assume that, conditional on $(h(i), X^U_{i,1:T_i})$, unemployment shocks are independent over t and across workers. Let $\pi(h)$ denote the population share of risk type h.

Consumption declines

The consumption-decline block follows the finite-mixture specification in the main text (equation 7),

$$\Delta_{i,t}^{FD} = \sum_{k=1}^{K} \mathbb{1}[k(i) = k] \left(\delta_0^k + \delta_1^k U_{i,t}\right) + X_{i,t}^{C\prime}\beta + \varepsilon_{i,t}, \tag{C.2}$$

where δ_0^k and δ_1^k are type-specific intercepts and consumption declines, respectively, and β collects the coefficients on observables. Conditional on consumption type k, observables, and unemployment status, we assume that

$$\varepsilon_{i,t} \mid \left(k(i) = k, X_{i,t}^C, U_{i,t}\right) \sim N\left(0, \sigma_k^2\right),$$

and that the shocks $\varepsilon_{i,t}$ are independent over t and across workers, conditional on $(k(i), X_{i,1:T_i}^C, U_{i,1:T_i})$.

Independence across blocks. Finally, we assume that, conditional on the latent types and observables, the unemployment and consumption shocks are independent

$$\begin{aligned}
u_{i,t} \perp u_{i,s} &\quad \mid X_{i,1:T_i}^U, h(i), \ \forall t \neq s, \\
\varepsilon_{i,t} \perp \varepsilon_{i,s} &\quad \mid X_{i,1:T_i}^C, k(i), \ \forall t \neq s, \\
(\varepsilon_{i,1:T_i}) \perp (u_{i,1:T_i}) &\quad \mid X_{i,1:T_i}^U, X_{i,1:T_i}^C, h(i), k(i),
\end{aligned}$$

where $u_{i,t}$ denotes the latent logistic error underlying equation (C.1). This assumption, combined with equations (C.1)–(C.2), implies that the joint likelihood factorizes into an unemployment-risk component and a consumption component, conditional on type.

C.2. Likelihood

Let

$$\begin{aligned}
\chi_U &= \left\{\pi(h), \eta_0^h, \gamma\right\}_{h=1}^H, \\
\chi_C &= \left\{\delta_0^k, \delta_1^k, \sigma_k^2, \beta\right\}_{k=1}^K
\end{aligned}$$

denote the parameters of the unemployment-risk and consumption blocks, respectively. In addition, let $\pi(h, k)$ denote the joint population share of risk type h

and consumption type k, satisfying

$$\sum_{h=1}^{H}\sum_{k=1}^{K}\pi(h,k)=1, \qquad \pi(h,k)\geq 0 \;\; \forall h,k,$$

with marginal shares $\pi(h)=\sum_{k=1}^{K}\pi(h,k)$ and $\pi(k)=\sum_{h=1}^{H}\pi(h,k)$.

Likelihood contributions conditional on type. For a given worker i and consumption type k, the conditional likelihood contribution of the consumption data is

$$\begin{aligned}\ell_i^C(k;\chi_C) &\equiv \ell_i^C(\Delta_i^{FD}, U_i, X_i^C \mid k(i)=k;\chi_C)\\ &= \prod_{t=1}^{T_i} f\Big(\Delta_{i,t}^{FD} \,\Big|\, \underbrace{\delta_0^k+\delta_1^k U_{i,t}+X_{i,t}^{C\prime}\beta}_{\mu^k(U_{i,t},X_{i,t}^C)}, \sigma_k^2\Big)^{o_{i,t}^C},\end{aligned} \tag{C.3}$$

where $f(\cdot \mid \mu, \sigma^2)$ denotes the normal density with mean μ and variance σ^2.

For a given worker i and risk type h, the conditional likelihood contribution of the employment data is

$$\begin{aligned}\ell_i^U(h;\chi_U) &\equiv \ell_i^U(U_i, X_i^U \mid h(i)=h;\chi_U)\\ &= \prod_{t=1}^{T_i}\Big(g\big(\eta_0^h+X_{i,t}^{U\prime}\gamma\big)^{U_{i,t}}\big[1-g\big(\eta_0^h+X_{i,t}^{U\prime}\gamma\big)\big]^{1-U_{i,t}}\Big)^{o_{i,t}^U}.\end{aligned} \tag{C.4}$$

Conditional on $(h(i)=h, k(i)=k)$, the joint likelihood contribution for individual i factorizes:

$$\ell_i(h,k;\chi_U,\chi_C)=\ell_i^U(h;\chi_U)\,\ell_i^C(k;\chi_C). \tag{C.5}$$

Observed-data likelihood. Because the latent types are unobserved, the observed-data likelihood for individual i sums over all (h,k) pairs:

$$\begin{aligned}\ell_i(\pi;\chi_U,\chi_C) &\equiv p(\Delta_i^{FD}, U_i \mid X_i^C, X_i^U;\pi,\chi_U,\chi_C)\\ &= \sum_{h=1}^{H}\sum_{k=1}^{K}\pi(h,k)\,\ell_i^U(h;\chi_U)\,\ell_i^C(k;\chi_C).\end{aligned} \tag{C.6}$$

The sample log-likelihood is then

$$\mathcal{L}(\pi; \chi_U, \chi_C) = \sum_{i=1}^{N} \ln \ell_i(\pi; \chi_U, \chi_C).$$

Conditional consumption likelihood given employment outcomes. For estimation, it is helpful to consider the conditional likelihood of the consumption process given the employment history and covariates:

$$\begin{aligned} \ell_i^{C|U}(\pi; \chi_U, \chi_C) &\equiv p\big(\Delta_i^{FD} \mid U_i, X_i^C, X_i^U; \pi, \chi_U, \chi_C\big) \\ &= \frac{\sum_{h=1}^{H} \sum_{k=1}^{K} \pi(h,k)\, \ell_i^U(h; \chi_U)\, \ell_i^C(k; \chi_C)}{\sum_{b=1}^{H} \sum_{a=1}^{K} \pi(b,a)\, \ell_i^U(b; \chi_U)}. \end{aligned} \tag{C.7}$$

Define the individual-specific mixing weights

$$\alpha_i\big(k; U_i, X_i^U\big) \equiv \frac{\sum_{h=1}^{H} \pi(h,k)\, \ell_i^U(h; \chi_U)}{\sum_{b=1}^{H} \sum_{a=1}^{K} \pi(b,a)\, \ell_i^U(b; \chi_U)}, \tag{C.8}$$

so that equation (C.7) can be written as

$$\ell_i^{C|U}(\pi; \chi_U, \chi_C) = \sum_{k=1}^{K} \alpha_i\big(k; U_i, X_i^U\big)\, \ell_i^C(k; \chi_C). \tag{C.9}$$

In general, $\alpha_i(k; U_i, X_i^U)$ depends on the full employment history U_i and the covariates X_i^U. Estimating the consumption model in isolation via

$$\prod_{i=1}^{N} \sum_{k=1}^{K} \pi(k)\, \ell_i^C(k; \chi_C)$$

implicitly imposes $\alpha_i(k; U_i, X_i) \equiv \pi(k)$ for all i, k, which holds only under additional restrictions (e.g., $h(i)$ independent of $k(i)$). The joint model (equation C.6) therefore provides a coherent way to allow for correlation between unobserved heterogeneity in unemployment risk and consumption declines.

C.3. Estimation procedure

Estimation proceeds in two stages, as described in the main text. In the first stage, we estimate the unemployment-risk mixture and obtain $\hat{\chi}_U$ by maximizing the marginal likelihood of workers' unemployment histories. In the second stage, conditional on $\hat{\chi}_U$, we estimate the consumption parameters χ_C and the joint type probabilities

$\pi(h,k)$ by maximum likelihood using an EM algorithm.[5]

Stage 1: Unemployment-risk mixture. We estimate the parameters of the risk block, $\chi_U \equiv \{\pi(h), \eta_0^h, \gamma\}_{h=1}^H$, by maximizing

$$\hat{\chi}_U \in \arg\max_{\chi_U} \prod_{i=1}^N \sum_{h=1}^H \pi(h)\, \ell_i^U\left(h;\chi_U\right), \tag{C.10}$$

subject to $\sum_{h=1}^H \pi(h) = 1$ and $\pi(h) \geq 0$ for all h.

In the EM algorithm, the E-step computes the posterior risk-type probabilities

$$\begin{aligned}\tau_i(h) &\equiv \Pr\left(h(i) = h \mid U_i, X_i^U; \pi, \chi_U\right) \\ &= \frac{\pi(h)\, \ell_i^U\left(h;\chi_U\right)}{\sum_{b=1}^H \pi(b)\, \ell_i^U\left(b;\chi_U\right)}.\end{aligned} \tag{C.11}$$

The M-step updates the type probabilities and the logit parameters. The optimal mixture weights are

$$\pi^{\text{new}}(h) = \frac{1}{N}\sum_{i=1}^N \tau_i(h),$$

and the logit parameters solve

$$(\{\eta_0^h\}_{h=1}^H, \gamma)^{\text{new}} = \arg\max_{\{\eta_0^h\}_h, \gamma} \sum_{i=1}^N \sum_{h=1}^H \tau_i(h)\, \ln \ell_i^U\left(h;\chi_U\right),$$

which corresponds to a weighted logistic regression with type-specific intercepts and a common slope vector.

Stage 2: Consumption mixture and joint type probabilities. Given $\hat{\chi}_U$ from Stage 1, we estimate χ_C and the joint type probabilities $\pi(h,k)$ by maximizing the plug-in likelihood

$$(\{\pi(h,k)\}_{h,k}, \chi_C) \in \arg\max_{\{\pi(h,k)\}_{h,k}, \chi_C} \prod_{i=1}^N \sum_{h=1}^H \sum_{k=1}^K \pi(h,k)\, \ell_i^U\left(h;\hat{\chi}_U\right) \ell_i^C\left(k;\chi_C\right), \tag{C.12}$$

[5] As in Lewis et al. (2026), we initialize the EM algorithm from a large number of random starting values and retain the solution with the highest log-likelihood.

subject to

$$\sum_{k=1}^{K} \pi(h,k) = \hat{\pi}(h) \quad \forall h, \qquad \sum_{h=1}^{H}\sum_{k=1}^{K} \pi(h,k) = 1, \qquad \pi(h,k) \geq 0 \ \ \forall h,k. \tag{C.13}$$

The constraints (C.13) ensure that the marginal risk-type shares in the joint distribution coincide with the Stage 1 estimates.

As a preliminary step, we estimate a consumption-only mixture that ignores correlation between $h(i)$ and $k(i)$,

$$(\{\tilde{\pi}(k)\}_{k=1}^{K}, \tilde{\chi}_C) \in \arg\max_{\{\pi(k)\}_k, \chi_C} \prod_{i=1}^{N} \sum_{k=1}^{K} \pi(k)\, \ell_i^C(k;\chi_C), \tag{C.14}$$

subject to $\sum_{k=1}^{K} \pi(k) = 1$ and $\pi(k) \geq 0$. These estimates are used to initialize estimation of the joint model:

$$\pi^{(0)}(h,k) = \hat{\pi}(h)\,\tilde{\pi}(k), \qquad \chi_C^{(0)} = \tilde{\chi}_C.$$

The EM algorithm for equation (C.12) proceeds as follows. In the E-step, for each individual we compute the posterior probability of each joint type:

$$\begin{aligned} p_i(h,k) &\equiv \Pr\left(k(i)=k, h(i)=h \mid \Delta_i^{FD}, U_i, X_i^C, X_i^U; \pi, \chi_C, \hat{\chi}_U\right) \\ &= \frac{\pi(h,k)\,\ell_i^U(h;\hat{\chi}_U)\,\ell_i^C(k;\chi_C)}{\sum_{b=1}^{H}\sum_{a=1}^{K} \pi(b,a)\,\ell_i^U(b;\hat{\chi}_U)\,\ell_i^C(a;\chi_C)}. \end{aligned} \tag{C.15}$$

In the M-step, we update the joint type probabilities and the consumption parameters. The constrained update for the joint type probabilities is

$$\pi(h,k)^{\text{new}} = \hat{\pi}(h) \cdot \frac{\sum_{i=1}^{N} p_i(h,k)}{\sum_{i=1}^{N}\sum_{a=1}^{K} p_i(h,a)}, \tag{C.16}$$

which enforce the constraints (C.13) exactly. The consumption parameters are updated by maximizing the expected complete-data log-likelihood,

$$\chi_C^{\text{new}} = \arg\max_{\chi_C} \sum_{i=1}^{N}\sum_{h=1}^{H}\sum_{k=1}^{K} p_i(h,k)\, \ln \ell_i^C(k;\chi_C), \tag{C.17}$$

which corresponds to estimating K Gaussian regressions with type-specific parameters using weights

$$p_i^C(k) \equiv \sum_{h=1}^{H} p_i(h,k)$$

for each consumption type k.

We iterate the E- and M-steps until convergence in the log-likelihood (C.12). The resulting estimates $(\hat{\pi}(h,k), \hat{\chi}_U, \hat{\chi}_C)$ are used in the main text to construct worker-level unemployment probabilities and consumption declines as

$$1 - \hat{e}_{i,t} = \sum_{h=1}^{H}\sum_{k=1}^{K} \hat{p}_i(h,k)\ \Pr\left(U_{i,t} = 1 \mid h(i) = h, X_{i,t}^{U}; \hat{\chi}_U\right), \quad \widehat{(\Delta c/c)}_i \approx -\widehat{\Delta \ln c}_i = -\sum_{h=1}^{H}\sum_{k=1}^{K} \hat{p}_i(h,k)\,\hat{\delta}_1^k.$$

Inference. We obtain standard errors for all parameters and welfare objects using a nonparametric bootstrap. Specifically, we draw 400 bootstrap samples by resampling households with replacement, re-estimate the full model in each sample, and compute the distribution of mixture parameters, type probabilities, and welfare statistics. Within each replication, we order risk types from low to high unemployment risk and consumption types from large to small consumption declines, relabeling the associated type probabilities accordingly. We report bootstrap standard errors based on these 400 replications.

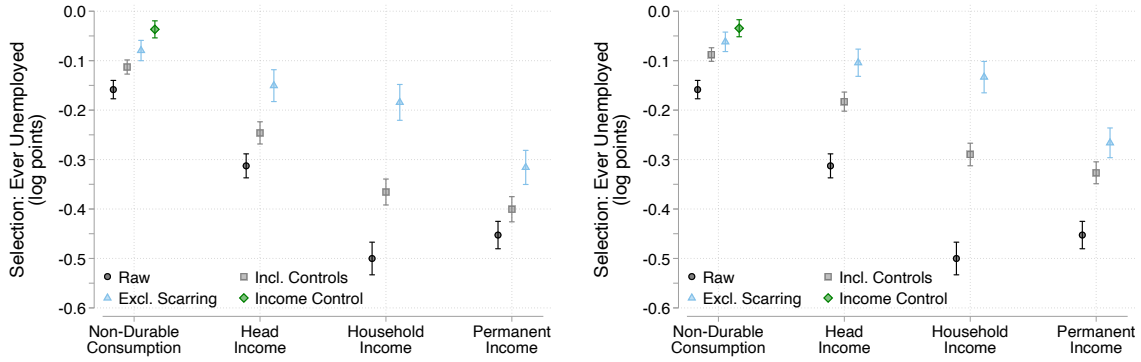
D. Additional Empirical Results

D.1. Robustness of Negative Selection Estimates to Alternative Controls

In our baseline empirical test of negative selection, we condition on a limited set of household characteristics. We exclude education and race controls because including them would net out the selection arising from average differences across education and race groups. Here, we assess the robustness to the inclusion of additional controls. Figure D.1 shows that average consumption differences remain similar when these controls are added. The right panel, which focuses on comparisons within gender, education, and race groups,shows some attenuation, reflecting the fact that we control for additional household characteristics relevant to consumption and labor market outcomes. Nevertheless, the quantitative impact of including these additional controls is modest.

D.2. Posterior-implied Consumption Decline Confidence Bands

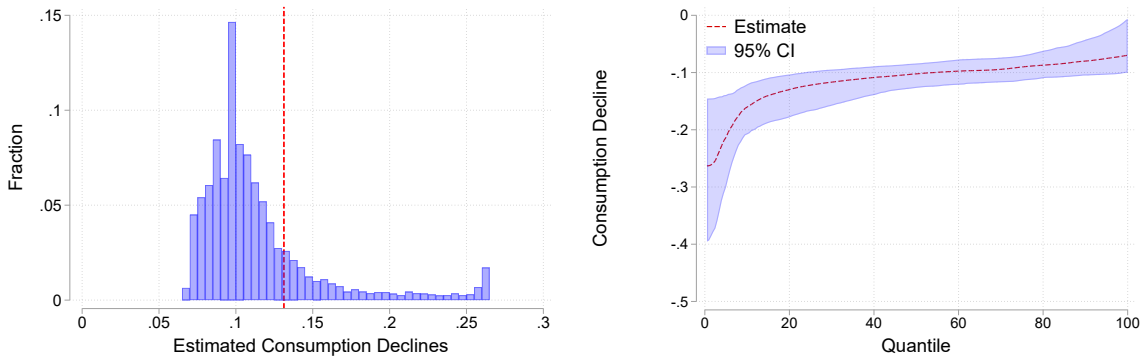
Panel (A) of Figure D.2 plots the distribution of posterior-implied consumption declines across workers (replicating Figure 3) and the right panel plots quantiles of the distribution along with 95% confidence intervals.



(A) Baseline (B) Additional Controls

FIGURE D.1. Negative Selection Among the Unemployed

Notes: All regressions drop periods during which the reference person is unemployed. For all outcomes, we report the raw, controls, and excl. scarring specifications. The raw specification includes no controls. The remaining specifications control for a cubic in age, indicators for family size and marital status, and year fixed effects. Estimates in panel (B) also control for indicators for gender, education, and race. The excl. scarring specification further restricts the sample to periods before the first observed unemployment spell. For nondurable consumption only, we also report an "income controls" specification that additionally controls for a cubic in household income.



(A) Distribution (B) Quantiles and CI

FIGURE D.2. Distribution of Estimated Consumption Declines

Notes: Panel (A) plots the distribution of posterior-implied unemployment-induced consumption declines under our baseline nondurable-consumption measure. Panel (B) reports the corresponding quantiles and 95% bootstrap confidence intervals. Worker-level declines are constructed from the Gaussian mixture linear regression by combining the estimated type-specific unemployment coefficients δ_1^k with each worker's posterior type probabilities $p_i(k)$. The sample is defined as in the text. The homogeneous estimate (dashed vertical line in left panel) is obtained by imposing $K = 1$.

D.3. Consumption Decline Estimates Under Alternative Consumption Measures

In addition to our baseline consumption measure, we assess robustness to alternative expenditure series. Table D.1 shows that estimated average consumption decline are similar across measure.

	Baseline	Alternative Measures			
	Non-durable	Food Expenditure	Services	Total (Non-Housing)	Total (Incl. Housing)
$\mathbb{1}\left[U_{i,t}=1\right]$	−0.131***	−0.148***	−0.145***	−0.129***	−0.149***
	(0.012)	(0.017)	(0.018)	(0.014)	(0.014)
Controls					
Age	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*
Year	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*
Family Size	*Yes*	*Yes*	*Yes*	*Yes*	*Yes*
N	37, 143	36, 880	37, 126	37, 167	37, 170

TABLE D.1. Consumption Declines by Alternative Consumption Measures ($\Delta c/c$)

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The first column reports the estimated average consumption decline from the homogeneous specification (equation (6)) for our baseline consumption measure. The remaining columns report estimated declines for alternative consumption measures constructed from PSID expenditures. Standard errors are clustered at the household level. The number of observations differs across regressions due to item non-response. Covariates include a cubic in age, indicators for year, and the change in log of family size.

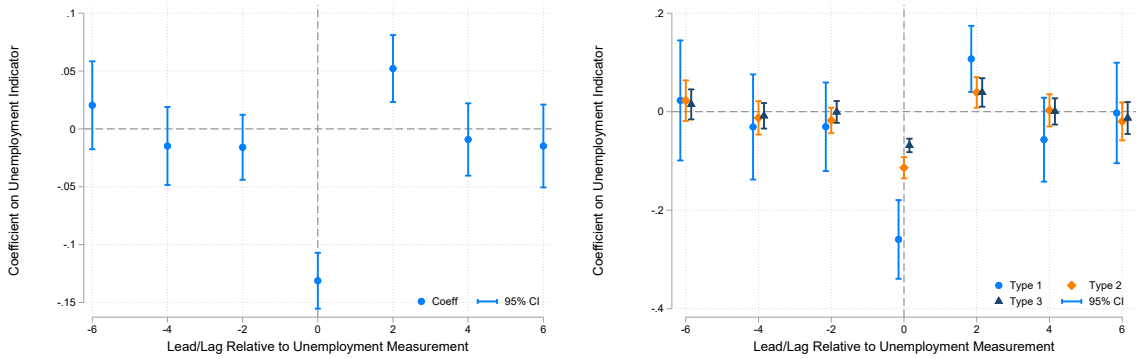
D.4. Pre-trends and Anticipation Effects

To assess the identifying assumption that, conditional on type, consumption growth for workers who remain employed provides a valid counterfactual for those who become unemployed, we follow the lead–lag exercise in Hendren (2017). Specifically, we estimate variants of the homogeneous consumption-drop specification in equation (6) that replace the contemporaneous unemployment indicator $U_{i,t}$ with leads and lags $U_{i,t-k}$ for $k \in \{-6, -4, -2, 0, 2, 4, 6\}$:

$$\Delta^{FD}_{i,t} \equiv \ln c_{i,t} - \ln c_{i,t-2} = \delta_0 + \delta_1 U_{i,t-k} + X^{C\prime}_{i,t}\beta + \varepsilon_{i,t}. \tag{D.1}$$

Negative values of k capture leads, while positive values correspond to lags. We estimate this specification (i) in the homogeneous model and (ii) using posterior type probabilities from the joint mixture model as weights to obtain type-specific profiles.[6]

[6]We obtain type-specific profiles by estimating the lead–lag regressions with each observation weighted by the worker's posterior probability of belonging to each latent type, treating these posteriors as given. The resulting generalized least squares estimator corresponds to the M-step in the mixture estimation.



(A) Homogeneous estimates (B) Type-specific estimates

FIGURE D.3. Impact of Unemployment on Consumption Growth

Notes: Each point plots the estimated effect of unemployment on two-year log consumption growth at a given event time k, with vertical bars showing 95% confidence intervals. Event time $k = 0$ corresponds to unemployment at t for workers employed at $t - 2$; negative (positive) values of k are leads (lags) relative to this unemployment event. For each value of k, the sample is restricted to workers employed in the period immediately before the unemployment measure used in that regression. All specifications control for the same covariates as in the baseline homogeneous regression. Panel (A) reports homogeneous estimates from equation (D.1) with the unemployment indicator replaced by lead/lag indicators $U_{i,t-k}$. Panel (B) reports type-specific estimates obtained by re-estimating these regressions using posterior type probabilities from the joint mixture model as weights, treating the posteriors as given. Standard errors are clustered at the household level.

Figure D.3 plots the estimated coefficients and 95% confidence intervals. We find little evidence of differential consumption trends prior to job loss in either the homogeneous or type-specific estimates: the lead coefficients are close to zero and statistically indistinguishable from zero, and do not display a systematic pre-trend. We observe a sizable decline in the period when unemployment occurs, and some "bounce back" in consumption two years after the spell, consistent with partial recovery.

Hendren (2017) documents an anticipatory decline in food consumption one year prior to unemployment. Such an adjustment can occur within our two-year consumption window, while our lead coefficients concern earlier intervals and show no systematic pre-trend. Overall, the lead–lag patterns provide evidence in favor of the parallel-trends assumption underlying our consumption-drop estimates.

D.5. Alternative Unemployment Definitions

We examine robustness to alternative definitions of job loss and to an IV strategy using plant closures. The PSID asks respondents who leave a job to report the reason for separation. We use these responses to construct an indicator for separations due to plant closure or firm shutdown (a proxy for mass layoffs).

Table D.2 reports estimates from the homogeneous consumption-decline specification under several alternative treatments of unemployment. Column (1) reproduces the baseline specification using our main unemployment indicator. Column (2) instruments this indicator with the plant-closure indicator. Columns (3)–(4) use alternative measures of job loss based on plant closures and broader "displacement" definitions that include workers who report being fired or laid off, following Stephens (2002) and Charles and Stephens (2004). The IV and displacement-based estimates of the consumption decline are slightly smaller in magnitude than the baseline, but imprecisely estimated, and none is statistically distinguishable from the baseline estimate. As in prior PSID work, we find only modest differences between plant closures and the broader displaced group. These results suggest that selection into unemployment along dimensions captured by separation reason is unlikely to be a first-order source of bias in our estimated consumption drops.

	Unemployed at Interview		Alternative Definitions	
	Baseline	IV	Plant Closed	Displaced
$\mathbb{1}\left[U_{i,t}=1\right]$	−0.131***	−0.091**		
	(0.012)	(0.042)		
$\mathbb{1}\left[U_{i,t}=1\right] \times \mathbb{1}\left[Closed(t,t-2)_{i,t}=1\right]$			−0.087**	
			(0.040)	
$\mathbb{1}\left[U_{i,t}=1\right] \times \mathbb{1}\left[Displaced(t,t-2)_{i,t}=1\right]$				−0.116***
				(0.019)
N	37,143	37,143	37,143	37,143

TABLE D.2. Consumption Declines by Alternative Displacement Measures

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (1) reports the estimated average consumption decline from the homogeneous specification using our baseline unemployment definition. Column (2) instruments this unemployment indicator with an indicator for plant closures. The Kleibergen-Paap F-statistic is 1.8×10^5. The remaining columns report estimated consumption declines using alternative job-loss definitions. Standard errors are clustered at the household level. Covariates include a cubic in age, indicators for year, and the change in log family size.

D.6. Validation of the Mixture Model

To validate our finite-mixture specification, we compare the distribution of consumption drops it implies to a quantile-regression benchmark estimated on an independent holdout sample. We randomly split workers into two equally sized subsamples, assigning all observations for a given worker to the same subsample.

In the first, or training, subsample, we estimate the joint risk–consumption mixture model and recover posterior probabilities over consumption types, which imply a distribution of unemployment-induced consumption declines. In the second, or holdout, subsample, we estimate conditional quantile regressions for consumption growth to recover a flexible estimate of the distribution of consumption drops among unemployed workers, along with 95% confidence bands.

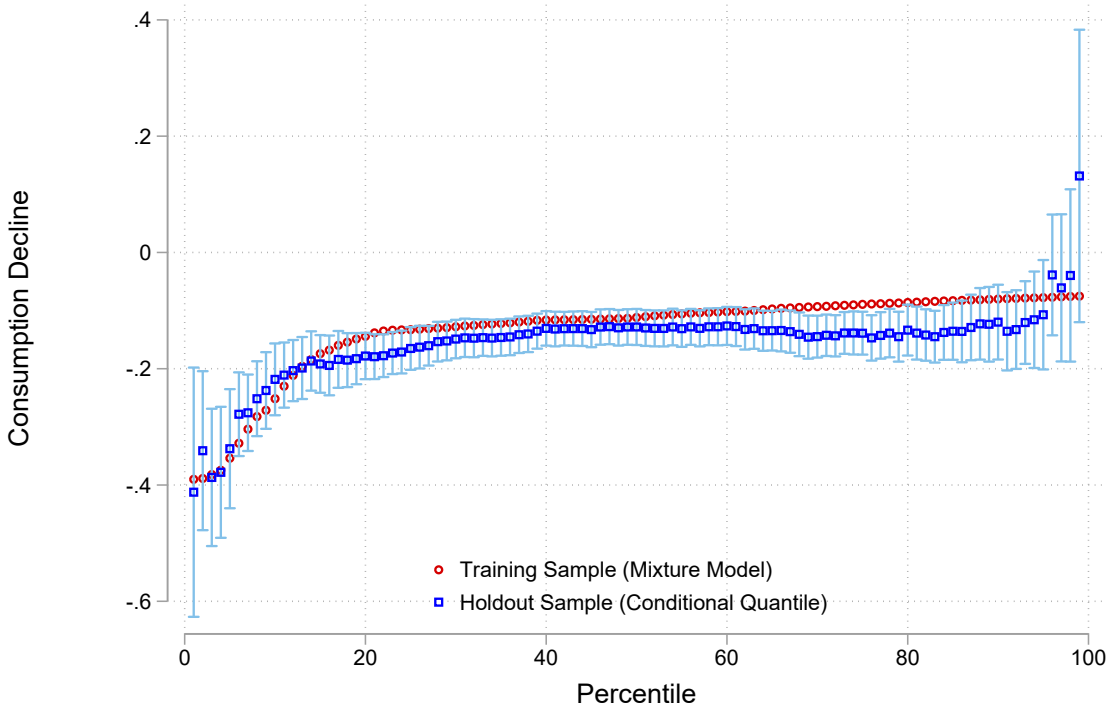


FIGURE D.4. Distribution of Consumption Declines: Prediction Sample vs Holdout

Notes: The red circles plot percentiles of the estimated consumption declines implied by the finite-mixture model, using a random training sample comprising 50% of observations, reweighted by ex ante unemployment probabilities. The blue squares plot the corresponding quantiles in the remaining holdout sample, estimated by quantile regression; shaded bands show 95% bootstrap confidence intervals. Consumption declines are defined as $\Delta^{FD}_{i,t} = \ln c_{i,t} - \ln c_{i,t-2}$.

Figure D.4 plots percentiles of the consumption-decline distribution implied by the mixture model in the training sample, reweighted by ex ante unemployment probabilities so that the implied distribution is comparable to the ATT distribution targeted by the holdout quantile regression. It also plots the corresponding quantile-regression estimates and confidence intervals from the holdout sample. Across most of the distribution, the mixture-based percentiles lie well within the 95% confidence bands of the quantile-regression benchmark and are typically close to their midpoints. The quantile regressions imply somewhat larger declines over

part of the middle and upper-middle distribution, and the upper tail is estimated imprecisely. We interpret this exercise as providing reassuring external validation that the finite-mixture specification does a good job of recovering the distribution of consumption responses to unemployment.

D.7. Robustness of Consumption Declines to Unemployment Spell Length

A potential concern with our consumption-based estimates is that the PSID's biennial structure may under-represent short unemployment spells and their associated consumption responses. In our baseline specification, we classify a worker as in unemployment at period t when we observe them employed at $t-2$ but unemployed at t, so that $\Delta_{i,t}^{FD}$ compares pre- and post-unemployment consumption. Spells that begin and end between these two interview dates appear as employment at both $t-2$ and t and therefore do not enter our "unemployed" group, tilting the sample toward longer spells for which measured consumption declines may be larger.

	Baseline	Post-2003	No Spell Two Calendar Years Prior ($t-2$)	No Spell In Previous Calendar Year ($t-1$)	Neither
$\mathbb{1}\left[U_{i,t}=1\right]$	−0.131***	−0.133***	−0.132***	−0.117***	−0.110***
	(0.012)	(0.013)	(0.014)	(0.017)	(0.018)
N	37,143	33,449	31,701	31,459	30,485

TABLE D.3. Consumption Declines under Alternative Unemployment Spell Definitions

Notes: * $p<0.10$, ** $p<0.05$, *** $p<0.01$. Column (1) reports the estimated average consumption decline from the homogeneous specification (equation (6)) in our full baseline sample. Column (2) repeats this specification using only post–2003 data, for which we observe whether the respondent experienced unemployment in the previous 12 months and in the preceding 12–24 months. The remaining columns report estimates from alternative specifications that impose sample restrictions based on these unemployment-history indicators, as described in the text. Standard errors are clustered at the household level. Covariates include a cubic in age, indicators for year, and the change in log family size.

To assess the importance of this issue, we exploit additional information in the post–2003 PSID waves on unemployment between interviews. In these waves, respondents report whether they experienced any unemployment spell in the previous 12 months and in the preceding 12–24 months. We first replicate the homogeneous specification using only post–2003 waves (column 2 of Table D.3). We then re-estimate the same specification under alternative sample restrictions that drop observations reporting additional unemployment outside the contemporaneous interview state. Specifically, we consider three alternative

restrictions based on the two unemployment-history indicators, excluding in turn: (i) observations with unemployment reported in the preceding 12–24 months; (ii) observations with unemployment reported in the previous 12 months; and (iii) observations with unemployment reported in either window. Columns 3–5 of Table D.3 report the resulting estimates.

The estimates in Table D.3 are similar across specifications. Restricting attention to shorter spells yields slightly smaller (but statistically indistinguishable) consumption drops relative to the post–2003 baseline. This pattern suggests that any bias from the biennial sampling frame is modest and, if anything, would lead us to slightly overstate the insurance value of UI in our baseline estimates.

D.8. Observable Heterogeneity in Consumption Declines

Our approach to estimating unemployment-induced consumption declines does not require specifying observable determinants of heterogeneity ex ante. Ex post, however, we can relate predicted declines to observables. Table D.4 reports the results of this exercise.

We focus on a proxy for a household's capacity to self-insure, measured while employed. This enables us to assess whether predicted consumption declines are larger for households that appear less able to smooth unemployment shocks.[7] To proxy for the availability of private savings, we include indicators for each tercile of the liquid-wealth distribution. Following Carroll and Samwick (1997), we construct liquid wealth as the sum of cash (including checking accounts), savings, stocks held outside retirement accounts, and bonds and government treasuries—assets that are relatively liquid and can be used to smooth shocks in the short run.

Table D.4 reports coefficient estimates from a regression of predicted worker-level consumption declines on terciles of liquid wealth or permanent income using within-worker averages (yielding one observation per worker). We find that households with greater apparent ability to self-insure, i.e., higher liquid assets or higher permanent income, have smaller posterior-predicted consumption declines when unemployed. These relationships are economically and statistically significant. At the same time, these observables explain only a modest fraction of total variation in predicted decline: a regression including both partitions has an R^2 below 0.1. This suggests that observable proxies such as liquid wealth and permanent income capture meaningful variation in self-insurance capacity, but do

[7] As Lewis et al. (2026) emphasize in the context of marginal propensities to consume, a key advantage of using latent types to capture heterogeneity is that it implicitly allows many observables to enter jointly, without the rapid loss of statistical power that arises from the fully interacted specifications.

not fully summarize the heterogeneity recovered by the mixture model. This may reflect both latent heterogeneity in exposure to unemployment risk and self-insurance capacity, for example, due to differences in beliefs, preferences, or shock processes, and imperfect measurement of self-insurance capacity, especially because liquid assets are measured while employed. Importantly for our purposes, our latent-type approach does not rely on these observables for identification and is therefore less reliant on omitted or mismeasured proxies for self-insurance capacity.

	Liquid Wealth	Permanent Income
Group 1	−0.128***	−0.127***
	(0.001)	(0.001)
Group 2	−0.113***	−0.113***
	(0.001)	(0.001)
Group 3	−0.108***	−0.109***
	(0.001)	(0.001)

TABLE D.4. Consumption Declines ($\Delta c/c$) and Proxies for Access to Self-Insurance

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. For each household, we compute the posterior-weighted unemployment-induced consumption decline by averaging the type-specific declines $\hat{\delta}_1^k$ using the worker's posterior consumption-type probabilities $p_i^C(k)$. We then regress these predicted declines on proxies for access to self-insurance. The estimation sample is the same as in the main text, and covariates are measured during periods of employment. Liquid wealth is defined using the "very liquid assets" concept in Carroll and Samwick (1997). Groups are ordered from lowest to highest terciles. Standard errors clustered at the household level are conditional on fitted mixture estimates and reported in parentheses.

We also compare our mixture-based estimates to "interacted observable" specifications. Specifically, we estimate versions of equation (7) that partition the sample into terciles of age, liquid wealth, or permanent income, and interact unemployment with group indicators.[8] Table D.5 reports the resulting group-specific consumption declines. The first column reproduces our baseline mixture estimates; the remaining columns show interacted specifications based on terciles of observables. Within each column, groups are ordered from largest to smallest estimated consumption decline.

The interacted specifications display intuitive patterns: consumption declines are generally smaller for older households and for households with higher liquid

[8]Table D.4 projects individual-level posterior-predicted consumption declines on panel-average observables, whereas the interacted observable specification is a panel regression. We therefore also include an analysis by age terciles.

wealth or permanent income. These estimates show that observable characteristics capture economically meaningful variation in unemployment-induced consumption responses. However, the range of estimated consumption declines across observable terciles is smaller than the range across latent consumption types. A similar pattern is documented by Lewis et al. (2026) for marginal propensities to consume.

	Mixture	Age	Liquid Wealth	Permanent Income
Group 1	−0.262***	−0.157***	−0.152***	−0.141***
	(0.042)	(0.022)	(0.017)	(0.016)
Group 2	−0.113***	−0.154***	−0.117***	−0.113***
	(0.011)	(0.022)	(0.024)	(0.023)
Group 3	−0.069***	−0.077***	−0.089***	−0.101***
	(0.007)	(0.020)	(0.024)	(0.032)

TABLE D.5. Interacted Observable Models for Consumption Declines ($\Delta c/c$) and Proxies of Access to Self-Insurance

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Each column reports estimated consumption declines for different groupings of households. Column (1) reproduces our baseline Gaussian mixture linear regression estimates. The remaining columns partition the sample into terciles based on age, liquid wealth, or permanent income, and interact unemployment with these group indicators. Liquid wealth is measured at the preceding interview, two years earlier, when the worker was employed. Groups are ordered from largest to smallest consumption decline in each column. Robust standard errors are reported in parentheses; for the mixture model, they condition on estimated posterior group assignments, as discussed in the text. Covariates include a cubic in age, indicators for year, and the change in log of family size.

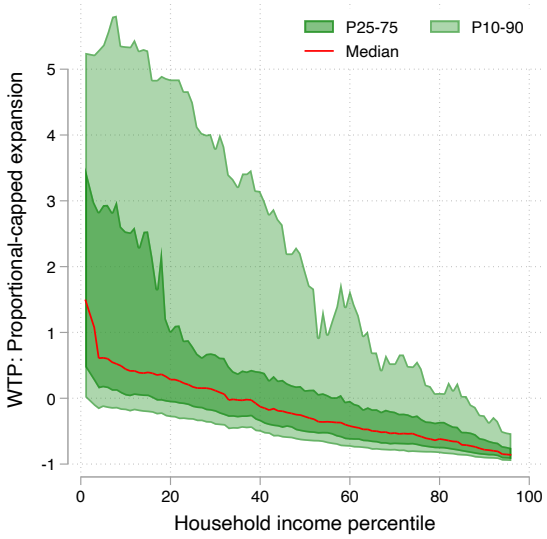
E. Willingness-to-Pay: Additional Results

E.1. Alternative Scaling of Willingness-to-Pay

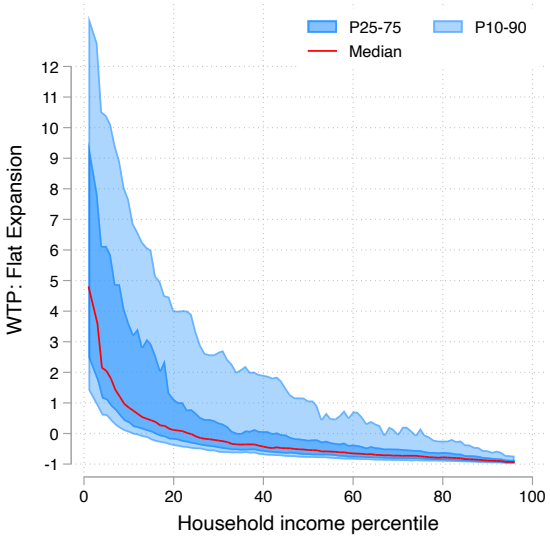
In the main text, we report individual WTP per $1 of mechanical increase in the worker's own expected benefit payments. This normalization provides a natural individual-level measure of how strongly each worker values a reform relative to their own expected entitlement. An alternative is to express individual surplus per $1 of aggregate mechanical increase in expected benefit payments. To do so, we rescale individual WTP by

$$\frac{(1-e_i)\phi_{\mathcal{B}}(y_i)}{\int_{i'}(1-e_{i'})\phi_{\mathcal{B}}(y_{i'})\,di'}.$$

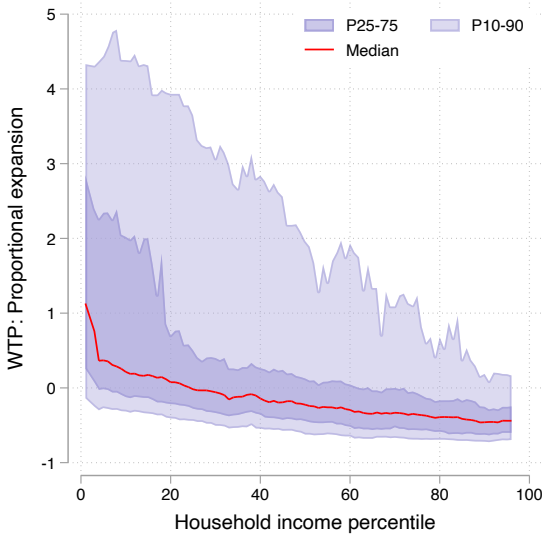
This combines each worker's normalized valuation with the mechanical incidence of the reform.



(A) Proportional-Capped



(B) Flat



(C) Proportional

FIGURE E.1. Willingness-to-Pay for UI Expansion by Income and Reform Type: Alternative Scaling

Notes: Each panel summarizes the distribution of workers' WTP for a UI expansion, reporting the median and 10th, 25th, 75th, and 90th percentiles. The measures are constructed from baseline WTP (equation (3)), rescaled by the worker's mechanical expected benefit increase relative to the aggregate mechanical increase in benefit payments. Thus, values are expressed per $1 of aggregate mechanical increase in expected benefit payments. Worker incomes are winsorized at $10,000 and $200,000. Panel (A) shows results for a proportional-capped reform reflecting the current U.S. UI structure. Panels (B) and (C) show results for flat and proportional expansions, respectively. Quantiles are smoothed using a uniform rolling window of ± 2 percentiles.

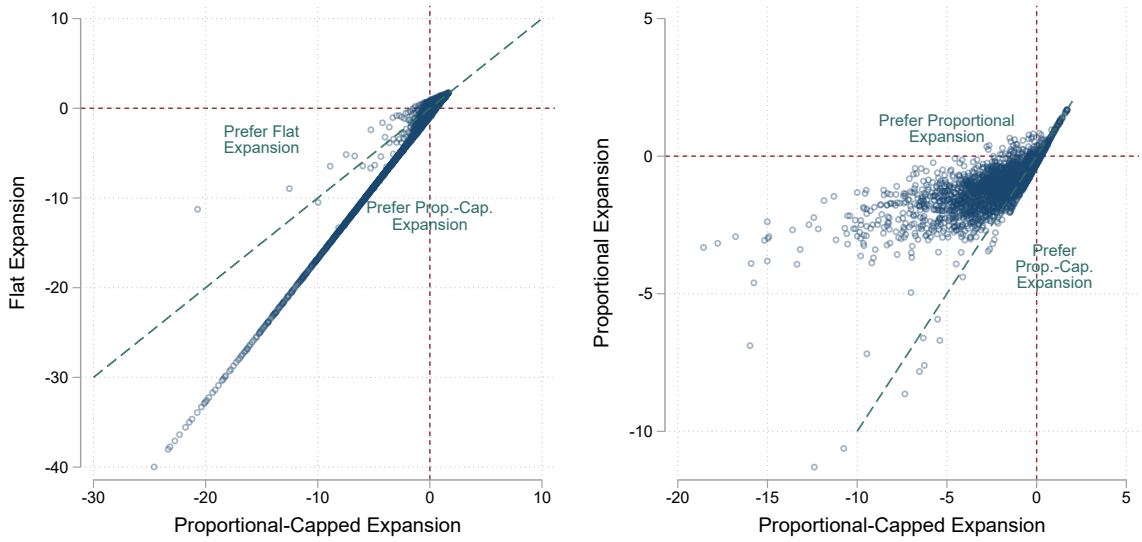
Figure E.1 reports the resulting WTP distributions. Because the tax incidence function is held fixed across the reforms we consider, $\phi_{\mathcal{T}}(y) = 1$, this rescaling

preserves within-worker rankings across reforms. It therefore changes the magnitudes and visual income gradients, but not the qualitative conclusions about individual reform rankings.

E.2. Worker Preferences Across Alternative Reforms

Figure E.2 illustrates how WTP for UI expansions covaries across different reform types. Panel (A) compares workers' WTP under the proportional-capped reform and a flat expansion; panel (B) compares the proportional-capped and proportional expansion. To visualize the joint distribution, we bin workers by WTP under the proportional-capped reform and, within those bins, by WTP under the alternative reform, and then plot average WTP in each cell. Points on the 45-degree line correspond to workers with similar valuations under the two reforms; points above or below the line indicate different rankings of reforms. As in Figure 4, WTP is expressed per $1 of mechanical increase in the worker's own expected benefit payments, so the comparisons rank reforms under this scaling.

The figure shows that workers differ not only in whether they support expanding UI, but also in which expansion they prefer. The Northeast and Southwest quadrants contain workers with the same sign of WTP under both reforms, while the Northwest and Southeast quadrants contain workers whose support depends on reform design. More generally, position relative to the 45-degree line reveals pairwise reform rankings: in panel (A), points above the line correspond to workers who prefer a flat expansion to the proportional-capped reform, while in panel (B), points above the line correspond to workers who prefer a proportional expansion. Overall, 82% of workers prefer expanding the proportional-capped system to a flat expansion, and 52% prefer it to a proportional expansion.



(A) Flat (B) Proportional

FIGURE E.2. Individual Preferences over Reforms

Notes: Panel (A) compares worker-level WTP under the proportional-capped reform and a flat expansion; panel (B) compares the proportional-capped and proportional reforms. To visualize the joint distribution, we first partition the sample into 50 equally sized groups based on WTP under the proportional-capped reform and then, within each group, into 50 equally sized bins based on WTP under the alternative reform; each point plots average WTP in the resulting cell. Points on the 45-degree line correspond to workers with similar valuations under the two reforms; points above or below the line indicate different reform rankings. WTP is expressed per $1 of mechanical increase in an individual's expected benefit payments.

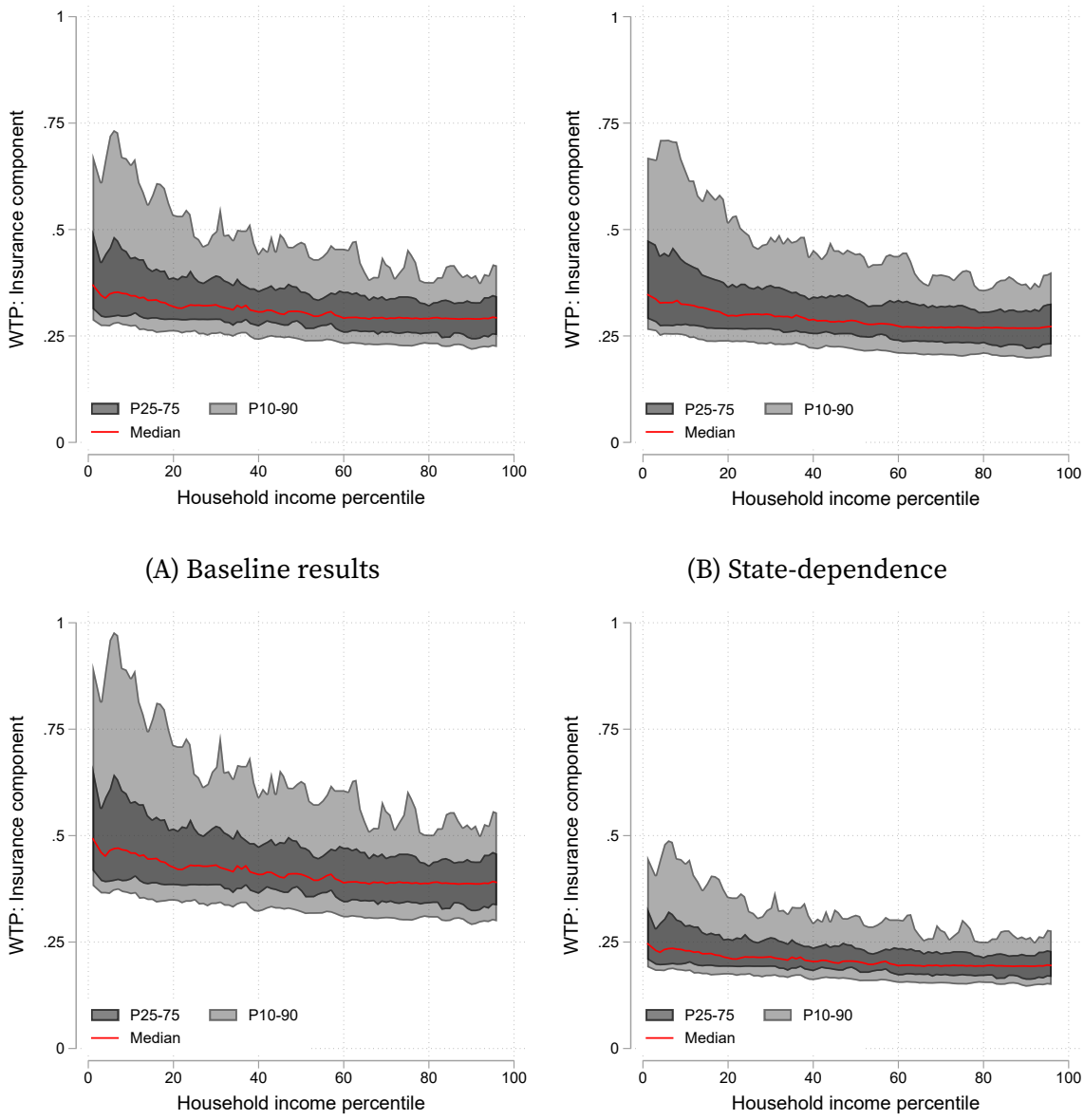
E.3. Sensitivity to Preference Specification

The willingness-to-pay results in Figure 4 assume a coefficient of risk aversion of 3 and state-independent utility. Figure E.3 presents sensitivity analyses exploring robustness to these assumptions. Since both risk-aversion and state dependence affect only the insurance component of WTP, we focus on that component.

Panel (A) reproduces the baseline distribution of the insurance component of WTP by household income (Figure 4A). Panel (B) introduces state dependence by assuming that consumption prices are 1.5% lower in unemployment, following Kaplan and Menzio (2015) and Campos and Reggio (2020).[9] This assumption affects the insurance component through two offsetting channels (see Appendix A.2). Lower prices increase the marginal value of a dollar in unemployment, increasing insurance value. However, because observed expenditure understates true consumption for the unemployed, measured expenditure-based consumption drops overstate insurance

[9] Kaplan and Menzio (2015) find that married U.S. households with one unemployed member pay 1.5% lower prices; Campos and Reggio (2020) reports similar findings for Spanish households.

values. The latter effect dominates, modestly lowering the distribution relative to the baseline.



(A) Baseline results

(B) State-dependence

(C) High risk aversion

(D) Low risk aversion

FIGURE E.3. Willingness-to-Pay for Actuarially Fair UI Expansion by Income

Notes: Each panel summarizes the distribution of workers' WTP for an actuarially fair UI expansion (i.e,, the insurance component of overall WTP), reporting the median and the 10th, 25th, 75th, and 90th percentiles. We compute worker-level WTP using estimates of individual consumption declines. Panel (A) repeats our baseline results (Figure 4A). Panel (B) assumes the price of consumption is 1.5% lower in unemployment. Panels (C) and (D) assume coefficients of relative risk aversion of 4 and 2, respectively. Quantiles are smoothed using a uniform rolling window of ± 1 percentiles.

Panels (C) and (D) show the distribution assuming coefficients of relative risk aversions of 4 and 2, respectively. As expected, the insurance component scales

proportionately with risk aversion. However, because WTP for realistic reforms is shaped primarily by cross-subsidization (Figure 4), the overall distribution of total WTP is relatively insensitive to the calibrated level of risk aversion.

F. Normative Framework: Theory

F.1. Section 5.1 Derivations

Social welfare is given by

$$W(\theta) = \int_i \omega_i V_i(\theta) di,$$

where individual expected utility, V_i, is defined in equation (1), and indexed by the reform parameter θ.

We consider the reform parameterized by equation (2). Its impact on social welfare is:

$$\frac{dW}{d\theta} = \int_i (1-e_i)\phi_{\mathcal{B}}(y_i)\omega_i u_i^{l\prime}(c_i^l) di - \left(\frac{d\tau}{db}\right) \int_i e_i \phi_{\mathcal{T}}(y_i)\omega_i u_i^{h\prime}(c_i^h) di\,.$$

We focus on a budget-neutral reform, which implies that $\frac{d\tau}{db}$ is given by equation (9), repeated here:

$$\frac{d\tau}{db} = \frac{\int_i (1-e_i)\phi_{\mathcal{B}}(y_i)\, di}{\int_i e_i \phi_{\mathcal{T}}(y_i)\, di} \left(1 + \frac{1}{\int_i (1-e_i)\phi_{\mathcal{B}}(y_i)\, di} \left(\int_i \frac{d(1-e_i)}{d\theta} (\mathcal{T}(y_i) + \mathcal{B}(y_i))\, di\right)\right).$$

Note that we can re-express the fiscal externality (i.e., the portion of the tax adjustment needed to cover behavioral responses),

$$\begin{aligned}
\text{FE} \equiv & \frac{1}{\int_i (1-e_i)\phi_{\mathcal{B}}(y_i)\, di} \left(\int_i \frac{d(1-e_i)}{d\theta} (\mathcal{T}(y_i) + \mathcal{B}(y_i))\, di\right) \\
= & \int_i \frac{(1-e_i)\phi_{\mathcal{B}}(y_i)}{\int_{i'} (1-e_{i'})\phi_{\mathcal{B}}(y_{i'})\, di'} \frac{1}{\phi_{\mathcal{B}}(y_i)} \frac{\mathcal{B}(y_i)}{1-e_i} \frac{d(1-e_i)}{d\theta} \frac{\mathcal{T}(y_i) + \mathcal{B}(y_i)}{\mathcal{B}(y_i)} di \\
= & \mathbb{E}^l \left[\epsilon^{(1-e,b)} \frac{\mathcal{T}(y) + \mathcal{B}(y)}{\mathcal{B}(y)}\right],
\end{aligned}$$

where the final line corresponds to equation (10), and uses the definition of the risk-weighted expectation (equation (8)) and the individual behavioral elasticity $\epsilon_i^{(1-e,b)} \equiv \frac{1}{\phi_{\mathcal{B}}(y_i)} \frac{d(1-e_i)}{db} \frac{\mathcal{B}(y_i)}{1-e_i}$. Note, in the definition of this elasticity db should be understood as the budget-neutral reform path.

Using equations (8) and (9), we can then re-write $dW/d\theta$:

$$\frac{dW}{d\theta} = \left(\int_i (1-e_i)\phi_{\mathcal{B}}(y_i)\,di\right)\left(\mathbb{E}^l\left[\omega u^{l\prime}(c^l)\right] - \mathbb{E}^h\left[\omega u^{h\prime}(c^h)\right](1+\text{FE})\right).$$

Define the money-metric social welfare impact as:

$$\frac{dW^{MM}}{d\theta} = \frac{dW/d\theta}{\int_i (1-e_i)\phi_{\mathcal{B}}(y_i)\,di} \Big/ \frac{-\partial W/\partial\tau}{\int_i e_i\phi_{\mathcal{T}}(y_i)\,di}.$$

Division by $\int_i(1-e_i)\phi_{\mathcal{B}}(y_i)\,di$ expresses the change in social welfare per \$1 of mechanical increase in benefit expenditure. The scaling by the inverse of $\frac{-\partial W/\partial\tau}{\int_i e_i\phi_{\mathcal{T}}(y_i)\,di}$ expresses the social welfare change in money-metric terms, relative to the value of an unfunded tax cut. Note that $-\frac{\partial W}{\partial\tau} = \int_i e_i\phi_{\mathcal{T}}(y_i)\,di\mathbb{E}^h\left[\omega u^{h\prime}(c^h)\right]$. As our implementation sets $\phi_{\mathcal{T}}(y_i) = 1$, $-\frac{\partial W}{\partial\tau}$ corresponds to the social value of a \$1 increase in high-state incomes.

Hence, we obtain equation (12), repeated here:

$$\frac{dW^{MM}}{d\theta} = \frac{\mathbb{E}^l[\omega u^{l\prime}(c^l)] - \mathbb{E}^h[\omega u^{h\prime}(c^h)]}{\mathbb{E}^h[\omega u^{h\prime}(c^h)]} - \text{FE}.$$

To decompose the resource-reallocation term, note that:

$$\begin{aligned}
\mathbb{E}^l[\omega u^{l\prime}(c^l)] - \mathbb{E}^h[\omega u^{h\prime}(c^h)] &= \mathbb{E}^l\left[\omega\left(u^{l\prime}(c^l) - \left(\frac{e\phi_{\mathcal{T}}(y)}{(1-e)\phi_{\mathcal{B}}(y)}\Big/\frac{\int_{i'} e_{i'}\phi_{\mathcal{T}}(y_{i'})di'}{\int_{i'}(1-e_{i'})\phi_{\mathcal{B}}(y_{i'})di'}\right)u^{h\prime}(c^h)\right)\right] \\
&= \mathbb{E}^l\left[\omega u^{h\prime}(c^h)\left(\frac{u^{l\prime}(c^l) - u^{h\prime}(c^h)}{u^{h\prime}(c^h)} + \left(1 - \frac{e\phi_{\mathcal{T}}(y)}{(1-e)\phi_{\mathcal{B}}(y)}\Big/\frac{\int_{i'} e_{i'}\phi_{\mathcal{T}}(y_{i'})di'}{\int_{i'}(1-e_{i'})\phi_{\mathcal{B}}(y_{i'})di'}\right)\right)\right] \\
&= \mathbb{E}^l[\omega u^{h\prime}(c^h)]\mathbb{E}^l\left[\lambda\left(\frac{u^{l\prime}(c^l) - u^{h\prime}(c^h)}{u^{h\prime}(c^h)} + \left(1 - \frac{e\phi_{\mathcal{T}}(y)}{(1-e)\phi_{\mathcal{B}}(y)}\Big/\frac{\int_{i'} e_{i'}\phi_{\mathcal{T}}(y_{i'})di'}{\int_{i'}(1-e_{i'})\phi_{\mathcal{B}}(y_{i'})di'}\right)\right)\right],
\end{aligned}$$

where $\lambda_i \equiv \frac{\omega_i u_i^{h\prime}(c_i^h)}{\mathbb{E}^l[\omega u^{h\prime}(c^h)]}$. Hence we obtain the decomposition in equation (13),

$$\text{RR} = \bar{\lambda}\mathbb{E}^l\left[\lambda\left(\frac{u^{l\prime}(c^l) - u^{h\prime}(c^h)}{u^{h\prime}(c^h)} + 1 - \frac{e\phi_{\mathcal{T}}(y)}{(1-e)\phi_{\mathcal{B}}(y)}\Big/\frac{\int_{i'} e_{i'}\phi_{\mathcal{T}}(y_{i'})di'}{\int_{i'}(1-e_{i'})\phi_{\mathcal{B}}(y_{i'})di'}\right)\right],$$

where $\bar{\lambda} \equiv \frac{\mathbb{E}^l[\omega u^{h\prime}(c^h)]}{\mathbb{E}^h[\omega u^{h\prime}(c^h)]}$ and $\frac{dW^{MM}}{d\theta} = \text{RR} - \text{FE}$. Note that:

$$\mathbb{E}^l\left[\lambda\left(1 - \frac{e\phi_{\mathcal{T}}(y)}{(1-e)\phi_{\mathcal{B}}(y)}\Big/\frac{\int_{i'} e_{i'}\phi_{\mathcal{T}}(y_{i'})di'}{\int_{i'}(1-e_{i'})\phi_{\mathcal{B}}(y_{i'})di'}\right)\right] = 1 - \mathbb{E}^h[\lambda] = 1 - \frac{1}{\bar{\lambda}}.$$

Hence we have the decomposition in equation (14):

$$\text{RR} = \bar{\lambda}\mathbb{E}^l\left[\lambda\left(\frac{u^{l\prime}(c^l) - u^{h\prime}(c^h)}{u^{h\prime}(c^h)}\right)\right] + (\bar{\lambda} - 1).$$

F.2. Extension to General Model

Under the general model outlined in Appendix A.1, the impact of the reform on social welfare is given by:

$$\frac{dW}{d\theta} = -\left(\frac{d\tau}{db}\right)\int_i \omega_i \int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{T}}(y_i)\xi_{i,t}(\varphi_{i,t})\frac{\partial u_i^h(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}))}{\partial c_{i,t}(\varphi_{i,t})} dF_{i,t}(\varphi_{i,t})\, dt\, di$$
$$+ \int_i \omega_i \int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{B}}(y_i)(1 - \xi_{i,t}(\varphi_{i,t}))\frac{\partial u_i^l(c_{i,t}(\varphi_{i,t}), x_{i,t}(\varphi_{i,t}))}{\partial c_{i,t}(\varphi_{i,t})} dF_{i,t}(\varphi_{i,t})\, dt\, di,$$

where budget-neutrality requires:

$$\frac{d\tau}{db} = \frac{\int_i \int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{B}}(y_i)\left(1 - \xi_{i,t}(\varphi_{i,t})\right) dF_{i,t}(\varphi_{i,t})\, dt\, di}{\int_i \int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{T}}(y_i)\xi_{i,t}(\varphi_{i,t}) dF_{i,t}(\varphi_{i,t})\, dt\, di} \times \left(1 + \frac{1}{\int_i \int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{B}}(y_i)\left(1 - \xi_{i,t}(\varphi_{i,t})\right) dF_{i,t}(\varphi_{i,t})\, dt\, di}\right.$$
$$\left.\int_i \frac{d}{d\theta}\left(\int_0^1 \int_{\varphi_{i,t}} (1 - \xi_{i,t}(\varphi_{i,t})) dF_{i,t}(\varphi_{i,t})\, dt\right)(\mathcal{T}(y_i) + \mathcal{B}(y_i))di\right)$$
$$= \frac{\int_i \int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{B}}(y_i)\left(1 - \xi_{i,t}(\varphi_{i,t})\right) dF_{i,t}(\varphi_{i,t})\, dt\, di}{\int_i \int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{T}}(y_i)\xi_{i,t}(\varphi_{i,t}) dF_{i,t}(\varphi_{i,t})\, dt\, di} \times \left(1 + \underbrace{\mathbb{E}^l\left[\epsilon \times \frac{\mathcal{T}(y_i) + \mathcal{B}(y_i)}{\mathcal{B}(y_i)}\right]}_{\text{FE}}\right).$$

ϵ_i is given by

$$\epsilon_i = \frac{d}{db}\left(\int_0^1 \int_{\varphi_{i,t}} (1 - \xi_{i,t}(\varphi_{i,t})) dF_{i,t}(\varphi_{i,t})\, dt\right)\frac{1}{\phi_{\mathcal{B}}(y_i)}\frac{\mathcal{B}(y_i)}{\int_0^1 \int_{\varphi_{i,t}} (1 - \xi_{i,t}(\varphi_{i,t})) dF_{i,t}(\varphi_{i,t})\, dt},$$

and the low-state risk-weighted expectation over $x_i(\varphi_{i,t})$ is given by

$$\mathbb{E}^l[x] \equiv \int_i \int_0^1 \int_{\varphi_{i,t}} \left(\frac{\phi_{\mathcal{B}}(y_i)(1 - \xi_{i,t}(\varphi_{i,t}))}{\int_{i'} \int_0^1 \int_{\varphi_{i',t}} \phi_{\mathcal{B}}(y_{i'})(1 - \xi_{i',t}(\varphi_{i',t})) dF_{i',t}(\varphi_{i',t}) dt di'}\right) x_i(\varphi_{i,t}) dF_{i,t}(\varphi_{i,t})\, dt\, di,$$

with the high-state risk-weighted expectation defined analogously (see equation (8)).

Under the standard assumption that when a worker finds a job they remain employed thereafter, $D_i \equiv \int_0^1 \int_{\varphi_{i,t}} (1 - \xi_{i,t}(\varphi_{i,t})) dF_{i,t}(\varphi_{i,t}) dt$ is the worker's expected unemployment duration, ϵ_i is the elasticity of worker i's expected unemployment duration with respect to the benefit, where the derivative is taken along the budget-neutral reform path indexed by $\phi_{\mathcal{B}}(y_i)db$, and the low-state risk-weighted

expectation weights individuals by their expected benefit rise, which depends on both expected unemployment duration and benefit adjustment, $\phi_{\mathcal{B}}(y_i)$.

To express the social welfare impact in money-metric terms per \$1 of mechanical increase in benefit expenditure, relative to an unfunded tax cut, we rescale $dW/d\theta$ according to:

$$\frac{dW^{MM}}{d\theta} = \frac{dW/d\theta}{\int_i \int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{B}}(y_i) \left(1 - \xi_{i,t}(\varphi_{i,t})\right) dF_{i,t}(\varphi_{i,t})\, dt\, di} \Big/ \frac{-\partial W/\partial \tau}{\int_i \int_0^1 \int_{\varphi_{i,t}} \phi_{\mathcal{T}}(y_i) \xi_{i,t}(\varphi_{i,t}) dF_{i,t}(\varphi_{i,t})\, dt\, di}.$$

After algebra analogous to that in the preceding subsection, we obtain:

$$\frac{dW^{MM}}{d\theta} = \frac{\mathbb{E}^l \left[\omega \frac{\partial u^l(c,x)}{\partial c}\right] - \mathbb{E}^h \left[\omega \frac{\partial u^h(c,x)}{\partial c}\right]}{\mathbb{E}^h \left[\omega \frac{\partial u^h(c,x)}{\partial c}\right]} - \text{FE}\,.$$

F.3. Relationship with Marginal Value of Public Funds

Our normative analysis considers a reform to UI generosity that raises benefits and is financed by a flat increase in payroll taxation. We focus on this combined reform because the structure of the U.S. UI system links UI benefits to a hypothecated tax. However, it is instructive to consider the benefit increase and financing tax change separately. Doing so clarifies the relationship between our analysis and the marginal value of public funds (MVPF) framework (see Hendren and Sprung-Keyser 2020).

The MVPF for an unfunded benefit increase is:

$$\text{MVPF}^{\mathcal{B}} = \frac{\mathbb{E}^l[\text{WTP}^{\mathcal{B}}]}{1 + \text{FE}^{\mathcal{B}}},$$

where the willingness-to-pay is

$$\text{WTP}_i^{\mathcal{B}} = \left(\frac{u_i^{l\prime}(c_i^l) - u_i^{h\prime}(c_i^h)}{u_i^{h\prime}(c_i^h)} + 1 \right),$$

and $\text{FE}^{\mathcal{B}}$ denotes the associated fiscal externality from behavioral responses.

Willingness-to-pay to avoid a flat tax increase is 1, and the MVPF for the tax rise is

$$\text{MVPF}^{\mathcal{T}} = \frac{1}{1 + \text{FE}^{\mathcal{T}}},$$

where $\text{FE}^{\mathcal{T}}$ is the fiscal externality associated with the payroll tax increase.

To assess whether the combined reform is socially valuable, we compare the

welfare-weighted MVPF of the benefit expansion to that of the financing tax increase. Weighting each MVPF by the incidence-weighted average marginal social welfare weights of affected groups, we require:

$$\mathbb{E}^l\left[g\frac{\text{WTP}^{\mathcal{B}}}{\mathbb{E}^l[\text{WTP}^{\mathcal{B}}]}\right]\frac{\mathbb{E}^l[\text{WTP}^{\mathcal{B}}]}{1+\text{FE}^{\mathcal{B}}} > \mathbb{E}^h[g]\frac{1}{1+\text{FE}^{\mathcal{T}}}.$$

Rewriting this inequality, we obtain:

$$\frac{\mathbb{E}^l[g]}{\mathbb{E}^h[g]}\mathbb{E}^l\left[\frac{g}{\mathbb{E}^l[g]}\text{WTP}^{\mathcal{B}}\right] - \frac{1+\text{FE}^{\mathcal{B}}}{1+\text{FE}^{\mathcal{T}}} > 0$$

$$\frac{\mathbb{E}^l[g]}{\mathbb{E}^h[g]}\mathbb{E}^l\left[\frac{g}{\mathbb{E}^l[g]}\left(\frac{u^{l\prime}(c^l)-u^{h\prime}(c^h)}{u^{h\prime}(c^h)}\right)\right] + \left(\frac{\mathbb{E}^l[g]}{\mathbb{E}^h[g]}-1\right) - \left(\frac{1+\text{FE}^{\mathcal{B}}}{1+\text{FE}^{\mathcal{T}}}-1\right) > 0.$$

Recognizing that $\bar{\lambda} = \frac{\mathbb{E}^l[g]}{\mathbb{E}^h[g]}$, $\lambda_i = \frac{g_i}{\mathbb{E}^l[g]}$ and that the fiscal externality of the combined reform in the main text is $\text{FE} = \frac{1+\text{FE}^{\mathcal{B}}}{1+\text{FE}^{\mathcal{T}}} - 1$, shows that the left-hand side is exactly the welfare condition in equations (12) and (14).[10]

G. Normative Framework: Implementation

G.1. Measuring the Fiscal Externality

The fiscal externality term decomposes as

$$\text{FE} = \mathbb{E}^l\left[\epsilon^{(1-e,b)}\right] + \mathbb{E}^l\left[\epsilon^{(1-e,b)}\frac{\mathcal{T}(y)}{\mathcal{B}(y)}\right],$$

where $\mathcal{T}(y_i)$ denotes the net payment to the government in the high state, and $\mathcal{B}(y_i)$ the net transfer received in the low state. We refer to the first component as the direct fiscal externality from UI expansion, arising from higher benefit payments, and to the second component as the indirect effect operating through changes in income tax liabilities.

In the model, the behavioral elasticity is defined along the reform path $d\theta = (db, d\tau)$, where the marginal change in low-state benefits for individual i is $db\,\phi_{\mathcal{B}}(y_i)$:

$$\epsilon_i^{(1-e,b)} \equiv \frac{1}{\phi_{\mathcal{B}}(y_i)}\frac{d(1-e_i)}{db}\frac{\mathcal{B}(y_i)}{1-e_i}.$$

[10] In practice, a natural approximation is $\text{FE}^{\mathcal{B}} \approx \text{FE}$ and $\text{FE}^{\mathcal{T}} \approx 0$, since the combined reform's fiscal externality is likely primarily driven by the benefit expansion rather than the small financing-side payroll tax increase. However, the equivalence above does not rely on this.

In the empirical implementation, however, we use estimates of the elasticity with respect to gross UI payments. Writing low-state net benefits as

$$\mathcal{B}(y_i) = ui_i - T_i^l,$$

where ui_i denotes gross UI payments and T_i^l net tax liability in the unemployment state, define the corresponding empirical elasticity as

$$\epsilon_i^{(1-e,ui)} \equiv \frac{1}{\phi_{\mathcal{B}}(y_i)} \frac{d(1-e_i)}{db} \frac{ui_i}{1-e_i} = \epsilon_i^{(1-e,b)} \frac{ui_i}{\mathcal{B}(y_i)}.$$

Using this relationship, the fiscal externality can be rewritten as

$$\text{FE} = \mathbb{E}^l \left[\epsilon^{(1-e,ui)} \right] + \mathbb{E}^l \left[\epsilon^{(1-e,ui)} \frac{T^h - T^l}{ui} \right],$$

which separates the direct fiscal externality from higher UI payments from the indirect fiscal externality operating through differences in state-specific tax liabilities. In our implementation, T_i^l excludes UI benefit payments, which enter separately through ui_i.[11]

The first component depends on risk weights (in the expectation), which we construct using our unemployment risk model and observed data on worker earnings, along with the behavioral elasticity with respect to UI benefits. We draw on elasticity estimates from Chetty (2008), who reports how the unemployment duration elasticity varies across wealth quartiles (see Table 2, p. 204 of Chetty (2008) and footnote 23 in the main paper). In the PSID, we measure wealth as total household net worth excluding housing equity, business wealth, and vehicles. We form unweighted quartiles over all worker-wave observations in our estimation sample and assign Chetty's corresponding quartile elasticity to each observation. We then average the assigned elasticities across waves within worker. This calibration assumes that relative wealth rank captures comparable variation in benefit responsiveness across the two samples.[12]

To account for benefit exhaustion, we follow Schmieder and Von Wachter (2016), who show that under a constant-hazard approximation, the fiscal cost of a benefit increase depends on both total non-employment responses and the fraction of time

[11] We abstract from the marginal taxation of additional UI payments at a given unemployment duration, treating gross and net benefit increments as equal.

[12] This calibration approximates the reform-path elasticity using benefit-only responses, treating the response to the associated financing-tax adjustment as negligible (see Appendix F.3). We also abstract from effects of UI generosity on unemployment entry and use estimates of the duration response conditional on job loss.

spent receiving benefits. Let D denote average non-employment duration and D_B average benefit duration (both measured in weeks). Then the fiscal externality can be written as

$$\text{FE} = \frac{D}{D_B}\left(\mathbb{E}^l\left[\epsilon^{(1-e,ui)}\right]\cdot\xi + \mathbb{E}^l\left[\epsilon^{(1-e,ui)}\frac{T^h - T^l}{ui}\right]\right), \tag{G.1}$$

where

$$\xi \equiv 1 - (1+sP)e^{-sP}$$

captures how benefit increases affect compensated duration relative to total duration, with P denoting potential benefit duration and s the unemployment exit hazard. We set $s = 1/D$ and use $D = 24.3$, $D_B = 15.8$ from Chetty (2008), and $P = 26$. We treat the estimated type-specific unemployment consumption declines as representative of those during benefit receipt.

The indirect component depends on risk weights, behavioral elasticities, and the state-specific tax position of households. We compute these tax positions using the NBER TAXSIM (version 32) U.S. tax simulator, implementing 2019 federal and state tax schedules.

For employed observations, we use TAXSIM to calculate annual household tax liabilities in employment and under a hypothetical D-week unemployment spell with UI receipt for D_B weeks. We account for household composition and spousal earnings, assuming joint filing for married households. In equation (G.1), we express the tax difference $T_i^h - T_i^l$ as the resulting annual tax-liability difference divided by D, and ui_i as the weekly UI benefit, equal to 50% of weekly earnings up to the cap.

G.2. Average Implementation

Our full implementation of the flat reform entails evaluating the expression

$$\frac{\mathbb{E}^l[g_y\tilde{g}]}{\mathbb{E}^h[g_y\tilde{g}]}\mathbb{E}^l\left[\frac{g_y\tilde{g}}{\mathbb{E}^l[g_y\tilde{g}]}\gamma\frac{\Delta c}{c^h}\right] + \left(\frac{\mathbb{E}^l[g_y\tilde{g}]}{\mathbb{E}^h[g_y\tilde{g}]} - 1\right) - \frac{D}{D_B}\left(\mathbb{E}^l\left[\epsilon^{(1-e,ui)}\right]\cdot\xi + \mathbb{E}^l\left[\epsilon^{(1-e,ui)}\frac{T^h - T^l}{ui}\right]\right),$$

which is obtained by combining equations (4), (12), (14), (15) and (G.1). Recall that the welfare weights include a cross-income (inverse-optimum) component g_y and a within-income component $\tilde{g}_i = \frac{(c_i^h)^{-\gamma}}{\mathbb{E}[(c^h)^{-\gamma}|y_i]}$.

In the average implementation, we treat the employed and unemployed populations as representative individuals with model-implied average values. We abstract from within-income differences in social marginal welfare weights by setting $\tilde{g}_i = 1$, so that $g_i = g_{y_i}$, and we evaluate other terms in the social welfare

expression by replacing worker-level objects with their model-implied state-specific averages. Specifically, for $s \in \{h, l\}$

$$\widetilde{\mathbb{E}}^h[x] = \int_i \left(\frac{e_i}{\int_{i'} e_{i'} di'} \right) x_i \, di \qquad \widetilde{\mathbb{E}}^l[x] = \int_i \left(\frac{1 - e_i}{\int_{i'} (1 - e_{i'}) di'} \right) x_i \, di,$$

where e_i is the worker's estimated high-state probability. We use these state-specific averages for the income-based welfare weight (g_y), consumption drop ($\Delta c / c^h$), behavioral elasticity ($\epsilon^{(1-e,ui)}$), UI payments (ui), and difference in state-specific tax liabilities ($T^h - T^l$). This leads to the simplified expression:

$$\frac{\widetilde{\mathbb{E}}^l[g_y]}{\widetilde{\mathbb{E}}^h[g_y]} \gamma \widetilde{\mathbb{E}}^l \left[\frac{\Delta c}{c^h} \right] + \left(\frac{\widetilde{\mathbb{E}}^l[g_y]}{\widetilde{\mathbb{E}}^h[g_y]} - 1 \right) - \frac{D}{D_B} \left(\widetilde{\mathbb{E}}^l \left[\epsilon^{(1-e,ui)} \right] \cdot \xi + \widetilde{\mathbb{E}}^l \left[\epsilon^{(1-e,ui)} \right] \frac{\widetilde{\mathbb{E}}^l \left[T^h - T^l \right]}{\widetilde{\mathbb{E}}^l \left[ui \right]} \right).$$

The average implementation captures differences in average welfare weights between the employed and unemployed populations, but it abstracts from within-group heterogeneity. As a result, it is invariant across reform structure, since differences between reforms stem from heterogeneity in incidence within the low state.

H. Additional Welfare Results

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Resource reallocation			Fiscal externality			Total effect
	Risk protection	Cross-subsidization	Total (1)+(2)	Direct	Indirect	Total (4)+(5)	(3)-(6)
H=2, K=3 (Baseline)	0.483	0.190	0.673	0.268	0.309	0.577	0.096
H=1, K=3 (-1 H)	0.449	0.130	0.579	0.264	0.313	0.577	0.002
H=3, K=3 (+1 H)	0.478	0.192	0.670	0.262	0.337	0.599	0.071
H=2, K=2 (-1 K)	0.474	0.190	0.663	0.268	0.309	0.577	0.086
H=2, K=4 (+1 K)	0.542	0.191	0.732	0.268	0.309	0.577	0.155

TABLE H.1. Robustness of Social Value of Expanding UI Generosity by Number of Latent Types

Notes: Column (7) reports the total social welfare impact of a UI expansion, decomposed in columns (1)–(6). H denotes the number of unemployment-risk types and K the number of consumption types; the baseline results in the paper correspond to $H = 2, K = 3$. Entries correspond to an expansion of the current U.S. UI system under baseline social preferences and are expressed in dollars per $1 mechanical increase in benefit expenditure.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Resource reallocation			Fiscal externality			Total effect
	Risk protection	Cross-subsidization	Total (1)+(2)	Direct	Indirect	Total (4)+(5)	(3)-(6)
Baseline	0.483	0.190	0.673	0.268	0.309	0.577	0.096
Age	0.462	0.130	0.591	0.264	0.313	0.577	0.015
Liquid Wealth	0.448	0.130	0.578	0.264	0.313	0.577	0.001
Permanent Income	0.443	0.130	0.573	0.264	0.313	0.577	-0.004

TABLE H.2. Robustness of Social Value of Expanding UI Generosity by Observable Only Heterogeneity

Notes: Column (7) reports the total social welfare impact of a UI expansion, decomposed in columns (1)–(6). Entries correspond to an expansion of the current U.S. UI system under baseline social preferences and are expressed per $1 of mechanical increase in benefit expenditure.

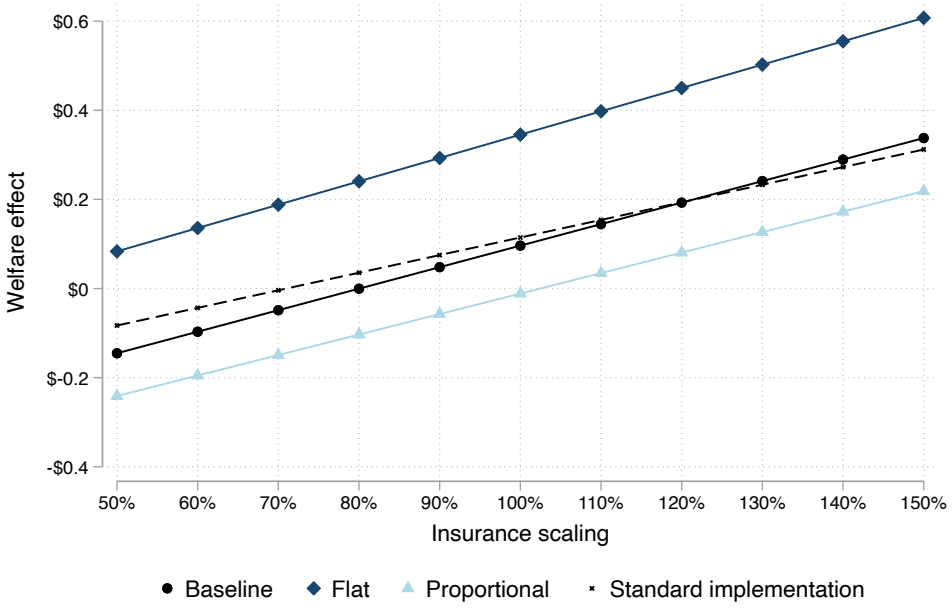


FIGURE H.1. Sensitivity of Welfare Effects to Insurance Value

Notes: Each line plots the net social value per $1 of mechanical increase in benefit expenditure as a function of a scaling factor s applied to the individual-level insurance component of WTP, holding the cross-subsidization and fiscal-externality components fixed. Values of $s < 100\%$ correspond to proportionally smaller consumption drops (or lower risk aversion), and $s > 100\%$ to larger drops (or higher risk aversion). The figure reports results for the baseline reform (solid circles), a flat benefit expansion (squares), a proportional expansion (triangles), and the standard homogeneous implementation (dashed).

H.1. Heterogeneous Take-Up and Benefit Exhaustion

Our baseline implementation does not allow take-up rates or benefit exhaustion to vary across unemployed workers beyond an average exhaustion adjustment. However, the framework can incorporate richer measures of expected benefit receipt when such evidence is available. In this appendix, we illustrate this by undertaking a simple sensitivity analysis that allows both take-up and benefit exhaustion to vary over the wage distribution.

Let L_i denote the probability that individual i takes up UI when unemployed, and let

$$\rho_i \equiv \frac{D_{B,i}}{D_i}$$

denote the share of unemployment duration during which benefits are received. With heterogeneous incomplete take-up and exhaustion, the mechanical incidence of a benefit expansion is proportional to

$$L_i \rho_i (1 - e_i) \phi_{\mathcal{B}}(y_i),$$

rather than $(1 - e_i)\phi_{\mathcal{B}}(y_i)$. We recompute the resource-reallocation terms replacing the baseline benefit-incidence weights with these adjusted weights.

We calibrate the take-up and exhaustion gradients using evidence from Lachowska et al. (2025), who report UI claiming and benefit duration by pre-separation wages based on administrative data for Washington state. We map their wage gradients into our data using deciles of the wage distribution. For take-up, we assign rates across wage deciles of

$$0.20,\ 0.25,\ 0.30,\ 0.35,\ 0.375,\ 0.40,\ 0.40,\ 0.425,\ 0.425,\ 0.35\,.$$

For benefit exhaustion, we assign average benefit durations across wage deciles of

$$22,\ 21,\ 20,\ 19,\ 18,\ 18,\ 17,\ 16,\ 16,\ 18$$

weeks. We then use the constant-hazard relationship

$$D_{B,i} = D_i \left(1 - \exp(-P/D_i)\right),$$

with potential benefit duration $P = 26$, to recover the implied covered share $\rho_i = D_{B,i}/D_i$. This yields duration ratios across wage deciles of

$$0.29,\ 0.36,\ 0.42,\ 0.48,\ 0.53,\ 0.53,\ 0.59,\ 0.65,\ 0.65,\ 0.53\,.$$

Thus, lower-wage workers are assigned lower take-up rates and, under the constant-hazard mapping, lower covered shares of unemployment duration.

Table H.3 reports the resulting resource-reallocation terms. Allowing for heterogeneous take-up alone reduces the resource-reallocation gain from \$0.67 to \$0.62 per \$1 of mechanical increase in benefit expenditure. Allowing for heterogeneous exhaustion alone produces a similar reduction. Combining both adjustments reduces the resource-reallocation gain to \$0.57. The majority of this reduction is through the cross-subsidization component, which falls from \$0.19 in the baseline to \$0.13 when both adjustments are applied. These results indicate that take-up and exhaustion gradients attenuate the extent of cross-income redistribution generated by the reform, but leave the qualitative conclusion unchanged: accounting for heterogeneous incidence remains important for evaluating UI expansions.

TABLE H.3. Resource-Reallocation Gains with Heterogeneous Take-Up and Exhaustion

	(1)	(2)	(3)
	Risk protection	Cross-subsidization	Total (1)+(2)
Baseline	0.48	0.19	0.67
Heterogeneous take-up	0.46	0.16	0.62
Heterogeneous exhaustion	0.46	0.16	0.62
Heterogeneous take-up + exhaustion	0.44	0.13	0.57

Notes: The first row reports our baseline results. The heterogeneous take-up adjustment assigns UI take-up rates by wage decile. The heterogeneous exhaustion adjustment assigns benefit durations by wage decile and maps them into covered duration shares using a constant-hazard model with potential benefit duration $P = 26$ weeks. Entries correspond to an expansion of the current U.S. UI system under baseline social preferences and are expressed per $1 of mechanical increase in benefit expenditure.

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